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Started 15th June 2026

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Chapter 1

Construction of the Real Numbers

Definition 1.1. *Let S be a set. A **binary operation** on S is a function $S \times S \to S$. A **unary operation** on S is a function $S \to S$.*

Let S be a set and let $* : S \times S \to S$ be a binary operation. If $x, y \in S$, the correct way to write the result of the operation to the pair $\langle x, y \rangle$ would be $*(\langle x, y \rangle)$. However since this is a bit cumbersome one may write $x * y$ instead. Similarly if $\neg : S \to S$ is a unary operation and $x \in S$ one may choose to write $\neg x$ over $\neg(x)$.

1.1 Axioms for the Natural numbers

1.1.1 Peano Postulates

Axiom: Peano Postulates
There exists a set $\mathbb{N}$ with an element $1 \in \mathbb{N}$ and a function $s : \mathbb{N} \to \mathbb{N}$ that satisfy the following three properties.

- There is no $n \in \mathbb{N}$ such that $s(n) = 1$.
- The function s is injective.
- Let $G \subseteq \mathbb{N}$ be a set. Suppose that $1 \in G$, and that if $g \in G$ then $s(g) \in G$. Then $G = \mathbb{N}$.

It will be shown that the set $\mathbb{N}$ is unique. We can therefore give it a name.

Definition 1.2. *The set of **natural numbers** denoted $\mathbb{N}$, is the set the existence of which is given in the Peano Postulates.*

1.1.2 Consequences of Peano Postulates

Lemma 1.3. *Let $a \in \mathbb{N}$. Suppose that $a \neq 1$. Then there is a unique $b \in \mathbb{N}$ such that $a = s(b)$. (where existence of s and $\mathbb{N}$ are outlined by the peano axioms)*

Proof. First we prove existence. Let

$$G = \{1\} \cup \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$$

(which exists by the subset, pairing, and union axioms) We will prove that $G = \mathbb{N}$. To do this we need to show $1 \in G$, $G \subseteq \mathbb{N}$ and $\forall g \in G(s(g) \in G)$. The first two are straightforward. For the third suppose $g \in G$; we have two cases. First suppose $g \in \{1\}$; then $g = 1 \in \mathbb{N}$. Since $s : \mathbb{N} \to \mathbb{N}$, we then have $s(1) \in \mathbb{N}$. So by definition $s(1) \in \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$ and $s(g) = s(1) \in G$. Now suppose $g \in \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$; then by definition $g \in \mathbb{N}$ and $s(g) \in \mathbb{N}$ exists. By definition $s(g) \in \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$ so $s(g) \in G$. By the third peano postulate we then can assert that $G = \mathbb{N}$. So $a \in \mathbb{N} = G$. Since $a \neq 1$ we must then have $a \in \{c \in \mathbb{N} \mid \exists b \in \mathbb{N}(s(b) = c)\}$ and by definition, $\exists b \in \mathbb{N}(s(b) = a)$.

Now for uniqueness. Let m and n be arbitrary elements of $\mathbb{N}$ such that $a = s(m)$ and $a = s(n)$. Then $\langle m, a\rangle \in s$ and $\langle n, a\rangle \in s$. Since s is injective we must have $n = m$. □

Theorem 1.4. *Let H be a set, Let $e \in H$ and let $k : H \to H$ be a function. Then there is a unique function $f : \mathbb{N} \to H$ such that $f(1) = e$ and $f \circ s = k \circ f$.*

Proof. Letting H, e, and k be arbitrary such that $e \in H$ and $k : H \to H$. We will show existence and uniqueness separately. We start with uniqueness.

Let g and h be arbitrary such that $g : \mathbb{N} \to H$, $g(1) = e$, and $g \circ s = k \circ g$ and $h : \mathbb{N} \to H$, $h(1) = e$, and $h \circ s = k \circ h$. We will first define V as the unique set (guaranteed by the subset and extensionality axioms) such that

$$\forall a(a \in V \leftrightarrow a \in \mathbb{N} \wedge g(a) = h(a))$$

and show that $V = \mathbb{N}$. By peano's postulates, we need to show $V \subseteq \mathbb{N}$, $1 \in V$, and $\forall n \in V(s(n) \in V)$. The first is straightforward. For the second requirement, we know by peano's postulates that $1 \in \mathbb{N}$, so by our assumptions $g(1)$ and $h(1)$ exist and $g(1) = e = h(1)$. So $1 \in V$.

For the third requirement, let n be arbitrary and suppose that $n \in V$. Then by definition $n \in \mathbb{N}$ and $g(n) = h(n)$. Since $n \in \mathbb{N} = \text{dom}(h \circ s)$, we have, by our assumptions,

$$h(s(n)) = h \circ s(n) = k \circ h(n) = k(h(n)) = k(g(n)) = k \circ g(n) = g \circ s(n) = g(s(n))$$

we know by definition $\text{ran}(s) \subseteq \mathbb{N}$, so $s(n) \in \mathbb{N}$ and again by definition $s(n) \in V$. We can conclude that $V = \mathbb{N}$.

Suppose $\langle x, y\rangle \in g$, then $x \in \text{dom}g = \mathbb{N} = V$. So by definition $y = g(x) = h(x)$, and $\langle x, y\rangle \in h$. Suppose $\langle x, y\rangle \in h$, then $x \in \text{dom}h = \mathbb{N} = V$. So by definition $y = h(x) = g(x)$, and $\langle x, y\rangle \in g$.
(next page)

Now for existence. We need to come up with a function from $\mathbb{N}$ into H such that $f(1) = e$ and $f \circ s = k \circ f$. We first define C, where

$$\forall X(X \in C \leftrightarrow X \subseteq \mathbb{N} \times H \wedge \langle 1, e\rangle \in X \wedge \forall n \forall y(\langle n, y\rangle \in X \rightarrow \langle s(n), k(y)\rangle \in X))$$

(guaranteed by the subset and extensionality axioms.) We propose that $f = \bigcap C$ is the desired function. To ensure its existence, we need to show that C is nonempty. We will show that $\mathbb{N} \times H \in C$. For the first requirement, clearly $\mathbb{N} \times H \subseteq \mathbb{N} \times H$. Next, since $1 \in \mathbb{N}$ and $e \in H$ then $\langle 1, e\rangle \in \mathbb{N} \times H$. Finally, let n and y be arbitrary such that $\langle n, y\rangle \in \mathbb{N} \times H$. Since $s : \mathbb{N} \rightarrow \mathbb{N}$ and $k : H \rightarrow H$, we must then have $s(n) \in \mathbb{N}$ and $k(y) \in H$. So $\langle s(n), k(y)\rangle \in \mathbb{N} \times H$. $f = \bigcap C$ is therefore a valid candidate.

Before we prove the functionality of f, we first state a few facts about f. First we note that $f \in C$. Let arbitrary $x \in \bigcap C$. Since C is nonempty there exists $X \in C$, and by definition $x \in X \subseteq \mathbb{N} \times H$. So $\bigcap C \subseteq \mathbb{N} \times H$. Next let arbitrary $X \in C$. By definition $\langle 1, e\rangle \in X$, so $\langle 1, e\rangle \in \bigcap C$. Finally let n and y be arbitrary and suppose $\langle n, y\rangle \in \bigcap C$. We need to show that $\langle s(n), k(y)\rangle \in \bigcap C$. Let arbitrary $X \in C$. We have $\langle n, y\rangle \in X$. Since $X \in C$, by definition we must then have $\langle s(n), k(y)\rangle \in X$. We are done.

Next we note that $\forall X \in C(f \subseteq X)$. This is straightforward to prove. Finally we show

$$\begin{aligned}\forall n \forall y([\langle n, y\rangle \in f \wedge \langle n, y\rangle \neq \langle 1, e\rangle] \\ \rightarrow \exists m \exists u(\langle m, u\rangle \in f \wedge \langle n, y\rangle = \langle s(m), k(u)\rangle))\end{aligned}$$

Let n and y be arbitrary such that $\langle n, y\rangle \in f$ and $\langle n, y\rangle \neq \langle 1, e\rangle$. Suppose, for contradiction, that the statement after the implication is false. Then $\forall m \forall u(\langle m, u\rangle \in f \rightarrow \langle n, y\rangle \neq \langle s(m), k(u)\rangle)$. The set $f - \{\langle n, y\rangle\}$ exists by the subset axiom; we will show that this set is a member of C. We know that $f \in C$, so $f \subseteq \mathbb{N} \times H$, and it is straightforward to show that $(f - \{\langle n, y\rangle\}) \subseteq \mathbb{N} \times H$. Since $\langle 1, e\rangle \neq \langle n, y\rangle$ and $\langle 1, e\rangle \in f$, we also have $\langle 1, e\rangle \in f - \{\langle n, y\rangle\}$. Finally, let arbitrary $\langle a, b\rangle \in f - \{\langle n, y\rangle\}$. Then $\langle a, b\rangle \in f \in C$ and $\langle s(a), k(b)\rangle \in f$. Since $\langle a, b\rangle \in f$, we must have, by our assumption, $\langle n, y\rangle \neq \langle s(a), k(b)\rangle$ and $\langle s(a), k(b)\rangle \in f - \{\langle n, y\rangle\}$. So $f - \{\langle n, y\rangle\} \in C$. It is straightforward to show that $f \nsubseteq f - \{\langle n, y\rangle\}$, contradicting the fact that $\forall X \in C(f \subseteq X)$.

We now prove functionality. We will do this by defining G:

$$\forall z(z \in G \leftrightarrow z \in \mathbb{N} \wedge \exists! x \in H(\langle z, x\rangle \in f))$$

(which exists by the subset axiom) and using the peano postulates to show that $G = \mathbb{N}$.

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We defined G:

$$\forall z(z \in G \leftrightarrow z \in \mathbb{N} \wedge \exists! x \in H(\langle z, x\rangle \in f))$$

To show that $G = \mathbb{N}$, we need to show that $G \subseteq \mathbb{N}$, $1 \in G$, and $\forall n \in G(s(n) \in G)$. The first requirement is straightforward. For the second, we already know by definition that $1 \in \mathbb{N}$. Since $f \in C$ we know by definition that $\langle 1, e\rangle \in f$, where $e \in H$. Let arbitrary $x' \in H$ such that $\langle 1, x'\rangle \in f$. Suppose for contradiction that $e \neq x'$. Then we have $\langle 1, e\rangle \neq \langle 1, x'\rangle$; so by the third statement proven earlier there exists m, u such that $\langle m, u\rangle \in f$ and $\langle 1, x'\rangle = \langle s(m), k(u)\rangle$. This means that $1 = s(m)$, where since $f \subseteq \mathbb{N} \times H$, $m \in \mathbb{N}$. This contradicts the peano postulates, which assert that $\neg\exists n \in \mathbb{N}(s(n) = 1)$.

For the third and final requirement, let n be an arbitrary element of G. Then by definition $n \in \mathbb{N}$ and there exists a unique x such that $\langle n, x\rangle \in f$. By definition we know $s(n) \in \mathbb{N}$, and since $f \in C$, we have $\langle s(n), k(x)\rangle \in f$, this verifies existence. Now let x' be arbitrary such that $\langle s(n), x'\rangle \in f$. By peano's postulates we know that $s(n) \neq 1$, so $\langle s(n), x'\rangle \neq \langle 1, e\rangle$. By the third statement proven earlier there exists m, u such that $\langle m, u\rangle \in f$ and $\langle s(n), x'\rangle = \langle s(m), k(u)\rangle$. So $s(n) = s(m)$ and $x' = k(u)$. Since s is injective we must have $m = n$. So $\langle n, u\rangle \in f$. By uniqueness of x we then have $u = x$. So $x' = k(u) = k(x)$, proving uniqueness. $s(n) \in G$.

We have shown that $G = \mathbb{N}$. It follows from this that $f : \mathbb{N} \to H$. For functionality, let $x \in \mathrm{dom} f$. Then there exists y such that $\langle x, y\rangle \in f$. Since $f \subseteq \mathbb{N} \times H$ we have $x \in \mathbb{N} = G$, and by definition y is unique. Next to show $\mathrm{dom} f = \mathbb{N}$, first let arbitrary $x \in \mathrm{dom} f$, similarly it immediately follows that $x \in \mathbb{N}$. Now let $x \in \mathbb{N}$. Then $x \in G$ and by definition there exists $y \in H$ such that $\langle x, y\rangle \in f$, meaning $x \in \mathrm{dom} f$. Finally to show $\mathrm{ran} f \subseteq H$, let $y \in \mathrm{ran} f$. Then there exists x such that $\langle x, y\rangle \in f \subseteq \mathbb{N} \times H$, so $y \in H$.

The fact that $\langle 1, e\rangle \in \bigcap C = f$ was already shown earlier. Finally we show $f \circ s = k \circ f$. Let $\langle x, y\rangle \in f \circ s$. Then $y = f \circ s(x) = f(s(x))$. Since $s(x) \neq 1$, $\langle s(x), y\rangle \neq \langle 1, e\rangle$ and as proven earlier there then exists m, u such that $\langle m, u\rangle \in f$ and $\langle s(x), y\rangle = \langle s(m), k(u)\rangle$. So $s(x) = s(m)$ and $y = k(u)$. Since s is injective $x = m$. We then have $\langle x, u\rangle \in f$ and $y = k(u) = k(f(x)) = k \circ f(x)$. so $\langle x, y\rangle \in k \circ f$. Next, suppose $\langle x, y\rangle \in k \circ f$. By definition there exists z such that $\langle x, z\rangle \in f$ and $\langle z, y\rangle \in k$. Since $f \in C$ we have $\langle s(x), k(z)\rangle \in f$. We then have $y = k(z) = f(s(x) = f \circ s(x)$ and $\langle x, y\rangle \in f \circ s$. We are done. $\square$

1.1.3 Addition on $\mathbb{N}$

Theorem 1.5. *There is a unique binary operation $+ : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ that satisfies the following two properties for all $n, m \in \mathbb{N}$:*

- $n + 1 = s(n)$
- $n + s(m) = s(n + m)$

Proof. We start with uniqueness. Let $+$ and $\oplus$ be arbitrary such that $+, \oplus : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$, $\forall m \in \mathbb{N} \forall n \in \mathbb{N}(+(\langle n, 1\rangle) = s(n) \wedge +(\langle n, s(m)\rangle) = s(+(\langle n, m\rangle)))$, and $\forall m \in \mathbb{N} \forall n \in \mathbb{N}(\oplus(\langle n, 1\rangle) = s(n) \wedge \oplus(\langle n, s(m)\rangle) = s(\oplus(\langle n, m\rangle)))$. We need to show $+ = \oplus$. We define G as

$$\forall x(x \in G \leftrightarrow x \in \mathbb{N} \wedge \forall n \in \mathbb{N}(+(\langle n, x\rangle) = \oplus(\langle n, x\rangle)))$$

(guaranteed by the subset axiom.) We will show that $G = \mathbb{N}$. To do this we need to show $G \subseteq \mathbb{N}$, $1 \in G$, and $\forall g \in G(s(g) \in G)$. The first requirement is straightforward. For the second requirement, we already know $1 \in \mathbb{N}$, and for arbitrary $n \in \mathbb{N}$, $+(\langle n, 1\rangle) = s(n) = \oplus(\langle n, 1\rangle)$.

For the third requirement, let arbitrary $g \in G$. Then by definition $g \in \mathbb{N}$ and $\forall n \in \mathbb{N}(+(\langle n, g\rangle) = \oplus(\langle n, g\rangle))$. Since $s : \mathbb{N} \to \mathbb{N}$ we know that $s(g) \in \mathbb{N}$ exists. Let n be arbitrary. By definition $+(\langle n, s(g)\rangle) = s(+(\langle n, g\rangle))$ and $\oplus(\langle n, s(g)\rangle) = s(\oplus(\langle n, g\rangle))$. By assumption we can assert that $+(\langle n, g\rangle) = \oplus(\langle n, g\rangle)$, so $+(\langle n, s(g)\rangle) = s(+(\langle n, g\rangle)) = s(\oplus(\langle n, g\rangle)) = \oplus(\langle n, s(g)\rangle)$. We conclude that $s(g) \in G$ and $G = \mathbb{N}$.

Let $\langle\langle x, y\rangle, z\rangle \in +$. By definition $y \in \mathbb{N} = G$ and $x \in \mathbb{N}$. So $+(\langle x, y\rangle) = \oplus(\langle x, y\rangle)$ and $z = \oplus(\langle x, y\rangle)$. So $\langle\langle x, y\rangle, z\rangle \in \oplus$. Now let $\langle\langle x, y\rangle, z\rangle \in \oplus$. A similar argument shows that $\langle\langle x, y\rangle, z\rangle \in +$.

Now for existence. We want to show

$$\exists + (+ : \mathbb{N} \times \mathbb{N} \to \mathbb{N} \wedge \forall m \in \mathbb{N} \forall n \in \mathbb{N}(+(\langle n, 1\rangle) = s(n) \\ \wedge +(\langle n, s(m)\rangle) = s(+(\langle n, m\rangle))))$$

Theorem 1.4 allows us to assert

$$\forall k \forall H \forall e([e \in H \wedge k : H \to H] \to \exists! f(f : \mathbb{N} \to H \wedge f(1) = e \wedge f \circ s = k \circ f))$$

It immediately follows that

$$\forall p \in \mathbb{N} \exists! f(f : \mathbb{N} \to \mathbb{N} \wedge f(1) = p \wedge f \circ s = s \circ f)$$

We will define $+$ as the set where

$$\forall z(z \in + \leftrightarrow \exists a \in \mathbb{N} \exists b \in \mathbb{N}(\exists f(f : \mathbb{N} \to \mathbb{N} \wedge f(1) = s(a) \wedge f \circ s = s \circ f \\ \wedge \exists c(\langle b, c\rangle \in f \wedge z = \langle\langle a, b\rangle, c\rangle))))$$

Existence follows from the subset axiom, where it can be shown that $+ \subseteq (\mathbb{N} \times \mathbb{N}) \times \mathbb{N}$.
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We defined + as

$$\forall z(z \in + \leftrightarrow \exists a \in \mathbb{N}\exists b \in \mathbb{N}(\exists f(f : \mathbb{N} \to \mathbb{N} \wedge f(1) = s(a) \wedge f \circ s = s \circ f \\ \wedge \exists c(\langle b, c\rangle \in f \wedge z = \langle\langle a, b\rangle, c\rangle))))$$

First we prove functionality. Let x be an arbitrary element of dom(+). Then there exists y such that $\langle x, y\rangle \in +$. Let y' be arbitrary such that $\langle x, y'\rangle \in +$. $\langle x, y\rangle \in +$ by definition means that there exists $a, b \in \mathbb{N}$, and $f : \mathbb{N} \to \mathbb{N}$ such that $f(1) = s(a)$ and $f \circ s = s \circ f$, and that there exists c such that $\langle b, c\rangle \in f$ and $\langle x, y\rangle = \langle\langle a, b\rangle, c\rangle$. $\langle x, y'\rangle \in +$ similarly implies the existence of $a_0, b_0 \in \mathbb{N}$ and $f_0 : \mathbb{N} \to \mathbb{N}$ such that $f_0(1) = s(a_0)$ and $f_0 \circ s = s \circ f_0$, and the existence of c_0 such that $\langle b_0, c_0\rangle \in f_0$ and $\langle x, y'\rangle = \langle\langle a_0, b_0\rangle, c_0\rangle$. Then $\langle a, b\rangle = x = \langle a_0, b_0\rangle$, so $a_0 = a$ and $b_0 = b$. It follows that $s(a_0) = s(a) \in \mathbb{N}$. This and

$$\forall p \in \mathbb{N}\exists! f(f : \mathbb{N} \to \mathbb{N} \wedge f(1) = p \wedge f \circ s = s \circ f)$$

implies that $f = f_0$. So we have $y = c = f(b) = f(b_0) = f_0(b_0) = c_0 = y'$.

Next we show that $\mathrm{dom}(+) = \mathbb{N} \times \mathbb{N}$. First let x be an arbitrary element of dom(+). Then there exists y such that $\langle x, y\rangle \in +$. By definition there then exists $a, b \in \mathbb{N}$ and there exists c such that $\langle x, y\rangle = \langle\langle a, b\rangle, c\rangle$. So $x = \langle a, b\rangle \in \mathbb{N} \times \mathbb{N}$.

Now suppose $x \in \mathbb{N} \times \mathbb{N}$. Then by definition there exists $a, b \in \mathbb{N}$ such that $x = \langle a, b\rangle$. $s(a) \in \mathbb{N}$ exists, so by theorem 1.4 there exists f such that $f : \mathbb{N} \to \mathbb{N}$, $f(1) = s(a)$, and $f \circ s = s \circ f$. Since $b \in \mathbb{N} = \mathrm{dom} f$, $f(b)$ exists and $\langle b, f(b)\rangle \in f$. Then by definition, $\langle\langle a, b\rangle, f(b)\rangle \in +$, so $x = \langle a, b\rangle \in \mathrm{dom}(+)$.

Next we show that $\mathrm{ran}(+) \subseteq \mathbb{N}$. Let arbitrary $y \in \mathrm{ran}(+)$. Then there exists x such that $\langle x, y\rangle \in +$. By definition there then exists $a, b \in \mathbb{N}$, $f : \mathbb{N} \to \mathbb{N}$, and c such that $\langle b, c\rangle \in f$ and $\langle x, y\rangle = \langle\langle a, b\rangle, c\rangle$. So we have $y = c = f(b) \in \mathbb{N}$.

Now we verify the two properties of addition, that is,

$$\forall m \in \mathbb{N}\forall n \in \mathbb{N}(+(\langle n, 1\rangle) = s(n) \wedge +(\langle n, s(m)\rangle) = s(+(\langle n, m\rangle)))$$

Let arbitrary $n, m \in \mathbb{N}$. We know that $+ : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$, so $\langle n, 1\rangle \in \mathbb{N} \times \mathbb{N} = \mathrm{dom}(+)$ and there exists y such that $\langle\langle n, 1\rangle, y\rangle \in +$. By definition there then exists $a, b \in \mathbb{N}$ and $f : \mathbb{N} \to \mathbb{N}$ such that $f(1) = s(a)$, and there exists c such that $\langle b, c\rangle \in f$ and $\langle\langle n, 1\rangle, y\rangle = \langle\langle a, b\rangle, c\rangle$. Then $n = a$, $1 = b$, and $y = c$. We then have $\langle 1, y\rangle \in f$ and $y = f(1) = s(a) = s(n)$. We conclude that $\langle\langle n, 1\rangle, s(n)\rangle \in +$ and $+(\langle n, 1\rangle) = s(n)$.

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Now for the second property. We know that $s(m) \in \mathbb{N}$. So $\langle n, s(m)\rangle \in \mathbb{N} \times \mathbb{N} = \text{dom}(+)$ and there exists y such that $\langle\langle n, s(m)\rangle, y\rangle \in +$. By definition there then exists $a, b \in \mathbb{N}$ such that $f : \mathbb{N} \to \mathbb{N}$, $f(1) = s(a)$, and $f \circ s = s \circ f$, and there exists c such that $\langle b, c\rangle \in f$ and $\langle\langle n, s(m)\rangle, y\rangle = \langle\langle a, b\rangle, c\rangle$. We then have $n = a$, $s(m) = b$, and $y = c = f(b) = f(s(m)) = s(f(m))$. We know that $\langle n, m\rangle \in \mathbb{N} \times \mathbb{N} = \text{dom}(+)$, so there exists y_0 such that $\langle\langle n, m\rangle, y_0\rangle \in +$. Again by definition there exists $a_0, b_0 \in \mathbb{N}$ and $f_0 : \mathbb{N} \to \mathbb{N}$ such that $f_0(1) = s(a_0)$ and $f_0 \circ s = s \circ f_0$ and there exists c_0 such that $\langle b_0, c_0\rangle \in f_0$ and $\langle\langle n, m\rangle, y_0\rangle = \langle\langle a_0, b_0\rangle, c_0\rangle$. Since it follows that $a_0 = n = a$ and $s(a_0) = s(a)$, theorem 1.4 allows us to assert $f = f_0$. Putting it all together, we have $+(\langle n, m\rangle) = y_0 = c_0 = f_0(b_0) = f_0(m) = f(m)$. We conclude that $+(\langle n, s(m)\rangle) = y = s(f(m)) = s(+(\langle n, m\rangle))$. □

1.1.4 Multiplication on $\mathbb{N}$

Theorem 1.6. *There is a unique binary operation $\cdot : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ that satisfies the following two properties for all $n, m \in \mathbb{N}$:*

- $n \cdot 1 = n$
- $n \cdot s(m) = (n \cdot m) + n$

Proof. We start with uniqueness. Let $\cdot, \odot$ be arbitrary. Suppose $\cdot : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ and $\forall m \in \mathbb{N} \forall n \in \mathbb{N}(\cdot(\langle n, 1\rangle) = n \wedge \cdot(\langle n, s(m)\rangle) = +(\langle\cdot(\langle n, m\rangle), n\rangle))$. Similarly suppose $\odot : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ and $\forall m \in \mathbb{N} \forall n \in \mathbb{N}(\odot(\langle n, 1\rangle) = n \wedge \odot(\langle n, s(m)\rangle) = +(\langle\odot(\langle n, m\rangle), n\rangle))$. We want to show that $\cdot = \odot$. We do this by defining G such that

$$\forall z(z \in G \leftrightarrow z \in \mathbb{N} \wedge \forall n \in \mathbb{N}(\cdot(\langle n, z\rangle) = \odot(\langle n, z\rangle)))$$

(where existence is assured by the subset axiom) and showing that $G = \mathbb{N}$.

We need to show $G \subseteq \mathbb{N}$, $1 \in G$, and $\forall g \in G(s(g) \in G)$. The first requirement is straightforward. For the second, we already know $1 \in \mathbb{N}$, and see that for arbitrary $n \in \mathbb{N}$ we have $\cdot(\langle n, 1\rangle) = n = \odot(\langle n, 1\rangle)$.

For the third requirement let arbitrary $g \in G$. Then by definition $g \in \mathbb{N}$ and $\forall n \in \mathbb{N}(\cdot(\langle n, g\rangle) = \odot(\langle n, g\rangle))$. We know that $s(g) \in \mathbb{N}$ exists. So since $\langle n, s(g)\rangle \in \mathbb{N} \times \mathbb{N} = \text{dom}(\cdot)$, we have, by our assumptions, $\cdot(\langle n, s(g)\rangle) = +(\langle\cdot(\langle n, g\rangle), n\rangle) = +(\langle\odot(\langle n, g\rangle), n\rangle) = \odot(\langle n, s(g)\rangle)$. So $s(g) \in G$ and $G = \mathbb{N}$.

To prove the desired result, let arbitrary $z \in \cdot \subseteq (\mathbb{N} \times \mathbb{N}) \times \mathbb{N}$. Then there exists $a_0, a_1, b \in \mathbb{N}$ such that $\langle\langle a_0, a_1\rangle, b\rangle = z \in \cdot$. So $b = \cdot(\langle a_0, a_1\rangle)$. Since $a_1 \in \mathbb{N} = G$, we can then conclude that $b = \cdot(\langle a_0, a_1\rangle) = \odot(\langle a_0, a_1\rangle)$. So $z = \langle\langle a_0, a_1\rangle, b\rangle \in \odot$. Next suppose $z \in \odot$. It then follows from a similar argument that $z \in \cdot$. We are done.

(next page)

Now for existence. We start by defining a function h for all q, where

$$\forall q \exists h \forall z(z \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N}(z = \langle m, x\rangle \wedge \langle\langle m, q\rangle, x\rangle \in +))$$

(which exists by the subset axiom, where $\mathbb{N} \times \mathbb{N}$ is a bounding set. Uniqueness for each q follows from extensionality.) We show that for all $q \in \mathbb{N}$, this h is a function from $\mathbb{N}$ into $\mathbb{N}$. That is

$$\forall q \in \mathbb{N}[\forall z(z \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N}(z = \langle m, x\rangle \wedge \langle\langle m, q\rangle, x\rangle \in +)) \rightarrow h : \mathbb{N} \rightarrow \mathbb{N}]$$

Let arbitrary $q \in \mathbb{N}$. Suppose that the statement before the implication is true. First to prove functionality. Let $a \in \text{dom}h$. Then there exists b such that $\langle a, b\rangle \in h$. Let b' be arbitrary such that $\langle a, b'\rangle \in h$. By our assumptions we must have $a, b, b' \in \mathbb{N}$, $+(\langle a, q\rangle) = b$, and $+(\langle a, q\rangle) = b'$. Since $+ : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ we must have $b = b'$. Next for the domain bound suppose arbitrary $a \in \text{dom}h$. Then there exists b such that $\langle a, b\rangle \in h$. It immediately follows that $a \in \mathbb{N}$. Suppose instead that $a \in \mathbb{N}$. Then $\langle a, q\rangle \in \mathbb{N} \times \mathbb{N} = \text{dom}(+)$. So $+(\langle a, q\rangle) \in \mathbb{N}$ exists and by definition, $\langle a, +(\langle a, q\rangle)\rangle \in h$. So $a \in \text{dom}h$. Finally for the range bound, let arbitrary $b \in \text{ran}h$. Then there exists a such that $\langle a, b\rangle \in h$. It immediately follows by definition that $b \in \mathbb{N}$. We have therefore shown

$$\forall q \in \mathbb{N} \exists! h(\forall z(z \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N}(z = \langle m, x\rangle \wedge \langle\langle m, q\rangle, x\rangle \in +)) \wedge h : \mathbb{N} \rightarrow \mathbb{N})$$

Theorem 1.4 states

$$\forall H \forall q \forall h([q \in H \wedge h : H \rightarrow H] \rightarrow \exists! g(g : \mathbb{N} \rightarrow H \wedge g(1) = q \wedge g \circ s = h \circ g))$$

which allows us to assert

$$\forall q \forall h([q \in \mathbb{N} \wedge h : \mathbb{N} \rightarrow \mathbb{N}] \rightarrow \exists! g(g : \mathbb{N} \rightarrow \mathbb{N} \wedge g(1) = q \wedge g \circ s = h \circ g))$$

(next page)

We showed

$$\forall q \in \mathbb{N} \exists! h(\forall z(z \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N}(z = \langle m, x\rangle \wedge \langle\langle m, q\rangle, x\rangle \in +)) \\ \wedge h : \mathbb{N} \to \mathbb{N})$$

and

$$\forall q \forall h([q \in \mathbb{N} \wedge h : \mathbb{N} \to \mathbb{N}] \to \exists! g(g : \mathbb{N} \to \mathbb{N} \wedge g(1) = q \wedge g \circ s = h \circ g))$$

Now we construct $\cdot$ as the set

$$\begin{aligned} &\forall z(z \in \cdot \leftrightarrow \exists a \in \mathbb{N} \exists b \in \mathbb{N}(\\ &\quad \exists h(\forall z_0(z_0 \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N}(z_0 = \langle m, x\rangle \wedge \langle\langle m, a\rangle, x\rangle \in +)) \\ &\quad\quad \wedge h : \mathbb{N} \to \mathbb{N} \wedge \exists g(g : \mathbb{N} \to \mathbb{N} \wedge g(1) = a \wedge g \circ s = h \circ g \\ &\quad\quad \wedge \exists c(\langle b, c\rangle \in g \wedge z = \langle\langle a, b\rangle, c\rangle)))))\end{aligned}$$

(existence follows from the subset axiom, where $(\mathbb{N} \times \mathbb{N}) \times \mathbb{N}$ is a bounding set.) We want to verify

$$\cdot : \mathbb{N} \times \mathbb{N} \to \mathbb{N} \wedge \forall m \in \mathbb{N} \forall n \in \mathbb{N}(\cdot(\langle n, 1\rangle) = n \wedge \cdot(\langle n, s(m)\rangle) = +(\langle \cdot(\langle n, m\rangle), n\rangle))$$

We start with functionality. Let arbitrary $p \in \mathrm{dom}(\cdot)$. Then by definition there exists q such that $\langle p, q\rangle \in \cdot$. This means that there exists $a, b \in \mathbb{N}$ and $h : \mathbb{N} \to \mathbb{N}$ such that $\forall z_0(z_0 \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N}(z_0 = \langle m, x\rangle \wedge \langle\langle m, a\rangle, x\rangle \in +))$. There also exists $g : \mathbb{N} \to \mathbb{N}$ such that $g(1) = a$ and $g \circ s = h \circ g$, and c such that $\langle b, c\rangle \in g$ and $\langle p, q\rangle = \langle\langle a, b\rangle, c\rangle$. Now suppose there exists $\langle p, q'\rangle \in \cdot$. Similarly there exists $a', b' \in \mathbb{N}$, $h' : \mathbb{N} \to \mathbb{N}$ such that $\forall z_0(z_0 \in h' \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N}(z_0 = \langle m, x\rangle \wedge \langle\langle m, a'\rangle, x\rangle \in +))$, $g' : \mathbb{N} \to \mathbb{N}$ such that $g'(1) = a'$ and $g' \circ s = h' \circ g'$, and c' such that $\langle b', c'\rangle \in g'$ and $\langle p, q'\rangle = \langle\langle a', b'\rangle, c'\rangle$. We have $\langle a', b'\rangle = p = \langle a, b\rangle$. So $a' = a$ and $b' = b$. Uniqueness, as proven, implies that $h = h'$ and $g = g'$. So $q' = c' = g'(b') = g(b') = g(b) = c = q$.

Next we show $\mathrm{dom}(\cdot) = \mathbb{N} \times \mathbb{N}$. Let arbitrary $p \in \mathrm{dom}(\cdot)$. Then by definition there exists q such that $\langle p, q\rangle \in \cdot$. So there exists $a, b \in \mathbb{N}$ and c such that $\langle p, q\rangle = \langle\langle a, b\rangle, c\rangle$. So $p = \langle a, b\rangle \in \mathbb{N} \times \mathbb{N}$. Now let $p \in \mathbb{N} \times \mathbb{N}$. Then by definition there exists $a, b \in \mathbb{N}$ such that $p = \langle a, b\rangle$. Since $a \in \mathbb{N}$, as proven there exists $h : \mathbb{N} \to \mathbb{N}$ such that $\forall z(z \in h \leftrightarrow \exists m \in \mathbb{N} \exists x \in \mathbb{N}(z = \langle m, x\rangle \wedge \langle\langle m, a\rangle, x\rangle \in +))$, and there exists $g : \mathbb{N} \to \mathbb{N}$ such that $g(1) = a$ and $g \circ s = h \circ g$. By definition we then have $\langle\langle a, b\rangle, g(b)\rangle \in \cdot$ and $p = \langle a, b\rangle \in \mathrm{dom}(\cdot)$.

For the range bound, let $q \in \mathrm{ran}(\cdot)$. Then there exists p such that $\langle p, q\rangle \in \cdot$. So there exists $a, b \in \mathbb{N}$, $g : \mathbb{N} \to \mathbb{N}$, and c such that $\langle b, c\rangle \in g$ and $\langle p, q\rangle = \langle\langle a, b\rangle, c\rangle$. So $q = c = g(b) \in \mathbb{N}$.

Finally we verify the two properties of multiplication. Let arbitrary $m, n \in \mathbb{N}$. We first show $\cdot(\langle n, 1\rangle) = n$. We know that $\langle\langle n, 1\rangle, \cdot(\langle n, 1\rangle)\rangle \in \cdot$. By definition there then exists $a, b \in \mathbb{N}$, $g : \mathbb{N} \to \mathbb{N}$ such that $g(1) = a$, and c such that $\langle b, c\rangle \in g$ and $\langle\langle n, 1\rangle, \cdot(\langle n, 1\rangle)\rangle = \langle\langle a, b\rangle, c\rangle$. So $\cdot(\langle n, 1\rangle) = c = g(b) = g(1) = a = n$. (next page)

Now we show $\cdot(\langle n, s(m)\rangle) = +(\langle \cdot(\langle n, m\rangle), n\rangle)$. $s(m) \in \mathbb{N}$ exists, so we have $\langle\langle n, s(m)\rangle, \cdot(\langle n, s(m)\rangle)\rangle \in \cdot$. Then there exists $a, b \in \mathbb{N}$, $h : \mathbb{N} \to \mathbb{N}$ such that $\forall z_0(z_0 \in h \leftrightarrow \exists m_0 \in \mathbb{N}\exists x \in \mathbb{N}(z_0 = \langle m_0, x\rangle \wedge \langle\langle m_0, a\rangle, x\rangle \in +))$, $g : \mathbb{N} \to \mathbb{N}$ such that $g(1) = a$ and $g \circ s = h \circ g$, and c such that $\langle b, c\rangle \in g$ and $\langle\langle n, s(m)\rangle, \cdot(\langle n, s(m)\rangle)\rangle = \langle\langle a, b\rangle, c\rangle$. We then have $\cdot(\langle n, s(m)\rangle) = c = g(b) = g(s(m)) = h(g(m))$ and $a = n$. Similarly we know that $\langle\langle n, m\rangle, \cdot(\langle n, m\rangle)\rangle \in \cdot$. So there exists $a_0, b_0 \in \mathbb{N}$, $h_0 : \mathbb{N} \to \mathbb{N}$ such that $\forall z_0(z_0 \in h_0 \leftrightarrow \exists m_0 \in \mathbb{N}\exists x \in \mathbb{N}(z_0 = \langle m_0, x\rangle \wedge \langle\langle m_0, a_0\rangle, x\rangle \in +))$, $g_0 : \mathbb{N} \to \mathbb{N}$ such that $g_0(1) = a_0$ and $g_0 \circ s = h_0 \circ g_0$, and c_0 such that $\langle b_0, c_0\rangle \in g_0$ and $\langle\langle n, m\rangle, \cdot(\langle n, m\rangle)\rangle = \langle\langle a_0, b_0\rangle, c_0\rangle$. This means that $a_0 = n = a$, and that by our earlier statements we can assert that $h = h_0$, and $g = g_0$. So $\cdot(\langle n, m\rangle) = c_0 = g_0(b_0) = g(m)$. We then have $\cdot(\langle n, s(m)\rangle) = h(g(m)) = h(\cdot(\langle n, m\rangle))$. From the definition of h, there then exists $m_0, x \in \mathbb{N}$ such that $\langle \cdot(\langle n, m\rangle), h(\cdot(\langle n, m\rangle))\rangle = \langle m_0, x\rangle$ and $\langle\langle m_0, a\rangle, x\rangle \in +$. This and the fact that $a = n$ allows us to conclude $+(\langle \cdot(\langle n, m\rangle), n\rangle) = h(\cdot(\langle n, m\rangle)) = \cdot(\langle n, s(m)\rangle)$. □

1.1.5 Properties of addition and multiplication on the naturals

Theorem 1.7. *Let $a, b, c \in \mathbb{N}$.*

1. *If $a + c = b + c$, then $a = b$ (Cancellation Law for Addition)*
2. *$(a + b) + c = a + (b + c)$ (Associative Law for Addition)*
3. *$1 + a = s(a) = a + 1$*
4. *$a + b = b + a$ (Commutative Law for Addition)*
5. *$a + b \neq 1$*
6. *$a + b \neq a$*
7. *$a \cdot 1 = a = 1 \cdot a$ (Identity Law for Multiplication)*
8. *$(a + b)c = ac + bc$ (Distributive Law)*
9. *$ab = ba$ (Commutative Law for Multiplication)*
10. *$c(a + b) = ca + cb$ (Distributive Law)*
11. *$(ab)c = a(bc)$ (Associative Law for Multiplication)*
12. *If $ac = bc$ then $a = b$ (Cancellation Law for Multiplication)*
13. *$ab = 1$ iff $a = 1 = b$*

(next page)

Proof. (1) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N}(a + c = b + c \to a = b)$$

To do this we define the set G as the set where

$$\forall c(c \in G \leftrightarrow c \in \mathbb{N} \wedge \forall a \in \mathbb{N} \forall b \in \mathbb{N}(a + c = b + c \to a = b))$$

(where existence follows from the subset axiom using $\mathbb{N}$ as the bounding set. Uniqueness follows from extensionality.) Then we show that $G = \mathbb{N}$. Showing $G \subseteq \mathbb{N}$ is straightforward; we need to show $1 \in G$ and $\forall g \in G(s(g) \in G)$. For the former, we already know $1 \in \mathbb{N}$. Let arbitrary $a, b \in \mathbb{N}$. Suppose that $a + 1 = b + 1$. By definition we then have $s(a) = s(b)$, so by injectivity $a = b$ and $1 \in G$.

Now the latter. Let arbitrary $g \in G$. As defined we then have $g \in \mathbb{N}$ and $\forall a \in \mathbb{N} \forall b \in \mathbb{N}(a + g = b + g \to a = b)$. $s(g) \in \mathbb{N}$ exists. Let arbitrary $a, b \in \mathbb{N}$ and suppose that $a + s(g) = b + s(g)$. Then by the definition of addition we have $s(a + g) = s(b + g)$ and, by injectivity, $a + g = b + g$. Since $g \in G$ we then have $a = b$. So $G = \mathbb{N}$. To prove the initial statement, let arbitrary $c \in \mathbb{N}$. Then from $\mathbb{N} = G$ the desired conclusion immediately follows. □

Proof. (2) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N}((a + b) + c = a + (b + c))$$

To do this we define G as the set where

$$\forall c(c \in G \leftrightarrow c \in \mathbb{N} \wedge \forall a \in \mathbb{N} \forall b \in \mathbb{N}((a + b) + c = a + (b + c)))$$

(where existence and uniqueness can be easily shown) and show that $G = \mathbb{N}$. Showing $G \subseteq \mathbb{N}$ is straightforward. We need to show $1 \in G$ and $\forall g \in G(s(g) \in G)$.

For the former, we already know $1 \in \mathbb{N}$. Let arbitrary $a, b \in \mathbb{N}$. By the definition of addition we have

$$(a + b) + 1 = s(a + b) = a + s(b) = a + (b + 1)$$

Now the latter. Let arbitrary $g \in G$. Then by definition $g \in \mathbb{N}$ and $\forall a \in \mathbb{N} \forall b \in \mathbb{N}((a + b) + g = a + (b + g))$. We know $s(g) \in \mathbb{N}$ exists. Let arbitrary $a, b \in \mathbb{N}$. By definition we then have

$$(a + b) + s(g) = s((a + b) + g) = s(a + (b + g)) = a + s(b + g) = a + (b + s(g))$$

So $\mathbb{N} = G$. To prove the initial statement, let arbitrary $c \in \mathbb{N} = G$. The desired conclusion immediately follows. □

(next page)

Proof. (3) We want to show

$$\forall a \in \mathbb{N}(1 + a = s(a) = a + 1)$$

To do this we define G as the set where

$$\forall a(a \in G \leftrightarrow a \in \mathbb{N} \wedge 1 + a = s(a) = a + 1)$$

(Showing existence and uniqueness is straightforward.) We show that $G = \mathbb{N}$. Showing $G \subseteq \mathbb{N}$ is straightforward. We need to show that $1 \in G$ and $\forall g \in G(s(g) \in G)$. For the former we already know that $1 \in \mathbb{N}$, and by definition we have $1 + 1 = s(1)$.

Now the latter. Let arbitrary $g \in G$. Then by definition $g \in \mathbb{N}$ and $1 + g = s(g) = g + 1$. We know $s(g) \in \mathbb{N}$ exists. We then have

$$1 + s(g) = s(1 + g) = s(s(g)) = s(g + 1) = (g + 1) + 1 = s(g) + 1$$

So $G = \mathbb{N}$. Proving the initial statement is straightforward. Let arbitrary $a \in \mathbb{N} = G$ and the desired conclusion immediately follows. □

Proof. (4) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(a + b = b + a)$$

To do this we define G as the set such that

$$\forall a(a \in G \leftrightarrow a \in \mathbb{N} \wedge \forall b \in \mathbb{N}(a + b = b + a))$$

(Showing existence and uniqueness is straightforward.) We show that $G = \mathbb{N}$. Showing $G \subseteq \mathbb{N}$ is straightforward. We need to show that $1 \in G$ and $\forall g \in G(s(g) \in G)$. For the former we already know $1 \in \mathbb{N}$. Letting arbitrary $b \in \mathbb{N}$, by (3) we have $1 + b = b + 1$.

Now the latter. Let arbitrary $g \in G$. Then by definition $g \in \mathbb{N}$ and $\forall b \in \mathbb{N}(g + b = b + g)$. $s(g) \in \mathbb{N}$ exists. Let arbitrary $b \in \mathbb{N}$. Then we have

$$s(g) + b = (g + 1) + b \overset{(2)}{=} g + (1 + b) \overset{(3)}{=} g + s(b) = s(g + b) = s(b + g) = b + s(g)$$

So $G = \mathbb{N}$. Showing the initial statement is straightforward. Let arbitrary $a \in \mathbb{N} = G$ and the desired conclusion immediately follows. □

(next page)

Proof. (5) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(a + b \neq 1)$$

Let arbitrary $a, b \in \mathbb{N}$. Suppose for contradiction that $a + b = 1$. We have 2 cases. First suppose $b = 1$. Then $a + b = a + 1 = s(a) = 1$, and the last equality is a contradiction of the peano postulates. Now suppose $b \neq 1$. Then by lemma 1.3 there exists $x \in \mathbb{N}$ such that $b = s(x)$. We then have $a + b = a + s(x) = s(a + x) = 1$, which is again a contradiction. □

Proof. (6) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(a + b \neq a)$$

We do this by defining G as the set such that

$$\forall a(a \in G \leftrightarrow a \in \mathbb{N} \wedge \forall b \in \mathbb{N}(a + b \neq a))$$

(Showing existence and uniqueness is straightforward.) We show that $G = \mathbb{N}$. Showing $G \subseteq \mathbb{N}$ is straightforward. We need to show $1 \in G$ and $\forall g \in G(s(g) \in G)$. For the former, we already know $1 \in \mathbb{N}$. Let arbitrary $b \in \mathbb{N}$. By (5) $1 + b \neq 1$.

Now the latter. Let arbitrary $g \in G$. Then $g \in \mathbb{N}$ and $\forall b \in \mathbb{N}(g + b \neq g)$. $s(g) \in \mathbb{N}$ exists. Let arbitrary $b \in \mathbb{N}$. For contradiction, suppose $s(g) + b = s(g)$. We then have

$$s(g) = s(g) + b \overset{(4)}{=} b + s(g) = s(b + g)$$

By injectivity of s and (4) we then have $g = b + g = g + b$. This contradicts $\forall b \in \mathbb{N}(g + b \neq g)$. So $G = \mathbb{N}$. Showing the initial statement is straightforward. Let arbitrary $a \in \mathbb{N} = G$ and the desired conclusion immediately follows. □

Proof. (7) We want to show

$$\forall a \in \mathbb{N}(a \cdot 1 = a = 1 \cdot a)$$

to do this we define G as the set such that

$$\forall a(a \in G \leftrightarrow a \in \mathbb{N} \wedge a \cdot 1 = a = 1 \cdot a)$$

(where existence and uniqueness can be easily shown.) We show that $G = \mathbb{N}$. Showing $G \subseteq \mathbb{N}$ is straightforward. We need to show $1 \in G$ and $\forall g \in G(s(g) \in G)$. For the former we already know $1 \in \mathbb{N}$, and $1 \cdot 1 = 1$.

Now the latter. Let arbitrary $g \in G$. Then $g \in \mathbb{N}$ and $g \cdot 1 = g = 1 \cdot g$. $s(g) \in \mathbb{N}$ exists. We then have

$$s(g) \cdot 1 = s(g) = s(g \cdot 1) = s(1 \cdot g) = (1 \cdot g) + 1 = 1 \cdot s(g)$$

so $G = \mathbb{N}$. Proving the initial statement is straightforward. Let arbitrary $a \in \mathbb{N} = G$ and the desired conclusion immediately follows. □

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Proof. (8) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N}((a+b)c = (ac) + (bc))$$

To do this we define G as the set such that

$$\forall c(c \in G \leftrightarrow c \in \mathbb{N} \wedge \forall a \in \mathbb{N} \forall b \in \mathbb{N}((a+b)c = (ac) + (bc))$$

(where proving existence and uniqueness is straightforward.) We show that $G = \mathbb{N}$. Showing $G \subseteq \mathbb{N}$ is straightforward. We need to show $1 \in G$ and $\forall g \in G(s(g) \in G)$. For the former, we already know $1 \in \mathbb{N}$. Let arbitrary $a, b \in \mathbb{N}$. We have

$$(a+b) \cdot 1 = a + b = (a \cdot 1) + (b \cdot 1)$$

Now the latter. Let arbitrary $g \in G$. Then by definition $g \in \mathbb{N}$ and $\forall a \in \mathbb{N} \forall b \in \mathbb{N}((a+b)g = (ag) + (bg))$. $s(g) \in \mathbb{N}$ exists. Let arbitrary $a, b \in \mathbb{N}$. We have

$$\begin{aligned}(a+b)s(g) &= ((a+b)g) + (a+b) = ((ag) + (bg)) + (a+b) \\ &\overset{(2)}{=} (ag) + ((bg) + (a+b)) \overset{(4,2)}{=} (ag) + ((bg+b) + a) \\ &= (ag) + ((bs(g)) + a) \overset{(4,2)}{=} ((ag) + a) + (bs(g)) \\ &= (as(g)) + (bs(g))\end{aligned}$$

So $G = \mathbb{N}$. Proving the initial statement is straightforward. Let arbitrary $c \in \mathbb{N} = G$ and the desired conclusion immediately follows. □

Proof. (9) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(ab = ba)$$

To do this we define G as the set such that

$$\forall a(a \in G \leftrightarrow a \in \mathbb{N} \wedge \forall b \in \mathbb{N}(ab = ba))$$

(where proving existence and uniqueness is straightforward.) We show that $G = \mathbb{N}$. Showing $G \subseteq \mathbb{N}$ is straightforward. We need to show $1 \in G$ and $\forall g \in G(s(g) \in G)$. For the former, we know $1 \in \mathbb{N}$. Let arbitrary $b \in \mathbb{N}$. Then by (7) we have $1 \cdot b = b \cdot 1$.

Now the latter. Let arbitrary $g \in G$. Then $g \in \mathbb{N}$ and $\forall b \in \mathbb{N}(gb = bg)$. $s(g) \in \mathbb{N}$ exists. Let arbitrary $b \in \mathbb{N}$. We have

$$s(g)b = (g+1)b \overset{(8)}{=} (gb) + (1 \cdot b) \overset{(7)}{=} (bg) + b = bs(g)$$

So $G = \mathbb{N}$. Proving the initial statement is straightforward. Let arbitrary $a \in \mathbb{N} = G$ and the desired conclusion immediately follows. □

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Proof. (10) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N}(c(a+b) = (ca) + (cb))$$

Let arbitrary $a, b, c \in \mathbb{N}$. We have

$$c(a+b) \overset{(9)}{=} (a+b)c \overset{(8)}{=} (ac) + (bc) \overset{(9)}{=} (ca) + (cb)$$ □

Proof. (11) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N}((ab)c = a(bc))$$

To do this we define G as the set such that

$$\forall b(b \in G \leftrightarrow b \in \mathbb{N} \wedge \forall a \in \mathbb{N} \forall c \in \mathbb{N}((ab)c = a(bc)))$$

(where proving existence and uniqueness is straightforward.) We show that $G = \mathbb{N}$. Showing $G \subseteq \mathbb{N}$ is straightforward. We need to show $1 \in G$ and $\forall g \in G(s(g) \in G)$. For the former, we already know $1 \in \mathbb{N}$. Let arbitrary $a, c \in \mathbb{N}$. We have

$$(a \cdot 1)c = ac \overset{(7)}{=} a(1 \cdot c)$$

Now the latter. Let arbitrary $g \in G$. Then $g \in \mathbb{N}$ and $\forall a \in \mathbb{N} \forall c \in \mathbb{N}((ag)c = a(gc))$. $s(g) \in \mathbb{N}$ exists. Let arbitrary $a, c \in \mathbb{N}$. We have

$$\begin{aligned}(as(g))c = ((ag) + a)c &\overset{(8)}{=} ((ag)c) + (ac) = (a(gc)) + (ac) \\ &\overset{(10)}{=} a((gc) + c) \overset{(9)}{=} a((cg) + c) = a(cs(g)) \\ &\overset{(9)}{=} a(s(g)c)\end{aligned}$$

So $G = \mathbb{N}$. Proving the initial statement is straightforward. Let arbitrary $b \in \mathbb{N} = G$ and the desired conclusion immediately follows. □

(next page)

Proof. (12) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N}(ac = bc \to a = b)$$

To do this we define G as the set such that

$$\forall a(a \in G \leftrightarrow a \in \mathbb{N} \wedge \forall b \in \mathbb{N} \forall c \in \mathbb{N}(ac = bc \to a = b))$$

(where proving existence and uniqueness is straightforward.) We show that $G = \mathbb{N}$. Showing $G \subseteq \mathbb{N}$ is straightforward. We need to show $1 \in G$ and $\forall g \in G(s(g) \in G)$.

For the former, we already know $1 \in \mathbb{N}$. Let arbitrary $b, c \in \mathbb{N}$ and suppose that $1 \cdot c = bc$. Suppose for contradiction that $b \neq 1$. Then by lemma 1.3 there exists $x \in \mathbb{N}$ such that $b = s(x)$. We then have

$$c \overset{(7)}{=} 1 \cdot c = bc = s(x)c = (x+1)c \overset{(8,7)}{=} (xc) + c \overset{(4,9)}{=} c + (cx)$$

which contradicts (6).

Now the latter. Let arbitrary $g \in G$. Then $g \in \mathbb{N}$ and $\forall b \in \mathbb{N} \forall c \in \mathbb{N}(gc = bc \to g = b)$. $s(g) \in \mathbb{N}$ exists. Let arbitrary $b, c \in \mathbb{N}$. Suppose $s(g)c = bc$. For contradiction, suppose $b = 1$. Then

$$c \overset{(7)}{=} 1 \cdot c = bc = s(g)c \overset{(9)}{=} cs(g) = (cg) + c \overset{(4)}{=} c + (cg)$$

contradicting (6). So we must have $b \neq 1$ and by lemma 1.3 there exists $x \in \mathbb{N}$ such that $b = s(x)$. Then

$$(cg) + c = cs(g) \overset{(9)}{=} s(g)c = bc = s(x)c = (x+1)c \overset{(8,7)}{=} (xc) + c$$

and by (1), $gc = cg = xc$. Since $g \in G$ we then have $g = x$ and $s(g) = s(x) = b$. So $G = \mathbb{N}$. Proving the initial statement is straightforward. Let arbitrary $a \in \mathbb{N} = G$ and the desired conclusion immediately follows. □

Proof. (13) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(ab = 1 \leftrightarrow a = 1 = b)$$

Let arbitrary $a, b \in \mathbb{N}$. First suppose $ab = 1$. For contradiction suppose $b \neq 1$. Then by lemma 1.3 there exists $x \in \mathbb{N}$ such that $b = s(x)$. So

$$1 = ab = as(x) = a(x+1) \overset{(10)}{=} (ax) + (a \cdot 1) = (ax) + a$$

which contradicts (5). So $b = 1$. We immediately have $1 = ab = a \cdot 1 = a$ and $a = 1 = b$. Now suppose $a = 1 = b$. We immediately have $ab = 1 \cdot 1 = 1$. □

1.1.6 Ordering on $\mathbb{N}$

Definition 1.8. *The relation $<$ on $\mathbb{N}$ is defined by $a < b$ if and only if there is some $p \in \mathbb{N}$ such that $a + p = b$ for all $a, b \in \mathbb{N}$.*

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(a < b \leftrightarrow \exists p \in \mathbb{N}(a + p = b))$$

The relation $\leq$ on $\mathbb{N}$ is defined by $a \leq b$ if and only if $a < b$ or $a = b$ for all $a, b \in \mathbb{N}$.

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(a \leq b \leftrightarrow (a < b \vee a = b))$$

We will write $a > b$ to mean the same thing as $b < a$. The same applies to $a \geq b$.

Theorem 1.9. *Let $a, b \in \mathbb{N}$. Suppose that $a < b$. Then there is a unique $p \in \mathbb{N}$ such that $a + p = b$.*

Proof. By definition there exists $p \in \mathbb{N}$ such that $a + p = b$. Let $p' \in \mathbb{N}$ be arbitrary such that $a + p' = b$. Then by part 4 of theorem 1.7 we have $p + a = b$ and $p' + a = b$, so $p + a = p' + a$. By part 1 of theorem 1.7 we have $p = p'$. □

Theorem 1.10. *Let $a, b, c \in \mathbb{N}$.*

1. *$a \leq a$, and $a \not< a$, and $a < a + 1$*
2. *$1 \leq a$*
3. *If $a < b$ and $b < c$, then $a < c$; if $a \leq b$ and $b < c$, then $a < c$; if $a < b$ and $b \leq c$, then $a < c$; if $a \leq b$ and $b \leq c$, then $a \leq c$*
4. *$a < b$ if and only if $a + c < b + c$*
5. *Precisely one of $a < b$ or $a = b$ or $a > b$ holds (Trichotomy Law)*
6. *$a < b$ if and only if $ac < bc$*
7. *$a \leq b$ or $b \leq a$*
8. *If $a \leq b$ and $b \leq a$, then $a = b$*
9. *It cannot be that $b < a < b + 1$*
10. *$a \leq b$ if and only if $a < b + 1$*
11. *$a < b$ if and only if $a + 1 \leq b$*

(next page)

Proof. (1) We want to show

$$\forall a \in \mathbb{N}(a \leq a \wedge a \not< a \wedge a < a + 1)$$

Let arbitrary $a \in \mathbb{N}$. Since $a = a$, by definition $a \leq a$. Next suppose for contradiction that $a < a$. Then there would exist $p \in \mathbb{N}$ such that $a + p = a$, contradicting part 6 of theorem 1.7. Finally since $1 \in \mathbb{N}$ and $a + 1 = a + 1$, by definition $a < a + 1$. □

Proof. (2) We want to show

$$\forall a \in \mathbb{N}(1 \leq a)$$

Let arbitrary $a \in \mathbb{N}$. We have 2 cases. First suppose $a = 1$; then by definition $1 \leq a$. Next suppose $a \neq 1$. Then by lemma 1.3 there exists $x \in \mathbb{N}$ such that $a = s(x) = x + 1$. By theorem 1.7 part 4 we have $a = 1 + x$. So by definition $1 < a$. □

Proof. (3) We want to show

$$\begin{aligned}\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N}([(a < b \wedge b < c) \rightarrow a < c] \wedge [(a \leq b \wedge b < c) \rightarrow a < c] \\ \wedge [(a < b \wedge b \leq c) \rightarrow a < c] \wedge [(a \leq b \wedge b \leq c) \rightarrow a \leq c])\end{aligned}$$

Let arbitrary $a, b, c \in \mathbb{N}$. First suppose $a < b \wedge b < c$. Then there exists $x, y \in \mathbb{N}$ such that $a + x = b$ and $b + y = c$. We then have $(a + x) + y = b + y = c$. By theorem 1.7 part 2 we have $a + (x + y) = c$ and therefore $a < c$.

Next suppose $a \leq b \wedge b < c$. There exists $y \in \mathbb{N}$ such that $b + y = c$. We have 2 cases. First suppose $a < b$. Then as proven earlier we have $a < c$. Next suppose $a = b$. Then we immediately have $a + y = c$ and $a < c$.

Next suppose $a < b \wedge b \leq c$. There exists $x \in \mathbb{N}$ such that $a + x = b$. We have 2 cases. First suppose $b < c$. Then again we immediately have $a < c$. Next suppose $b = c$. Then $a + x = c$ and $a < c$.

Finally suppose $a \leq b \wedge b \leq c$. We have 4 cases. First suppose $a < b \wedge b < c$. As proven $a < c$. Next suppose $a = b \wedge b < c$. As proven $a < c$. Next suppose $a < b \wedge b = c$. As proven $a < c$. Finally suppose $a = b$ and $b = c$. It immediately follows that $a = c$ and $a \leq c$. □

Proof. (4) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N}(a < b \leftrightarrow a + c < b + c)$$

Let arbitrary $a, b, c \in \mathbb{N}$. First suppose $a < b$. Then there exists $x \in \mathbb{N}$ such that $a + x = b$. Then $(a + x) + c = b + c$ and by theorem 1.7 parts 2 and 4 we have $(a + c) + x = b + c$. So $a + c < b + c$.

Next suppose $a + c < b + c$. Then there exists $x \in \mathbb{N}$ such that $(a + c) + x = b + c$. By theorem 1.7 parts 2 and 4 we have $(a + x) + c = b + c$, and by part 1 of that same theorem we have $a + x = b$. So $a < b$. □

(next page)

Proof. (5) We want to prove trichotomy

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}((a < b \vee a = b \vee b < a) \\ \wedge \neg((a < b) \wedge (a = b)) \wedge \neg((a < b) \wedge (b < a)) \wedge \neg((a = b) \wedge (b < a)))$$

Let arbitrary $a, b \in \mathbb{N}$. We start by proving the last three statements. First suppose for contradiction that $a < b$ and $a = b$. Then $a < a$, contradicting (1). Next suppose for contradiction that $a = b$ and $b < a$. A similar argument shows that this contradicts (1). Finally suppose for contradiction that $a < b \wedge b < a$. By (3) we have $a < a$, a contradiction.

Now for the first statement. We define G as the set such that

$$\forall a(a \in G \leftrightarrow a \in \mathbb{N} \wedge \forall b \in \mathbb{N}(a < b \vee a = b \vee b < a))$$

(proving existence and uniqueness of G is straightforward.) We show that $G = \mathbb{N}$. Showing $G \subseteq \mathbb{N}$ is straightforward. We need to show that $1 \in G$ and $\forall g \in G(s(g) \in G)$. For the former, we already know $1 \in \mathbb{N}$. Let arbitrary $b \in \mathbb{N}$. By (2) we have $1 \leq b$, meaning $1 < b$ or $1 = b$. Either case allows us to conclude that $1 \in G$.

Now the latter. Let arbitrary $g \in G$. Then $g \in \mathbb{N}$ and $\forall b \in \mathbb{N}(g < b \vee g = b \vee b < g)$. $s(g) \in \mathbb{N}$ exists. Let arbitrary $b \in \mathbb{N}$. We have 3 cases. First suppose $g < b$. Then there exists $x \in \mathbb{N}$ such that $g + x = b$. Suppose $x = 1$; then $s(g) = g + 1 = b$. Next suppose $x \neq 1$; then by lemma 1.3 there exists $y \in \mathbb{N}$ such that $x = s(y)$. So $b = g + x = g + s(y) = g + (y + 1)$. By theorem 1.7 parts 2 and 4 we have $(g+1)+y = b$ and $s(g) < b$. For the second case suppose $g = b$. Then by (1) we have $b = g < g + 1 = s(g)$. Finally suppose $b < g$. Then since $g < g + 1$ by (3) we have $b < g + 1 = s(g)$. We then have $G = \mathbb{N}$.

Showing the desired result is straightforward. For all arbitrary $a \in \mathbb{N}$, $a \in G$. The result immediately follows. □

Proof. (6) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N} \forall c \in \mathbb{N}(ac < bc \leftrightarrow a < b)$$

Let arbitrary $a, b, c \in \mathbb{N}$. First suppose $a < b$. Then there exists $x \in \mathbb{N}$ such that $a + x = b$. Then by theorem 1.7 part 8 we have $bc = (a + x)c = (ac) + (xc)$; so $ac < bc$.

Next suppose $ac < bc$. Suppose for contradiction that $a \not< b$. Then by (5) we have $a = b \vee b < a$. First suppose $a = b$. Then we immediately have $ac < ac$, which contradicts (1). Next suppose $b < a$. There exists $x, y \in \mathbb{N}$ such that $(ac) + x = bc$ and $b + y = a$. By theorem 1.7 parts 8 and 2 we have $bc = ((b + y)c) + x = ((bc) + (yc)) + x = (bc) + ((yc) + x)$, contradicting part 6 of that same theorem. □

(next page)

Proof. (7) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(a \leq b \vee b \leq a)$$

Let arbitrary $a, b \in \mathbb{N}$. First suppose $a \not< b$. Then by (5) $a = b \vee b < a$, meaning $b \leq a$. Next suppose $a < b$. Then immediately it follows that $a \leq b$. □

Proof. (8) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}((a \leq b \wedge b \leq a) \rightarrow a = b)$$

Let arbitrary $a, b \in \mathbb{N}$. Suppose that $a \leq b \wedge b \leq a$. We have four cases. Three of the cases: $a < b \wedge b < a$, $a = b \wedge b < a$, and $a < b \wedge b = a$ all contradict (5). The only possibility is the last case $a = b \wedge b = a$. □

Proof. (9) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(\neg(b < a \wedge a < b + 1))$$

Let arbitrary $a, b \in \mathbb{N}$. Suppose for contradiction that $b < a \wedge a < b + 1$. Then there exists $x, y \in \mathbb{N}$ such that $b+x = a$ and $a+y = b+1$. Then $(b+x)+y = b+1$. By theorem 1.7 parts 2 and 4 we have $(x + y) + b = 1 + b$. By part 1 of that theorem we then have $x + y = 1$, contradicting part 5 of that theorem. □

Proof. (10) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(a \leq b \leftrightarrow a < b + 1)$$

Let arbitrary $a, b \in \mathbb{N}$. First suppose $a \leq b$. Suppose for contradiction that $b + 1 \leq a$. Then by (3) we would have $b + 1 \leq b$. The case where $b + 1 = b$ is a contradiction to theorem 1.7 part 6. The case where $b + 1 < b$ implies the existence of $x \in \mathbb{N}$ such that $(b + 1) + x = b$; by theorem 1.7 part 2 we have $b + (1 + x) = b$, again contradicting part 6 of that theorem. We can then conclude that $\neg(b + 1 \leq a)$, meaning $\neg(b + 1 = a \vee b + 1 < a)$. By (5) we must then have $a < b + 1$.

Now for the other direction. Suppose $a < b + 1$. Supposing that $b < a$ immediately contradicts (9). Therefore by (5) we must have $a \leq b$. □

Proof. (11) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(a < b \leftrightarrow a + 1 \leq b)$$

Let arbitrary $a, b \in \mathbb{N}$. Suppose $a < b$. Supposing $b < a + 1$ immediately contradicts (9), so we must have, by (5), $a + 1 \leq b$.

Next suppose $a + 1 \leq b$. Suppose for contradiction that $b \leq a$, which would imply by (3) that $a + 1 \leq a$. The case where $a + 1 = a$ immediately contradicts theorem 1.7 part 6. The case where $a + 1 < a$ implies the existence of $x \in \mathbb{N}$ such that $(a+1)+x = a$; theorem 1.7 part 2 then implies $a+(1+x) = a$, again contradicting part 6 of that same theorem. We conclude that $\neg(b = a \vee b < a)$, so by (5) we have $a < b$. □

1.1.7 Well-ordering

Theorem 1.11. *(Well-Ordering Principle) Let $G \subseteq \mathbb{N}$ be a non-empty set. Then there is some $m \in G$ such that $m \leq g$ for all $g \in G$.*

Proof. We want to show

$$\forall G(G \subseteq \mathbb{N} \wedge G \neq \emptyset \rightarrow \exists m \in G(\forall g \in G(m \leq g)))$$

Let G be arbitrary; assume that $G \subseteq \mathbb{N}$ and $G \neq \emptyset$. Assume for contradiction that the implication is false. That is

$$\neg\exists m \in G(\forall g \in G(m \leq g))$$

We define H as the set such that

$$\forall a(a \in H \leftrightarrow a \in \mathbb{N} \wedge \forall n \in \mathbb{N}(n \leq a \rightarrow n \notin G))$$

(where existence and uniqueness can be easily shown). We can show that $H \cap G = \emptyset$. For contradiction suppose there exists x such that $x \in H$ and $x \in G$. By definition then $x \in \mathbb{N}$ and $\forall n \in \mathbb{N}(n \leq x \rightarrow n \notin G)$. Then since $x \leq x$ we must have $x \notin G$, a contradiction. We will show that $H = \mathbb{N}$. Showing $H \subseteq \mathbb{N}$ is straightforward. We need to show $1 \in H$ and $\forall h \in H(s(h) \in H)$.

We start with the former. Suppose for contradiction that $1 \notin H$. We know $1 \in \mathbb{N}$, so we must have $\neg\forall n \in \mathbb{N}(n \leq 1 \rightarrow n \notin G)$. This is logically equivalent to

$$\exists n \in \mathbb{N}(n \leq 1 \wedge n \in G)$$

That is, there exists $n \in \mathbb{N}$ such that $n \leq 1$ and $n \in G$. $n < 1$ is a contradiction as it would imply the existence of $x \in \mathbb{N}$ such that $n + x = 1$, contradicting theorem 1.7 part 5. We therefore must have $1 = n \in G$. Let arbitrary $g \in G \subseteq \mathbb{N}$. By theorem 1.10 part 2 we must have $n = 1 \leq g$. This contradicts our assumption that such an element does not exist, so $1 \in H$.

Now the latter. Let arbitrary $h \in H$. Then $h \in \mathbb{N}$ and $\forall n \in \mathbb{N}(n \leq h \rightarrow n \notin G)$. We know $s(h) \in \mathbb{N}$ exists. Suppose for contradiction that $s(h) \notin H$. Then as before there then exists $n \in \mathbb{N}$ such that $n \leq s(h)$ and $n \in G$. By the contrapositive of one of the consequences of $h \in H$, since $n \in G$ we then have $\neg(n \leq h)$, which by theorem 1.10 part 5 implies $h < n$. Since $n < s(h) = h + 1$ would then lead to a contradiction of theorem 1.10 part 9, we must have $n = s(h)$. Let arbitrary $g \in G$. Suppose for contradiction that $g < n = s(h) = h+1$. Then by theorem 1.10 part 10 we have $g \leq h$. Since $h \in H$ we then have $g \notin G$, a contradiction. So we must have $\neg(g < n)$, which by theorem 1.10 part 5 implies $n \leq g$. We know that $n \in G$, so this contradicts our earlier assumption that such an element must not exist. So $s(h) \in H$ and $H = \mathbb{N}$.

We know $H \cap G = \emptyset$. So $\mathbb{N} \cap G = \emptyset$. Suppose for contradiction that G is nonempty, meaning that there exists some $x \in G$. Then since $G \subseteq \mathbb{N}$ we would have $x \in \mathbb{N}$ and therefore $x \in \mathbb{N} \cap G = \emptyset$, a contradiction. So we must have $G = \emptyset$, contradicting a key assumption. □

1.2 Construction of the Integers

1.2.1 Construction

Definition 1.12. *The relation $\sim$ on $\mathbb{N} \times \mathbb{N}$ is defined by $\langle a, b\rangle \sim \langle c, d\rangle$ if and only if $a + d = b + c$ for all $\langle a, b\rangle, \langle c, d\rangle \in \mathbb{N} \times \mathbb{N}$. That is, the set such that*

$$\forall x(x \in \sim \leftrightarrow \exists \langle a, b\rangle \in \mathbb{N} \times \mathbb{N} \exists \langle c, d\rangle \in \mathbb{N} \times \mathbb{N}(x = \langle \langle a, b\rangle, \langle c, d\rangle\rangle \wedge a + d = b + c))$$

(Existence follows from using $(\mathbb{N} \times \mathbb{N}) \times (\mathbb{N} \times \mathbb{N})$ as a bounding set. Uniqueness from extensionality)

Lemma 1.13. *The relation $\sim$ is an equivalence relation on $\mathbb{N} \times \mathbb{N}$.*

Proof. Showing that $\sim \subseteq (\mathbb{N} \times \mathbb{N}) \times (\mathbb{N} \times \mathbb{N})$ is straightforward. We need to show reflexivity on $\mathbb{N} \times \mathbb{N}$, symmetry, and transitivity. We start with reflexivity; let arbitrary $\langle a, b\rangle \in \mathbb{N} \times \mathbb{N}$. We immediately have $a + b = b + a$ and $\langle a, b\rangle \sim \langle a, b\rangle$. For symmetry, let $\langle a, b\rangle, \langle c, d\rangle$ be arbitrary and suppose $\langle a, b\rangle \sim \langle c, d\rangle$. Then by definition both pairs are members of $\mathbb{N} \times \mathbb{N}$ and $a + d = b + c$. We then have $c + b = b + c = a + d = d + a$, which then means $\langle c, d\rangle \sim \langle a, b\rangle$.

Finally transitivity. Let $\langle a, b\rangle, \langle c, d\rangle, \langle e, f\rangle$ be arbitrary, suppose $\langle a, b\rangle \sim \langle c, d\rangle$ and $\langle c, d\rangle \sim \langle e, f\rangle$. This means that all three pairs are members of $\mathbb{N} \times \mathbb{N}$, $a + d = c + b$, and $c + f = e + d$. We then have

$$\begin{aligned}(a + f) + c &= a + (c + f) = a + (e + d) = (a + d) + e \\ &= (c + b) + e = (b + e) + c\end{aligned}$$

By cancellation we then have $a + f = b + e$. So $\langle a, b\rangle \sim \langle e, f\rangle$. □

Definition 1.14. *The set of **integers**, denoted $\mathbb{Z}$, is the quotient set $(\mathbb{N}\times\mathbb{N})/\sim$. That is,*

$$\forall x(x \in \mathbb{Z} \leftrightarrow \exists \langle a, b\rangle \in \mathbb{N} \times \mathbb{N}(x = [\langle a, b\rangle]))$$

where

$$\forall \langle a, b\rangle \in \mathbb{N} \times \mathbb{N} \forall x(x \in [\langle a, b\rangle] \leftrightarrow \langle a, b\rangle \sim x)$$

Each integer is an equivalence class. Existence of each equivalence class can be shown using $\mathrm{ran}(\sim)$ as a bounding set. The same can be done for $\mathbb{Z}$ using $\mathscr{P}(\mathrm{ran}(\sim))$. Uniqueness for both follows from extensionality.

Definition 1.15. *We define $\hat{0}, \hat{1}$ as $\hat{0} = [\langle 1, 1\rangle]$ and $\hat{1} = [\langle 1 + 1, 1\rangle]$.*

1.2.2 Operations on the integers

Definition 1.16. *The binary operation $+$ on $\mathbb{Z}$ is defined as $+ : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ where*

$$\forall x(x \in + \leftrightarrow \exists \langle a,b\rangle \in \mathbb{N}\times\mathbb{N}\exists \langle c,d\rangle \in \mathbb{N}\times\mathbb{N}(x = \langle\langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle,[\langle a+c,b+d\rangle]\rangle))$$

put simply, for all $[\langle a,b\rangle],[\langle c,d\rangle] \in \mathbb{Z}$,

$$[\langle a,b\rangle] + [\langle c,d\rangle] = [\langle a+c,b+d\rangle]$$

(A bounding set is $(\mathbb{Z}\times\mathbb{Z})\times\mathbb{Z}$. Uniqueness follows from extensionality.)

Proof. We start by showing the domain bound $\mathrm{dom}(+) = \mathbb{Z}\times\mathbb{Z}$. First suppose $x \in \mathrm{dom}(+)$. Then there exists y such that $\langle x,y\rangle \in +$ and $\langle a,b\rangle, \langle c,d\rangle \in \mathbb{N}\times\mathbb{N}$ such that $x = \langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle \in \mathbb{Z}\times\mathbb{Z}$. Next suppose $x \in \mathbb{Z}\times\mathbb{Z}$. Then there exists $x_0, x_1 \in \mathbb{Z}$ such that $x = \langle x_0, x_1\rangle$. There exists $\langle a,b\rangle, \langle c,d\rangle \in \mathbb{N}\times\mathbb{N}$ such that $x_0 = [\langle a,b\rangle]$ and $x_1 = [\langle c,d\rangle]$. We know $\langle a+c, b+d\rangle \in \mathbb{N}\times\mathbb{N}$ and $\langle\langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle,[\langle a+c,b+d\rangle]\rangle \in +$, so $x = \langle x_0,x_1\rangle = \langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle \in \mathrm{dom}(+)$.

Now the range bound $\mathrm{ran}(+) \subseteq \mathbb{Z}$. Let arbitrary $y \in \mathrm{ran}(+)$. Then there exists x such that $\langle x,y\rangle \in +$ and $a,b,c,d \in \mathbb{N}$ such that $y = [\langle a+c,b+d\rangle] \in \mathbb{Z}$.

Now we prove well definedness. That is, that addition between equivalence classes is independent of the particular elements used to represent each equivalence class:

$$\begin{aligned}\forall\langle a,b\rangle \in \mathbb{N}\times\mathbb{N}\forall\langle c,d\rangle \in \mathbb{N}\times\mathbb{N}\forall\langle a',b'\rangle \in \mathbb{N}\times\mathbb{N}\forall\langle c',d'\rangle \in \mathbb{N}\times\mathbb{N}(&\\ [\langle a,b\rangle] = [\langle a',b'\rangle] \wedge [\langle c,d\rangle] = [\langle c',d'\rangle]&\\ \to [\langle a+c,b+d\rangle] = [\langle a'+c',b'+d'\rangle])&\end{aligned}$$

Let arbitrary $\langle a,b\rangle, \langle c,d\rangle, \langle a',b'\rangle, \langle c',d'\rangle \in \mathbb{N}\times\mathbb{N}$ and suppose that $[\langle a,b\rangle] = [\langle a',b'\rangle]$ and $[\langle c,d\rangle] = [\langle c',d'\rangle]$. Then $\langle a,b\rangle \sim \langle a',b'\rangle$ and $a + b' = b + a'$. Similarly $c + d' = d + c'$. We then have, by associativity and commutativity of natural arithmetic:

$$\begin{aligned}(a+c)+(b'+d') &= (a+b')+(c+d')\\ &= (b+a')+(d+c') = (b+d)+(a'+c')\end{aligned}$$

and we can conclude that $\langle a+c,b+d\rangle \sim \langle a'+c',b'+d'\rangle$ and $[\langle a+c,b+d\rangle] = [\langle a'+c',b'+d'\rangle]$.

Finally we show functionality. Let arbitrary $x \in \mathrm{dom}(+)$. Then by definition there exists y such that $\langle x,y\rangle \in +$. There then exists $\langle a,b\rangle, \langle c,d\rangle \in \mathbb{N}\times\mathbb{N}$ such that $x = \langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle$ and $y = [\langle a+c,b+d\rangle]$. Let y' be arbitrary such that $\langle x,y'\rangle \in +$. Then similarly there exists $\langle a',b'\rangle, \langle c',d'\rangle \in \mathbb{N}\times\mathbb{N}$ such that $x = \langle[\langle a',b'\rangle],[\langle c',d'\rangle]\rangle$ and $y' = [\langle a'+c',b'+d'\rangle]$. We know $\langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle = \langle[\langle a',b'\rangle],[\langle c',d'\rangle]\rangle$. So since addition is well defined we have $y = [\langle a+c,b+d\rangle] = [\langle a'+c',b'+d'\rangle] = y'$. □

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Definition 1.17. *The binary operation $\cdot$ on $\mathbb{Z}$ is defined as $\cdot : \mathbb{Z}\times\mathbb{Z} \to \mathbb{Z}$ where*

$$\forall x(x \in \cdot \leftrightarrow \exists\langle a,b\rangle \in \mathbb{N}\times\mathbb{N}\exists\langle c,d\rangle \in \mathbb{N}\times\mathbb{N}(\\ x = \langle\langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle,[\langle ac+bd, ad+bc\rangle]\rangle))$$

(A bounding set is $(\mathbb{Z}\times\mathbb{Z})\times\mathbb{Z}$. Uniqueness follows from extensionality.)

Proof. We start with the domain bound $\mathrm{dom}(\cdot) = \mathbb{Z}\times\mathbb{Z}$. First suppose $x \in \mathrm{dom}(\cdot)$. Then there exists y such that $\langle x,y\rangle \in \cdot$, and $\langle a,b\rangle, \langle c,d\rangle \in \mathbb{N}\times\mathbb{N}$ such that $x = \langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle \in \mathbb{Z}\times\mathbb{Z}$. Next suppose $x \in \mathbb{Z}\times\mathbb{Z}$. Then there exists $x_0, x_1 \in \mathbb{Z}$ such that $x = \langle x_0, x_1\rangle$. There exists $\langle a,b\rangle, \langle c,d\rangle \in \mathbb{N}\times\mathbb{N}$ such that $x_0 = [\langle a,b\rangle]$ and $x_1 = [\langle c,d\rangle]$. We have $\langle\langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle,[\langle ac+bd, ad+bc\rangle]\rangle \in \cdot$. So $x = \langle x_0,x_1\rangle = \langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle \in \mathrm{dom}(\cdot)$.

Next the range bound $\mathrm{ran}(\cdot) \subseteq \mathbb{Z}$. Let arbitrary $y \in \mathrm{ran}(\cdot)$. Then there exists x such that $\langle x,y\rangle \in \cdot$ and $a,b,c,d \in \mathbb{N}$ such that $y = [\langle ac+bd, ad+bc\rangle] \in \mathbb{Z}$.

Next we show well definedness. That is

$$\forall\langle a,b\rangle \in \mathbb{N}\times\mathbb{N}\forall\langle c,d\rangle \in \mathbb{N}\times\mathbb{N}\forall\langle a',b'\rangle \in \mathbb{N}\times\mathbb{N}\forall\langle c',d'\rangle \in \mathbb{N}\times\mathbb{N}(\\ ([\langle a,b\rangle] = [\langle a',b'\rangle] \wedge [\langle c,d\rangle] = [\langle c',d'\rangle]) \\ \rightarrow [\langle ac+bd, ad+bc\rangle] = [\langle a'c'+b'd', a'd'+b'c'\rangle])$$

Let arbitrary $\langle a,b\rangle, \langle c,d\rangle, \langle a',b'\rangle, \langle c',d'\rangle \in \mathbb{N}\times\mathbb{N}$ and suppose $[\langle a,b\rangle] = [\langle a',b'\rangle]$ and $[\langle c,d\rangle] = [\langle c',d'\rangle]$. Then we have $\langle a,b\rangle \sim \langle a',b'\rangle$ and $a+b' = b+a'$. Similarly we have $c+d' = d+c'$. We want to show

$$(ac+bd)+(a'd'+b'c') = (ad+bc)+(a'c'+b'd')$$

See that

$$\begin{aligned} &((a+b')d+(b+a')c)+((c+d')b'+(d+c')a') \\ &= ((ad+b'd)+(bc+a'c))+((cb'+d'b')+(da'+c'a')) \\ &= ((ad+bc)+(b'd+a'c))+((a'c'+b'd')+(da'+cb')) \\ &= ((ad+bc)+(a'c'+b'd'))+((b'd+a'c)+(da'+cb')) \end{aligned}$$

Also see that

$$\begin{aligned} &((a+b')d+(b+a')c)+((c+d')b'+(d+c')a') \\ &= ((b+a')d+(a+b')c)+((d+c')b'+(c+d')a') \\ &= ((bd+a'd)+(ac+b'c))+((db'+c'b')+(ca'+d'a')) \\ &= ((ac+bd)+(da'+cb'))+((a'd'+c'b')+(a'c+b'd)) \\ &= ((ac+bd)+(a'd'+c'b'))+((b'd+a'c)+(da'+cb')) \end{aligned}$$

By cancellation we then have our desired result, which means

$$\langle ac+bd, ad+bc\rangle \sim \langle a'c'+b'd', a'd'+b'c'\rangle$$

and $[\langle ac+bd, ad+bc\rangle] = [\langle a'c'+b'd', a'd'+b'c'\rangle]$.
(next page)

Finally functionality. Let arbitrary $x \in \mathrm{dom}(\cdot)$. Then there exists y such that $\langle x, y\rangle \in \cdot$ and $\langle a, b\rangle, \langle c, d\rangle \in \mathbb{N} \times \mathbb{N}$ such that $x = \langle[\langle a, b\rangle], [\langle c, d\rangle]\rangle$ and $y = [\langle ac + bd, ad + bc\rangle]$. Let y' be arbitrary such that $\langle x, y'\rangle \in \cdot$. Then similarly there exists $\langle a', b'\rangle, \langle c', d'\rangle \in \mathbb{N} \times \mathbb{N}$ such that $x = \langle[\langle a', b'\rangle], [\langle c', d'\rangle]\rangle$ and $y' = [\langle a'c' + b'd', a'd' + b'c'\rangle]$. We have $\langle[\langle a, b\rangle], [\langle c, d\rangle]\rangle = \langle[\langle a', b'\rangle], [\langle c', d'\rangle]\rangle$, by well definedness we then have $y = [\langle ac + bd, ad + bc\rangle] = [\langle a'c' + b'd', a'd' + b'c'\rangle] = y'$. □

Definition 1.18. *The unary operation $-$ on $\mathbb{Z}$ is defined as $- : \mathbb{Z} \to \mathbb{Z}$ where*

$$\forall x(x \in - \leftrightarrow \exists \langle a, b\rangle \in \mathbb{N} \times \mathbb{N}(x = \langle[\langle a, b\rangle], [\langle b, a\rangle]\rangle))$$

(A bounding set is $\mathbb{Z} \times \mathbb{Z}$. Uniqueness follows from extensionality.)

Proof. We start with the domain bound $\mathrm{dom}(-) = \mathbb{Z}$. First let arbitrary $x \in \mathrm{dom}(-)$. Then there exists y such that $\langle x, y\rangle \in -$, and $\langle a, b\rangle \in \mathbb{N} \times \mathbb{N}$ such that $x = [\langle a, b\rangle] \in \mathbb{Z}$. Next suppose $x \in \mathbb{Z}$. Then there exists $\langle a, b\rangle \in \mathbb{N} \times \mathbb{N}$ such that $x = [\langle a, b\rangle]$. We have $\langle[\langle a, b\rangle], [\langle b, a\rangle]\rangle \in -$. So $x = [\langle a, b\rangle] \in \mathrm{dom}(-)$.

Next the range bound $\mathrm{ran}(-) \subseteq \mathbb{Z}$. Suppose $y \in \mathrm{ran}(-)$. Then there exists x such that $\langle x, y\rangle \in -$ and $a, b \in \mathbb{N}$ such that $y = [\langle b, a\rangle] \in \mathbb{Z}$.

Next we show well-defineness. That is

$$\forall \langle a, b\rangle \in \mathbb{N} \times \mathbb{N} \forall \langle a', b'\rangle \in \mathbb{N} \times \mathbb{N}([\langle a, b\rangle] = [\langle a', b'\rangle] \to [\langle b, a\rangle] = [\langle b', a'\rangle])$$

Letting arbitrary $\langle a, b\rangle, \langle a', b'\rangle \in \mathbb{N} \times \mathbb{N}$ and supposing that $[\langle a, b\rangle] = [\langle a', b'\rangle]$, we have $a + b' = b + a'$. See that this also means $\langle b, a\rangle \sim \langle b', a'\rangle$ and therefore $[\langle b, a\rangle] = [\langle b', a'\rangle]$.

Finally functionality. Let arbitrary $x \in \mathrm{dom}(-)$. Then there exists y such that $\langle x, y\rangle \in -$, and $\langle a, b\rangle \in \mathbb{N} \times \mathbb{N}$ such that $x = [\langle a, b\rangle]$ and $y = [\langle b, a\rangle]$. Let y' be arbitrary such that $\langle x, y'\rangle \in -$. Then there exists $\langle a', b'\rangle \in \mathbb{N} \times \mathbb{N}$ such that $x = [\langle a', b'\rangle]$ and $y' = [\langle b', a'\rangle]$. We have $[\langle a, b\rangle] = [\langle a', b'\rangle]$; by well-definedness we have $y = [\langle b, a\rangle] = [\langle b', a'\rangle] = y'$. □

1.2.3 Ordering

Definition 1.19. *The binary relation $<$ on $\mathbb{Z}$ is defined such that*

$$\forall x(x \in < \leftrightarrow \exists \langle a,b\rangle \in \mathbb{N}\times\mathbb{N}\exists\langle c,d\rangle \in \mathbb{N}\times\mathbb{N}(x = \langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle \wedge a+d<b+c))$$

($\mathbb{Z}\times\mathbb{Z}$ is a bounding set. Uniqueness follows from extensionality.)

Lemma 1.20.

$$\begin{gathered}\forall\langle a,b\rangle \in \mathbb{N}\times\mathbb{N}\forall\langle c,d\rangle \in \mathbb{N}\times\mathbb{N}\forall\langle a',b'\rangle \in \mathbb{N}\times\mathbb{N}\forall\langle c',d'\rangle \in \mathbb{N}\times\mathbb{N}(\\ ([\langle a,b\rangle] = [\langle a',b'\rangle] \wedge [\langle c,d\rangle] = [\langle c',d'\rangle])\\ \rightarrow a+d<b+c \leftrightarrow a'+d'<b'+c')\end{gathered}$$

Proof. Let arbitrary $\langle a,b\rangle, \langle c,d\rangle, \langle a',b'\rangle, \langle c',d'\rangle \in \mathbb{N}\times\mathbb{N}$. Suppose that $[\langle a,b\rangle] = [\langle a',b'\rangle]$ and $[\langle c,d\rangle] = [\langle c',d'\rangle]$, which then means $a+b' = b+a'$ and $c+d' = d+c'$. First suppose $a+d<b+c$. Then there exists x such that $(a+d)+x = b+c$, and we have

$$(((a+d)+x)+(b+a')) + (c+d') = ((b+c)+(a+b')) + (d+c')$$

Rearranging the left side:

$$\begin{aligned}&(((a+d)+x)+(b+a')) + (c+d')\\ &= ((a+d)+x) + ((b+c)+(a'+d'))\\ &= (a+d) + (x+((b+c)+(a'+d')))\\ &= (a+d) + ((b+c)+(x+(a'+d')))\\ &= ((a+d)+(b+c)) + ((a'+d')+x)\end{aligned}$$

and the right side:

$$\begin{aligned}&((b+c)+(a+b')) + (d+c')\\ &= (b+c) + ((a+b')+(d+c'))\\ &= (b+c) + ((a+d)+(b'+c'))\\ &= ((a+d)+(b+c)) + (b'+c')\end{aligned}$$

By cancellation we have $(a'+d')+x = b'+c'$ and $a'+d'<b'+c'$.
(next page)

Next suppose $a'+d' < b'+c'$. Then there exists x such that $a'+d'+x = b'+c'$. We have

$$(((a' + d') + x) + (a + b')) + (d + c') = ((b' + c') + (b + a')) + (c + d')$$

Rearranging the left side:

$$\begin{aligned}
&(((a' + d') + x) + (a + b')) + (d + c') \\
&= ((a' + d') + x) + ((a + b') + (d + c')) \\
&= ((a' + d') + x) + ((b' + c') + (a + d)) \\
&= (a' + d') + (x + ((b' + c') + (a + d))) \\
&= ((a' + d') + (b' + c')) + ((a + d) + x)
\end{aligned}$$

and the right hand side:

$$\begin{aligned}
&((b' + c') + (b + a')) + (c + d') \\
&= (b' + c') + ((b + a') + (c + d')) \\
&= (b' + c') + ((a' + d') + (b + c)) \\
&= ((a' + d') + (b' + c')) + (b + c)
\end{aligned}$$

So by cancellation $(a + d) + x = b + c$ and $a + d < b + c$. □

Lemma 1.21.

$$\forall\langle a, b\rangle \in \mathbb{N} \times \mathbb{N} \forall\langle c, d\rangle \in \mathbb{N} \times \mathbb{N}([\langle a, b\rangle] < [\langle c, d\rangle] \leftrightarrow a + d < b + c)$$

Proof. Let arbitrary $\langle a, b\rangle, \langle c, d\rangle \in \mathbb{N} \times \mathbb{N}$. First suppose $[\langle a, b\rangle] < [\langle c, d\rangle]$. Then by definition there exists $\langle a_0, b_0\rangle, \langle c_0, d_0\rangle \in \mathbb{N} \times \mathbb{N}$ such that $[\langle a, b\rangle] = [\langle a_0, b_0\rangle]$, $[\langle c, d\rangle] = [\langle c_0, d_0\rangle]$ and $a_0 + d_0 < b_0 + c_0$. It then immediately follows by lemma 1.20 that $a + d < b + c$.

The reverse direction is straightforward. Suppose $a + d < b + c$ and it immediately follows by definition that $[\langle a, b\rangle] < [\langle c, d\rangle]$. □

Proof. (range/domain bound and well-definedness) We start by showing $<$ is a relation on $\mathbb{Z}$, $<\subseteq \mathbb{Z} \times \mathbb{Z}$. Let arbitrary $x \in <$. Then by definition there exists $\langle a, b\rangle, \langle c, d\rangle \in \mathbb{N} \times \mathbb{N}$ such that $x = \langle[\langle a, b\rangle], [\langle c, d\rangle]\rangle \in \mathbb{Z} \times \mathbb{Z}$.

Next well-definedness. That is,

$$\begin{aligned}
\forall\langle a, b\rangle \in \mathbb{N} \times \mathbb{N} \forall\langle c, d\rangle \in \mathbb{N} \times \mathbb{N} \forall\langle a', b'\rangle \in \mathbb{N} \times \mathbb{N} \forall\langle c', d'\rangle \in \mathbb{N} \times \mathbb{N}(\\
([\langle a, b\rangle] = [\langle a', b'\rangle] \wedge [\langle c, d\rangle] = [\langle c', d'\rangle]) \\
\rightarrow ([\langle a, b\rangle] < [\langle c, d\rangle] \leftrightarrow [\langle a', b'\rangle] < [\langle c', d'\rangle]))
\end{aligned}$$

Letting arbitrary $\langle a, b\rangle, \langle c, d\rangle, \langle a', b'\rangle, \langle c', d'\rangle \in \mathbb{N} \times \mathbb{N}$. Suppose that $[\langle a, b\rangle] = [\langle a', b'\rangle]$ and $[\langle c, d\rangle] = [\langle c', d'\rangle]$, which then means $a+b' = b+a'$ and $c+d' = d+c'$. Suppose first that $[\langle a, b\rangle] < [\langle c, d\rangle]$, which by lemma 1.21 means $a+d < b+c$. It follows by lemma 1.20 that $a' + d' < b' + c'$ and by lemma 1.21 that $[\langle a', b'\rangle] < [\langle c', d'\rangle]$. An identical strategy proves the reverse direction. □

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Definition 1.22. *The binary relation $\leq$ on $\mathbb{Z}$ is defined such that*

$$\forall x(x \in \leq \leftrightarrow \exists \langle a,b\rangle \in \mathbb{N}\times\mathbb{N}\exists\langle c,d\rangle \in \mathbb{N}\times\mathbb{N}(x = \langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle \\ \wedge ([\langle a,b\rangle] < [\langle c,d\rangle] \vee [\langle a,b\rangle] = [\langle c,d\rangle])))$$

($\mathbb{Z}\times\mathbb{Z}$ is a bounding set. Uniqueness follows from extensionality.)

Lemma 1.23.

$$\forall\langle a,b\rangle \in \mathbb{N}\times\mathbb{N}\forall\langle c,d\rangle \in \mathbb{N}\times\mathbb{N}([\langle a,b\rangle] \leq [\langle c,d\rangle] \\ \leftrightarrow ([\langle a,b\rangle] < [\langle c,d\rangle] \vee [\langle a,b\rangle] = [\langle c,d\rangle]))$$

Proof. Let arbitrary $\langle a,b\rangle, \langle c,d\rangle \in \mathbb{N}\times\mathbb{N}$. First suppose $[\langle a,b\rangle] \leq [\langle c,d\rangle]$. Then by definition there exists $\langle a_0,b_0\rangle, \langle c_0,d_0\rangle \in \mathbb{N}\times\mathbb{N}$ such that $[\langle a,b\rangle] = [\langle a_0,b_0\rangle]$, $[\langle c,d\rangle] = [\langle c_0,d_0\rangle]$, and $[\langle a_0,b_0\rangle] < [\langle c_0,d_0\rangle]$ or $[\langle a_0,b_0\rangle] = [\langle c_0,d_0\rangle]$. In the first case suppose $[\langle a_0,b_0\rangle] < [\langle c_0,d_0\rangle]$. It then follows from the well-definedness of $<$ that $[\langle a,b\rangle] < [\langle c,d\rangle]$. In the other case suppose $[\langle a_0,b_0\rangle] = [\langle c_0,d_0\rangle]$. It immediately follows that $[\langle a,b\rangle] = [\langle a_0,b_0\rangle] = [\langle c_0,d_0\rangle] = [\langle c,d\rangle]$.

Now suppose $[\langle a,b\rangle] < [\langle c,d\rangle] \vee [\langle a,b\rangle] = [\langle c,d\rangle]$. It immediately follows by definition that $[\langle a,b\rangle] \leq [\langle c,d\rangle]$. □

Proof. (domain/range bound and well-definedness) First we show $\leq \subseteq \mathbb{Z}\times\mathbb{Z}$. Let arbitrary $x \in \leq$. Then there exists $\langle a,b\rangle, \langle c,d\rangle \in \mathbb{N}\times\mathbb{N}$ such that $x = \langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle \in \mathbb{Z}\times\mathbb{Z}$.

Next well-definedness. That is

$$\forall\langle a,b\rangle \in \mathbb{N}\times\mathbb{N}\forall\langle c,d\rangle \in \mathbb{N}\times\mathbb{N}\forall\langle a',b'\rangle \in \mathbb{N}\times\mathbb{N}\forall\langle c',d'\rangle \in \mathbb{N}\times\mathbb{N}(\\ ([\langle a,b\rangle] = [\langle a',b'\rangle] \wedge [\langle c,d\rangle] = [\langle c',d'\rangle]) \\ \rightarrow ([\langle a,b\rangle] \leq [\langle c,d\rangle] \leftrightarrow [\langle a',b'\rangle] \leq [\langle c',d'\rangle]))$$

Letting arbitrary $\langle a,b\rangle, \langle c,d\rangle, \langle a',b'\rangle, \langle c',d'\rangle \in \mathbb{N}\times\mathbb{N}$. Suppose that $[\langle a,b\rangle] = [\langle a',b'\rangle]$ and $[\langle c,d\rangle] = [\langle c',d'\rangle]$. First suppose $[\langle a,b\rangle] \leq [\langle c,d\rangle]$. It follows by lemma 1.23 that $[\langle a,b\rangle] < [\langle c,d\rangle] \vee [\langle a,b\rangle] = [\langle c,d\rangle]$. In the first case where $[\langle a,b\rangle] < [\langle c,d\rangle]$ it follows from the well-definedness of $<$ that $[\langle a',b'\rangle] < [\langle c',d'\rangle]$ and again by lemma 1.23 that $[\langle a',b'\rangle] \leq [\langle c',d'\rangle]$. For the other case where $[\langle a,b\rangle] = [\langle c,d\rangle]$ we immediately have $[\langle a',b'\rangle] = [\langle a,b\rangle] = [\langle c,d\rangle] = [\langle c',d'\rangle]$ and by lemma 1.23 $[\langle a',b'\rangle] \leq [\langle c',d'\rangle]$. The reverse implication follows by the same argument. □

1.2.4 Properties

Theorem 1.24. *Let $x, y, z \in \mathbb{Z}$.*

1. $(x + y) + z = x + (y + z)$ *(Associative Law for Addition)*
2. $x + y = y + x$ *(Commutative Law for Addition)*
3. $x + \hat{0} = x$ *(Identity Law for Addition)*
4. $x + (-x) = \hat{0}$ *(Inverses Law for Addition)*
5. $(xy)z = x(yz)$ *(Associative Law for Multiplication)*
6. $xy = yx$ *(Commutative Law for Multiplication)*
7. $x \cdot \hat{1} = x$ *(Identity Law for Multiplication)*
8. $x(y + z) = xy + xz$ *(Distributive Law)*
9. *If $xy = \hat{0}$, then $x = \hat{0}$ or $y = \hat{0}$* *(No Zero Divisors Law)*
10. *Precisely one of $x < y$ or $x = y$ or $x > y$ holds* *(Trichotomy Law)*
11. *If $x < y$ and $y < z$, then $x < z$* *(Transitive Law)*
12. *If $x < y$ then $x + z < y + z$* *(Addition Law for Order)*
13. *If $x < y$ and $z > \hat{0}$, then $xz < yz$* *(Multiplication Law for Order)*
14. $\hat{0} \neq \hat{1}$ *(Non-Triviality)*

Proof. (1) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} \forall z \in \mathbb{Z}((x + y) + z = x + (y + z))$$

Let arbitrary $x, y, z \in \mathbb{Z}$. Then there exists $a, b, c, d, e, f \in \mathbb{N}$ such that $x = [\langle a, b \rangle]$, $y = [\langle c, d \rangle]$, and $z = [\langle e, f \rangle]$. We have

$$\begin{aligned}
(x + y) + z &= ([\langle a, b \rangle] + [\langle c, d \rangle]) + [\langle e, f \rangle] \\
&= [\langle a + c, b + d \rangle] + [\langle e, f \rangle] \\
&= [\langle (a + c) + e, (b + d) + f \rangle] \\
&= [\langle a + (c + e), b + (d + f) \rangle] \\
&= [\langle a, b \rangle] + [\langle c + e, d + f \rangle] \\
&= [\langle a, b \rangle] + ([\langle c, d \rangle] + [\langle e, f \rangle]) \\
&= x + (y + z)
\end{aligned}$$

Where the fourth equality follows from associativity of $\mathbb{N}$. □

(next page)

Proof. (2) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}(x + y = y + x)$$

Let arbitrary $x, y \in \mathbb{Z}$. Then there exists $a, b, c, d \in \mathbb{N}$ such that $x = [\langle a, b \rangle]$ and $y = [\langle c, d \rangle]$. It follows that

$$\begin{aligned} x + y = [\langle a, b \rangle] + [\langle c, d \rangle] &= [\langle a + c, b + d \rangle] \\ &= [\langle c + a, d + b \rangle] = [\langle c, d \rangle] + [\langle a, b \rangle] = y + x \end{aligned}$$

□

Proof. (3) We want to show

$$\forall x \in \mathbb{Z}(x + \hat{0} = x)$$

Let arbitrary $x \in \mathbb{Z}$. Then there exists $a, b \in \mathbb{N}$ such that $x = [\langle a, b \rangle]$. We have $a + (b + 1) = b + (a + 1)$ and therefore $\langle a, b \rangle \sim \langle a + 1, b + 1 \rangle$. So

$$x = [\langle a, b \rangle] = [\langle a + 1, b + 1 \rangle] = [\langle a, b \rangle] + [\langle 1, 1 \rangle] = x + \hat{0}$$

□

Proof. (4) We want to show

$$\forall x \in \mathbb{Z}(x + (-x) = \hat{0})$$

Let arbitrary $x \in \mathbb{Z}$. Then there exists $a, b \in \mathbb{N}$ such that $x = [\langle a, b \rangle]$. Then by definition $-x = [\langle b, a \rangle]$. We have $x + (-x) = [\langle a, b \rangle] + [\langle b, a \rangle] = [\langle a + b, b + a \rangle]$. We also know $(a + b) + 1 = (b + a) + 1$, meaning $\langle a + b, b + a \rangle \sim \langle 1, 1 \rangle$. Putting it all together we have

$$x + (-x) = [\langle a + b, b + a \rangle] = [\langle 1, 1 \rangle] = \hat{0}$$

□

Proof. (5) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} \forall z \in \mathbb{Z}((xy)z = x(yz))$$

Let arbitrary $x, y, z \in \mathbb{Z}$. Then there exists $a, b, c, d, e, f \in \mathbb{N}$ such that $x = [\langle a, b \rangle]$, $y = [\langle c, d \rangle]$, and $z = [\langle e, f \rangle]$. We have

$$\begin{aligned} (xy)z &= [\langle ac + bd, ad + bc \rangle] \cdot [\langle e, f \rangle] \\ &= [\langle (ac + bd)e + (ad + bc)f, (ac + bd)f + (ad + bc)e \rangle] \\ &= [\langle ((ac)e + (bd)e) + ((ad)f + (bc)f), ((ac)f + (bd)f) + ((ad)e + (bc)e) \rangle] \\ &= [\langle (a(ce) + a(df)) + (b(de) + b(cf)), (a(cf) + a(de)) + (b(ce) + b(df)) \rangle] \\ &= [\langle a(ce + df) + b(cf + de), a(cf + de) + b(ce + df) \rangle] \\ &= [\langle a, b \rangle] \cdot [\langle ce + df, cf + de \rangle] \\ &= [\langle a, b \rangle] \cdot ([\langle c, d \rangle] \cdot [\langle e, f \rangle]) \\ &= x(yz) \end{aligned}$$

□

(next page)

Proof. (6) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}(xy = yx)$$

Let arbitrary $x, y \in \mathbb{Z}$. Then there exists $a, b, c, d \in \mathbb{N}$ such that $x = [\langle a, b \rangle]$ and $y = [\langle c, d \rangle]$. It follows that

$$xy = [\langle ac + bd, ad + bc \rangle] = [\langle ca + db, cb + da \rangle] = [\langle c, d \rangle] \cdot [\langle a, b \rangle] = yx \qquad \square$$

Proof. (7) We want to show

$$\forall x \in \mathbb{Z}(x \cdot \hat{1} = x)$$

Let arbitrary $x \in \mathbb{Z}$. Then there exists $a, b \in \mathbb{N}$ such that $x = [\langle a, b \rangle]$. We have

$$\begin{aligned} x \cdot \hat{1} = [\langle a, b \rangle] \cdot [\langle 1 + 1, 1 \rangle] &= [\langle a(1 + 1) + b \cdot 1, a \cdot 1 + b(1 + 1) \rangle] \\ &= [\langle (a + a) + b, a + (b + b) \rangle] \end{aligned}$$

We also have $a + (a + (b + b)) = b + ((a + a) + b)$, meaning $\langle a, b \rangle \sim \langle (a + a) + b, a + (b + b) \rangle$. Combining these two facts we have

$$x \cdot \hat{1} = [\langle (a + a) + b, a + (b + b) \rangle] = [\langle a, b \rangle] = x \qquad \square$$

Proof. (8) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} \forall z \in \mathbb{Z}(x(y + z) = xy + xz)$$

Let arbitrary $x, y, z \in \mathbb{Z}$. Then there exists $a, b, c, d, e, f \in \mathbb{N}$ such that $x = [\langle a, b \rangle]$, $y = [\langle c, d \rangle]$, and $z = [\langle e, f \rangle]$. We have

$$\begin{aligned} x(y + z) &= [\langle a, b \rangle] \cdot [\langle c + e, d + f \rangle] \\ &= [\langle a(c + e) + b(d + f), a(d + f) + b(c + e) \rangle] \\ &= [\langle (ac + bd) + (ae + bf), (ad + bc) + (af + be) \rangle] \\ &= [\langle ac + bd, ad + bc \rangle] + [\langle ae + bf, af + be \rangle] \\ &= [\langle a, b \rangle] \cdot [\langle c, d \rangle] + [\langle a, b \rangle] \cdot [\langle e, f \rangle] \\ &= xy + xz \end{aligned} \qquad \square$$

(next page)

Lemma 1.25. $\forall a \in \mathbb{N} \forall b \in \mathbb{N}([\langle a, b\rangle] = \hat{0} \leftrightarrow a = b)$

Proof. Let arbitrary $a, b \in \mathbb{N}$. First suppose that $[\langle a, b\rangle] = \hat{0} = [\langle 1, 1\rangle]$. Then $a+1 = b+1$ and by cancellation $a = b$. Next suppose $a = b$. Then $a+1 = b+1$ and $\langle a, b\rangle \sim \langle 1, 1\rangle$. We then have $[\langle a, b\rangle] = [\langle 1, 1\rangle] = \hat{0}$. □

Proof. (9) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}(xy = \hat{0} \to (x = \hat{0} \vee y = \hat{0}))$$

Let arbitrary $x, y \in \mathbb{Z}$ and suppose that $xy = \hat{0}$. The statement after the implication is logically equivalent to $x \neq \hat{0} \to y = \hat{0}$. Suppose $x \neq \hat{0}$. There exists $a, b, c, d \in \mathbb{N}$ such that $x = [\langle a, b\rangle]$ and $y = [\langle c, d\rangle]$. We then have $xy = [\langle ac + bd, ad + bc\rangle] = \hat{0}$; by the previous lemma $ac + bd = ad + bc$. That lemma also implies $a \neq b$; so by trichotomy of the natural numbers we have $a < b$ or $b < a$.

In the first case suppose $a < b$. Then there exists $g \in \mathbb{N}$ such that $a + g = b$. We then have

$$(ac + ad) + gd = ac + (a + g)d = ad + (a + g)c = (ac + ad) + gc$$

Cancellation of addition on the naturals allows us to conclude that $gd = gc$. Cancellation of multiplication allows us to conclude $c = d$. By the previous lemma this implies $y = [\langle c, d\rangle] = \hat{0}$.

In the other case suppose $b < a$. Then there exists $g \in \mathbb{N}$ such that $b+g = a$. We then have

$$(bc + bd) + gc = (b + g)c + bd = (b + g)d + bc = (bc + bd) + gd$$

Similarly again we conclude $gc = gd$ and $c = d$, which then implies $y = [\langle c, d\rangle] = \hat{0}$. □

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Proof. (10) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}((x < y \vee x = y \vee y < x) \wedge \\ \neg(x < y \wedge x = y) \wedge \neg(x < y \wedge y < x) \wedge \neg(x = y \wedge y < x))$$

Let arbitrary $x, y \in \mathbb{Z}$. Then there exists $a, b, c, d \in \mathbb{N}$ such that $x = [\langle a, b \rangle]$ and $y = [\langle c, d \rangle]$. We start with the latter three statements. First suppose $x < y$ and $x = y$. By lemma 1.21 $a + d < b + c$ and by definition $a + d = b + c$; this contradicts trichotomy for the naturals. Next suppose $x < y$ and $y < x$. Again this then implies $a + d < b + c$ and $b + c < a + d$ which contradicts trichotomy. Supposing $x = y$ and $y < x$ also contradicts trichotomy in a similar manner.

Finally the former statement. We know that $a+d, b+c \in \mathbb{N}$. By trichotomy we must have

$$a + d < b + c \vee a + d = b + c \vee b + c < a + d$$

which by lemma 1.21 and the definition of equality means

$$[\langle a, b \rangle] < [\langle c, d \rangle] \vee [\langle a, b \rangle] = [\langle c, d \rangle] \vee [\langle c, d \rangle] < [\langle a, b \rangle]$$

which is equivalent to our desired result. □

Proof. (11) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} \forall z \in \mathbb{Z}(x < y \wedge y < z \to x < z)$$

Let arbitrary $x, y, z \in \mathbb{Z}$. Suppose $x < y$ and $y < z$. Then by definition there exists $a, b, c, d, e, f \in \mathbb{N}$ such that $x = [\langle a, b \rangle], y = [\langle c, d \rangle]$, and $z = [\langle e, f \rangle]$. By lemma 1.21 we can assert that $a + d < b + c$ and $c + f < d + e$. By definition there then exists $g, g' \in \mathbb{N}$ such that $(a + d) + g = b + c$ and $(c + f) + g' = d + e$. The former, along with cancellation, associativity and commutativity, allows us to assert

$$((a + d) + g) + f = ((a + d) + f) + g = (b + c) + f$$

So $(a + d) + f < (b + c) + f$. Similarly the latter allows us to assert

$$((c + f) + g') + b = ((b + c) + f) + g' = (d + e) + b = b + (d + e)$$

So $(b + c) + f < b + (d + e)$. By transitivity of the naturals we then have

$$(a + d) + f < b + (d + e)$$

Then there exists $g_0 \in \mathbb{N}$ such that $((a + d) + f) + g_0 = b + (d + e)$. So we have

$$d + ((a + f) + g_0) = ((a + d) + f) + g_0 = b + (d + e) = d + (b + e)$$

which by cancellation means

$$(a + f) + g_0 = b + e$$

and $a + f < b + e$. By lemma 1.21 this means $[\langle a, b \rangle] < [\langle e, f \rangle]$ and $x < z$. □

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Proof. (12) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} \forall z \in \mathbb{Z} (x < y \to x + z < y + z)$$

Let arbitrary $x, y, z \in \mathbb{Z}$. By definition there exists $a, b, c, d, e, f \in \mathbb{N}$ such that $x = [\langle a, b \rangle], y = [\langle c, d \rangle]$, and $z = [\langle e, f \rangle]$. Suppose $x < y$. By lemma 1.21 we then have $a + d < b + c$. By theorem 1.10 part 4 and commutativity and associativity of the naturals, we can assert

$$(a + e) + (d + f) = (a + d) + (e + f) < (b + c) + (e + f) = (b + f) + (c + e)$$

which by lemma 1.21 and the definition of addition, implies

$$[\langle a, b \rangle] + [\langle e, f \rangle] = [\langle a + e, b + f \rangle] < [\langle c + e, d + f \rangle] = [\langle c, d \rangle] + [\langle e, f \rangle]$$

which is equivalent to our desired result. □

Proof. (13) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} \forall z \in \mathbb{Z} (x < y \wedge z > \hat{0} \to xz < yz)$$

Let arbitrary $x, y, z \in \mathbb{Z}$. By definition there exists $a, b, c, d, e, f \in \mathbb{N}$ such that $x = [\langle a, b \rangle], y = [\langle c, d \rangle]$, and $z = [\langle e, f \rangle]$. Suppose that $x < y$ and $z > \hat{0}$. By lemma 1.21 we have $a + d < b + c$ and $1 + f < 1 + e$. By theorem 1.10 part 4 we have $f < e$, so there exists $k \in \mathbb{N}$ such that $f + k = e$. By theorem 1.10 part 6 we have $k(a + d) < k(b + c)$. Part 4 of the same theorem allows us to then assert

$$k(a + d) + ((a + d)f + (b + c)f) < k(b + c) + ((a + d)f + (b + c)f)$$

The left hand side is equivalent to

$$\begin{aligned}
&k(a + d) + ((a + d)f + (b + c)f) \\
&= (ka + kd) + ((af + df) + (bf + cf)) \\
&= ((ka + kd) + (af + df)) + (bf + cf) \\
&= (a(k + f) + d(k + f)) + (bf + cf) \\
&= (ae + de) + (bf + cf) \\
&= (ae + bf) + (cf + de)
\end{aligned}$$

Similarly the right hand side is equivalent to

$$\begin{aligned}
&k(b + c) + ((a + d)f + (b + c)f) \\
&= (kb + kc) + ((af + df) + (bf + cf)) \\
&= (b(k + f) + c(k + f)) + (af + df) \\
&= (be + ce) + (af + df) \\
&= (af + be) + (ce + df)
\end{aligned}$$

(next page)

We can then conclude that $(ae+bf)+(cf+de) < (af+be)+(ce+df)$. By lemma 1.21 and the definition of multiplication on the integers we then have

$$[\langle a,b\rangle]\cdot[\langle e,f\rangle] = [\langle ae+bf, af+be\rangle] < [\langle ce+df, cf+de\rangle] = [\langle c,d\rangle]\cdot[\langle e,f\rangle]$$

Which is equivalent to our desired conclusion. □

Proof. (14) We want to show $\hat{0} \neq \hat{1}$. Suppose for contradiction that they are equal, meaning $[\langle 1,1\rangle] = [\langle 1{+}1,1\rangle]$. By definition we then have $1{+}1 = 1{+}(1{+}1)$, which by cancellation implies $1 = 1{+}1$, contradicting part 5 of theorem 1.7. □

1.2.5 Joining the naturals and the integers

Definition 1.26. *Let $x \in \mathbb{Z}$. x is **positive** if $x > \hat{0}$. x is **negative** if $x < \hat{0}$.*

Theorem 1.27. *Let $i : \mathbb{N} \to \mathbb{Z}$ be defined by $i(n) = [\langle n+1, 1\rangle]$ for all $n \in \mathbb{N}$.*

1. *The function $i : \mathbb{N} \to \mathbb{Z}$ is injective.*
2. *$i(\mathbb{N}) = \{x \in \mathbb{Z} \mid x > \hat{0}\}$*
3. *$i(1) = \hat{1}$*
4. *Let $a, b \in \mathbb{N}$. Then*
   (a) *$i(a+b) = i(a) + i(b)$*
   (b) *$i(ab) = i(a)i(b)$*
   (c) *$a < b$ if and only if $i(a) < i(b)$*

Proof. (1) Defining i as the set such that

$$\forall x(x \in i \leftrightarrow \exists a \in \mathbb{N}\exists b(x = \langle a, b\rangle \wedge b = [\langle a+1, 1\rangle]))$$

Existence of such a set can be shown using the subset axiom, with $\mathbb{N} \times \mathbb{Z}$ as a bounding set. We start by showing functionality. Let arbitrary $a \in \mathrm{dom}(i)$. By definition there exists b such that $\langle a, b\rangle \in i$; as defined this implies $a \in \mathbb{N}$ and $b = [\langle a+1, 1\rangle]$. Let b' be arbitrary such that $\langle a, b'\rangle \in i$. Again by definition we immediately have $b' = [\langle a+1, 1\rangle] = b$.

Next the domain bound $\mathrm{dom}(i) = \mathbb{N}$. Let a be arbitrary and suppose that $a \in \mathrm{dom}(i)$. By definition there then exists b such that $\langle a, b\rangle \in i$ and it immediately follows that $a \in \mathbb{N}$. Now suppose $a \in \mathbb{N}$. By definition $[\langle a+1, 1\rangle] \in \mathbb{Z}$ and $\langle a, [\langle a+1, 1\rangle]\rangle \in i$. So $a \in \mathrm{dom}(i)$.

Next the range bound $\mathrm{ran}(i) \subseteq \mathbb{Z}$. Let arbitrary $b \in \mathrm{ran}(i)$. Then by definition there exists a such that $\langle a, b\rangle \in i$. It immediately follows that $a \in \mathbb{N}$ and $b = [\langle a+1, 1\rangle] \in \mathbb{Z}$.

Finally we show injectivity. Let arbitrary $b \in \mathrm{ran}(i)$. By definition there exists a such that $\langle a, b\rangle \in i$; it also follows that $a \in \mathbb{N}$ and $b = [\langle a+1, 1\rangle]$. Let a' be arbitrary and suppose $\langle a', b\rangle \in i$; similarly we have $a' \in \mathbb{N}$ and $b = [\langle a'+1, 1\rangle]$. So $[\langle a+1, 1\rangle] = [\langle a'+1, 1\rangle]$, which implies $\langle a+1, 1\rangle \sim \langle a'+1, 1\rangle$ and $(a+1)+1 = 1+(a'+1)$. By commutativity and cancellation we then have $a = a'$. □

(next page)

Proof. (2) We show $\{i(n) \mid n \in \mathbb{N}\} = \{x \in \mathbb{Z} \mid x > \hat{0}\}$. More specifically, denoting the former as A and the latter B, they are the sets such that

$$\forall x(x \in A \leftrightarrow \exists n \in \mathbb{N}(x = i(n)))$$
$$\forall x(x \in B \leftrightarrow x \in \mathbb{Z} \wedge x > \hat{0})$$

Existence of A can be shown using the subset axiom with $\mathbb{Z}$ as a bounding set. Existence of B is immediately assured by the subset axiom. We want to show $A = B$.

Let x be arbitrary and suppose $x \in A$. Then there exists $n \in \mathbb{N}$ such that $x = i(n)$. So $\langle n, x\rangle \in i$ and $x \in \mathrm{ran}(i) \subseteq \mathbb{Z}$. By definition $x = [\langle n+1, 1\rangle]$. Since we know $(1+1)+n = 1+(n+1)$ it follows that $1+1 < 1+(n+1)$ and by lemma 1.21 that $\hat{0} = [\langle 1,1\rangle] < [\langle n+1,1\rangle] = x$. So $x \in B$.

Now suppose $x \in B$. Then $x \in \mathbb{Z}$ and $x > \hat{0}$. By definition there exists $a, b \in \mathbb{N}$ such that $x = [\langle a, b\rangle] > \hat{0}$. By lemma 1.21 $1+b < 1+a$, so there exists $k \in \mathbb{N}$ such that $(1+b)+k = 1+a$; associativity and cancellation allows us to conclude that $b+k = a$ it is apparent that $[\langle k+1, 1\rangle] \in \mathbb{Z}$ and that by definition $\langle k, [\langle k+1, 1\rangle]\rangle \in i$. We know $(1+b)+k = 1+a$; so $(k+1)+b = 1+a$, $\langle k+1, 1\rangle \sim \langle a, b\rangle$, and $x = [\langle a, b\rangle] = [\langle k+1, 1\rangle] = i(k)$, meaning $x \in A$. □

Proof. (3) We defined i such that

$$\forall x(x \in i \leftrightarrow \exists a \in \mathbb{N} \exists b(x = \langle a, b\rangle \wedge b = [\langle a+1, 1\rangle]))$$

By definition we have $\langle 1, [\langle 1+1, 1\rangle]\rangle \in i$. So $i(1) = [\langle 1+1, 1\rangle] = \hat{1}$. □

Proof. (4a) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(i(a+b) = i(a) + i(b))$$

Let arbitrary $a, b \in \mathbb{N}$. See that $\langle((a+b)+1)+1, 1+1\rangle \sim \langle(a+b)+1, 1\rangle$ since

$$(((a+b)+1)+1)+1 = (1+1)+((a+b)+1)$$

We therefore have the following string of equivalences

$$\begin{aligned} i(a+b) &= [\langle(a+b)+1, 1\rangle] = [\langle((a+b)+1)+1, 1+1\rangle] \\ &= [\langle(a+1)+(b+1), 1+1\rangle] = [\langle a+1, 1\rangle] + [\langle b+1, 1\rangle] \\ &= i(a) + i(b) \end{aligned}$$

□

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Proof. (4b) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(i(ab) = i(a)i(b))$$

Letting arbitrary $a, b \in \mathbb{N}$, see that $\langle (a+1)(b+1)+1, (a+1)+(b+1) \rangle \sim \langle ab+1, 1 \rangle$:

$$\begin{aligned} ((a+1)(b+1)+1)+1 &= (((a+1)b+(a+1))+1)+1 \\ &= (((ab+b)+(a+1))+1)+1 \\ &= ((a+1)+(b+1))+(ab+1) \end{aligned}$$

We therefore have

$$i(ab) = [\langle ab+1, 1 \rangle] = [\langle (a+1)(b+1)+1, (a+1)+(b+1) \rangle] = i(a)i(b) \quad \square$$

Proof. (4c) We want to show

$$\forall a \in \mathbb{N} \forall b \in \mathbb{N}(a < b \leftrightarrow i(a) < i(b))$$

Let arbitrary $a, b \in \mathbb{N}$. First suppose $a < b$. Then by theorem 1.10 part 4 we know $a+1 < b+1$; applying the same theorem again we have $(a+1)+1 < (b+1)+1 = 1+(b+1)$. By lemma 1.21 this implies $[\langle a+1, 1 \rangle] < [\langle b+1, 1 \rangle]$ and $i(a) < i(b)$.

Now instead suppose $i(a) < i(b)$. This implies $[\langle a+1, 1 \rangle] < [\langle b+1, 1 \rangle]$. By lemma 1.21 we then have $(a+1)+1 < 1+(b+1) = (b+1)+1$. Applying theorem 1.10 part 4 twice we then have $a < b$. $\square$

1.2.6 More properties

(Denoting $\hat{0}$ as 0 and $\hat{1}$ as 1)

Theorem 1.28. *Let $x, y, z \in \mathbb{Z}$.*

1. *If $x + z = y + z$, then $x = y$ (Cancellation Law for Addition)*
2. $-(-x) = x$
3. $-(x + y) = (-x) + (-y)$
4. $x \cdot 0 = 0$
5. $(-x)y = -xy = x(-y)$
6. *If $z \neq 0$ and $xz = yz$, then $x = y$ (Cancellation Law for Multiplication)*
7. *$x > 0$ if and only if $-x < 0$ and $x < 0$ if and only if $-x > 0$*
8. $0 < 1$
9. *If $x \leq y$ and $y \leq x$. Then $x = y$*
10. *If $x > 0$ and $y > 0$, then $xy > 0$. If $x > 0$ and $y < 0$, then $xy < 0$.*

Proof. (1) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} \forall z \in \mathbb{Z}(x + z = y + z \to x = y)$$

Let arbitrary $x, y, z \in \mathbb{Z}$. Suppose that $x + z = y + z$. The desired result follows from several parts of theorem 1.24:

$$\begin{aligned} x &\overset{(3)}{=} x + 0 \overset{(4)}{=} x + (z + (-z)) \overset{(1)}{=} (x + z) + (-z) = (y + z) + (-z) \\ &\overset{(1)}{=} y + (z + (-z)) \overset{(4)}{=} y + 0 \overset{(3)}{=} y \end{aligned}$$

□

Proof. (2) We want to show

$$\forall x \in \mathbb{Z}(-(-x) = x)$$

Let arbitrary $x \in \mathbb{Z}$. Part 4 of theorem 1.24 allows us to state

$$(-x) + (-(-x)) = 0$$

We then have, by the same theorem,

$$\begin{aligned} x &\overset{(3)}{=} x + 0 = x + ((-x) + (-(-x))) \overset{(1)}{=} (x + (-x)) + (-(-x)) \\ &\overset{(4)}{=} 0 + (-(-x)) \overset{(2,3)}{=} (-(-x)) \end{aligned}$$

□

(next page)

Proof. (3) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}(-(x+y) = (-x) + (-y))$$

Let arbitrary $x, y \in \mathbb{Z}$. By part 4 of theorem 1.24 we have

$$(x+y) + (-(x+y)) = 0$$

We then have, by the same theorem,

$$\begin{aligned}
(-x) + (-y) &\overset{(3)}{=} ((-x) + (-y)) + 0 = ((-x) + (-y)) + ((x+y) + (-(x+y))) \\
&\overset{(1)}{=} (((-x) + (-y)) + (x+y)) + (-(x+y)) \\
&\overset{(1,2)}{=} ((-x) + (((-y) + y) + x)) + (-(x+y)) \\
&\overset{(4)}{=} ((-x) + (0+x)) + (-(x+y)) \\
&\overset{(2,3)}{=} ((-x) + x) + (-(x+y)) \\
&\overset{(2,4)}{=} 0 + (-(x+y)) \\
&\overset{(2,3)}{=} -(x+y)
\end{aligned}$$

□

Proof. (4) We want to show

$$\forall x \in \mathbb{Z}(x \cdot 0 = 0)$$

Let arbitrary $x \in \mathbb{Z}$. We have, by theorem 1.24,

$$x \cdot 0 + 0 \overset{(3)}{=} x \cdot 0 \overset{(3)}{=} x(0+0) \overset{(8)}{=} x \cdot 0 + x \cdot 0$$

By commutativity and cancellation we have $0 = x \cdot 0$. □

Proof. (5) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}((-x)y = -(xy) = x(-y))$$

Let arbitrary $x, y \in \mathbb{Z}$. We first show $(-x)y = -(xy)$. By part 4 of this theorem and several parts of theorem 1.24 we have

$$xy + (-(xy)) \overset{(4)}{=} 0 = y \cdot 0 \overset{(4)}{=} y(x + (-x)) \overset{(8)}{=} yx + y(-x) \overset{(6)}{=} xy + (-x)y$$

where cancellation on addition immediately yields $-(xy) = (-x)y$. Similarly we have

$$xy + (-(xy)) \overset{(4)}{=} 0 = x \cdot 0 \overset{(4)}{=} x(y + (-y)) \overset{(8)}{=} xy + x(-y)$$

where it follows that $-(xy) = x(-y)$. □

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Proof. (6) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} \forall z \in \mathbb{Z}((z \neq 0 \wedge xz = yz) \rightarrow x = y)$$

Let arbitrary $x, y, z \in \mathbb{Z}$. Suppose that $z \neq 0$ and $xz = yz$. By part 5 of this theorem and several parts of theorem 1.24 we have

$$0 \overset{(4)}{=} xz + (-(xz)) = yz + (-(xz)) = yz + (-x)z \overset{(6,8)}{=} (y + (-x))z$$

Part 9 of theorem 1.24 then allows us to conclude $y + (-x) = 0$ or $z = 0$. By our earlier assumptions we must then have $y + (-x) = 0$. So by part 4 of theorem 1.24,

$$y + (-x) = 0 = x + (-x)$$

and by cancellation on addition $y = x$. □

Proof. (7) We want to show

$$\forall x \in \mathbb{Z}((x > 0 \leftrightarrow -x < 0) \wedge (x < 0 \leftrightarrow -x > 0))$$

Let arbitrary $x \in \mathbb{Z}$. First suppose $x > 0$. It follows from theorem 1.24 that

$$-x \overset{(3,2)}{=} 0 + (-x) \overset{(12)}{<} x + (-x) \overset{(4)}{=} 0$$

Next suppose $-x < 0$. Similarly,

$$0 \overset{(4)}{=} x + (-x) \overset{(2,12)}{<} x + 0 \overset{(3)}{=} x$$

Next suppose $x < 0$. We have

$$-x \overset{(3,2)}{=} 0 + (-x) \overset{(12)}{>} x + (-x) \overset{(4)}{=} 0$$

Finally suppose $-x > 0$. We have

$$0 \overset{(4)}{=} x + (-x) \overset{(2,12)}{>} x + 0 \overset{(3)}{=} x$$ □

(next page)

Proof. (8) We want to show $0 < 1$. We know by theorem 1.24 part 14 that $0 \neq 1$. So by part 10 (trichotomy) of that same theorem we either have $0 < 1$ or $0 > 1$. Suppose for contradiction that the latter is true. By part 7 of this theorem we immediately have $0 < -1$. By part 13 of theorem 1.24 we then have $0(-1) < (-1)(-1)$. By an earlier part of this theorem and theorem 1.24,

$$0 = 0(-1) < (-1)(-1) = 1(-(-1)) = 1 \cdot 1 = 1$$

Which contradicts trichotomy since we assumed $1 < 0$. We therefore must have $0 < 1$. □

Proof. (9) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}((x \leq y \wedge y \leq x) \rightarrow x = y)$$

Let arbitrary $x, y \in \mathbb{Z}$. Suppose that $x \leq y$ and $y \leq x$. Lemma 1.23 implies that $x < y \vee x = y$ and $y < x \vee y = x$. Supposing $x < y$, $y < x \vee y = x$ is a contradiction in either case. Therefore $x < y \vee x = y$ implies that it must be the case that $x = y$. □

Proof. (10) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}(([x > 0 \wedge y > 0] \rightarrow xy > 0) \wedge ([x > 0 \wedge y < 0] \rightarrow xy < 0))$$

Let $x, y \in \mathbb{Z}$ be arbitrary. First suppose $x > 0$ and $y > 0$. By theorem 1.24 part 13 we immediately have

$$xy > 0 \cdot y = 0$$

Where the equality follows from commutativity and part 4 of this theorem. Next suppose $x > 0$ and $y < 0$. Similarly we have

$$xy = yx < 0 \cdot x = 0$$ □

1.2.7 Discreteness

Theorem 1.29. *Let $x \in \mathbb{Z}$. Then there is no $y \in \mathbb{Z}$ such that $x < y < x + 1$.*

Proof. We want to show

$$\forall x \in \mathbb{Z}(\neg\exists y \in \mathbb{Z}(x < y < x + 1))$$

Let arbitrary $x \in \mathbb{Z}$. Suppose for contradiction that there exists some $y \in \mathbb{Z}$ such that $x < y < x + 1$. By theorem 1.24 we have

$$\begin{aligned} 1 = x + ((-x) + 1) < y + ((-x) + 1) &< (x + 1) + ((-x) + 1) \\ &= (1 + (x + (-x))) + 1 = 1 + 1 \end{aligned}$$

By theorem 1.28 part 8 and transitivity we have $0 < y + ((-x) + 1)$. It then follows from theorem 1.27 that there exists $n \in \mathbb{N}$ such that $i(n) = y+((-x)+1)$, where $i : \mathbb{N} \to \mathbb{Z}$. By the same theorem we have

$$i(1) = 1 < i(n) < 1 + 1 = i(1) + i(1) = i(1 + 1)$$

and

$$1 < n < 1 + 1$$

Which contradicts part 9 of theorem 1.10. □

Lemma 1.30. $\forall x \in \mathbb{Z}\forall y \in \mathbb{Z}(x > y \to x \geq y + 1)$

Proof. Let $x, y \in \mathbb{Z}$ be arbitrary. Suppose $x > y$. Suppose for contradiction that $x < y + 1$. This contradicts theorem 1.29 since $y < x < y + 1$. By trichotomy we must therefore $x \geq y + 1$. □

(next page)

Theorem 1.31. *Let $x, y \in \mathbb{Z}$. $xy = 1$ iff $x = 1 = y$ or $x = -1 = y$.*

Proof. We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}(xy = 1 \leftrightarrow (x = 1 = y \vee x = -1 = y))$$

Let arbitrary $x, y \in \mathbb{Z}$. Suppose that $xy = 1$. It follows from theorem 1.24 that $xy \neq 0$. By contradiction it can then be shown that $x \neq 0$ and $y \neq 0$. By trichotomy we must have either $x > 0$ or $x < 0$.

First consider the case where $x > 0$. Since $xy = 1 > 0$, by trichotomy on y it can be shown by contradiction that $y > 0$; and by the previous lemma $y \geq 1$. Suppose $x \neq 1$; by trichotomy we have either $x < 1$ or $x > 1$. The first case implies $0 < x < 1$, which contradicts theorem 1.29. In the second case by theorem 1.24 we would have $xy > y \geq 1$, which implies $xy > 1$, contradicting trichotomy. We must therefore have $x = 1$. It immediately follows that $1 = xy = 1 \cdot y = y$.

Next consider the case where $x < 0$. By trichotomy $y < 0 \vee y = 0 \vee y > 0$. The latter two cases are contradictions since $y = 0$ would imply $xy = 0 \neq 1$ and $y > 0$ would, by theorem 1.28, imply $xy < 0 < 1$. We therefore conclude that $y < 0$. Again by theorem 1.28, we have $-x > 0$ and $-y > 0$ and $(-x)(-y) = xy = 1$. Using an identical strategy to the previous case, it follows that $(-x) = 1 = (-y)$ and $x = -1 = y$.

Now for the reverse implication. Supposing $x = 1 = y$ it immediately follows that $xy = 1 \cdot 1 = 1$. Supposing $x = -1 = y$ we have $xy = (-1)(-1) = 1 \cdot 1 = 1$. □

1.2.8 More proofs

Theorem 1.32.

$$\forall a \in \mathbb{Z} \forall b \in \mathbb{Z}(ab > 0 \to ((a < 0 \wedge b < 0) \vee (a > 0 \wedge b > 0)))$$

Proof. Let arbitrary $a, b \in \mathbb{Z}$ such that $ab > 0$. We have three cases.

Case 1: $a = 0$. Then $ab = 0 \cdot b = 0$, contradicting $ab > 0$.

Case 2: $a < 0$. First suppose $b = 0$. Then $ab = a \cdot 0 = 0$, which is a contradiction. Next suppose $b > 0$. Then $a < 0$ implies $ab < 0 \cdot b = 0$, which is again a contradiction. By trichotomy we must then have $b < 0$.

Case 3: $a > 0$. First suppose $b = 0$. Then $ab = a \cdot 0 = 0$, a contradiction. Next suppose $b < 0$. This implies $ab < a \cdot 0 = 0$, which is a contradiction. By trichotomy we must then have $b > 0$. □

Theorem 1.33.

$$\forall a \in \mathbb{Z} \forall b \in \mathbb{Z}(ab < 0 \to ((a < 0 \wedge b > 0) \vee (a > 0 \wedge b < 0)))$$

Proof. Let arbitrary $a, b \in \mathbb{Z}$ such that $ab < 0$. We have three cases.

Case 1: $a = 0$. Then $ab = 0 \cdot b = 0$, a contradiction.

Case 2: $a < 0$. First suppose $b = 0$. Then $ab = a \cdot 0 = 0$, a contradiction. Next suppose $b < 0$. We know $(-a) > 0$ and $(-b) > 0$. So $ab = (-a)(-b) > 0$, a contradiction. It follows then that $b > 0$.

Case 3: $a > 0$. First suppose $b = 0$. Then $ab = 0$, a contradiction. Next suppose $b > 0$; then $ab > 0$, another contradiction. We must then have $b < 0$. □

Theorem 1.34.

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} \forall z \in \mathbb{Z}((x < y \wedge z < 0) \to xz > yz)$$

Proof. Let arbitrary $x, y, z \in \mathbb{Z}$ such that $x < y$ and $z < 0$. We know $-z > 0$. So $x(-z) < y(-z)$ and $-xz < -yz$. We then have

$$yz = -xz + (xz + yz) < -yz + (xz + yz) = xz$$ □

Theorem 1.35.

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} \forall z \in \mathbb{Z}((xz < yz \wedge z > 0) \to x < y)$$

Proof. Let arbitrary $x, y, z \in \mathbb{Z}$ such that $xz < yz \wedge z > 0$. Suppose for contradiction that $x \not< y$. By trichotomy we then have two cases:

Case 1: $x = y$. Which implies $xz = yz$, a contradiction.

Case 2: $x > y$. Since $z > 0$ we then have $xz > yz$, a contradiction. □

(next page)

Theorem 1.36.

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z} \forall z \in \mathbb{Z}((xz < yz \wedge z < 0) \rightarrow x > y)$$

Proof. Let arbitrary $x, y, z \in \mathbb{Z}$ such that $xz < yz$ and $z < 0$. Suppose for contradiction that $y \not< x$. We then have two cases:

Case 1: $x = y$. Which implies $xz = yz$, a contradiction.

Case 2: $x < y$. It follows from theorem 1.34 that $xz > yz$, another contradiction. □

Theorem 1.37.

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}(x < 0 \wedge y < 0 \rightarrow xy > 0)$$

Proof. Let arbitrary $x, y \in \mathbb{Z}$. Suppose $x < 0$ and $y < 0$. It follows that $-x > 0$ and $-y > 0$. So $xy = (-x)(-y) > 0$. □

1.2.9 Well-ordering on the positive integers

Theorem 1.38. *Let $G \subseteq \{x \in \mathbb{Z} \mid x > 0\}$ be a non-empty set. Then there is some $m \in G$ such that $m \leq g$ for all $g \in G$. That is*

$$\forall G((G \subseteq \{x \in \mathbb{Z} \mid x > 0\} \wedge G \neq \emptyset) \rightarrow \exists m \in G \forall g \in G(m \leq g))$$

Proof. Let G be arbitrary and suppose that $G \subseteq \{x \in \mathbb{Z} \mid x > 0\}$ and $G \neq \emptyset$. By theorem 1.27, the function $i : \mathbb{N} \rightarrow \mathbb{Z}$ is defined as $i(n) = [\langle n + 1, 1 \rangle]$ for all $n \in \mathbb{N}$. We can define G' as the set such that

$$\forall x(x \in G' \leftrightarrow x \in \mathbb{N} \wedge i(x) \in G)$$

It is straightforward to show that $G' \subseteq \mathbb{N}$. We need to show that $G' \neq \emptyset$. By theorem 1.27 we know $\{x \in \mathbb{Z} \mid x > 0\} = \{i(n) \mid n \in \mathbb{N}\}$. Since $G \neq \emptyset$ there exist $x \in G \subseteq \{x \in \mathbb{Z} \mid x > 0\} = \{i(n) \mid n \in \mathbb{N}\}$. So there exists $n \in \mathbb{N}$ such that $i(n) = x \in G$, and $n \in G'$.

We know $G' \subseteq \mathbb{N}$ and $G' \neq \emptyset$. So by well ordering on the naturals(theorem 1.11), there exists $m' \in G'$ such that $\forall g \in G'(m' \leq g)$. By definition we have $i(m') \in G$. Let $g \in G$ be arbitrary. Similar to before there exists $g' \in \mathbb{N}$ such that $i(g') = g$, so $g' \in G'$. It follows that $m' \leq g'$, and by theorem 1.27 that $i(m') \leq i(g') = g$. □

1.3 Construction of the Rationals

1.3.1 Construction

Definition 1.39. *Let $\mathbb{Z}^* = \mathbb{Z} - \{0\}$. The relation $\asymp$ on $\mathbb{Z} \times \mathbb{Z}^*$ is defined by $(x, y) \asymp (z, w)$ if and only if $xw = yz$, for all $(x, y), (z, w) \in \mathbb{Z} \times \mathbb{Z}^*$. That is, the set such that*

$$\forall a(a \in \asymp \leftrightarrow \exists \langle x, y\rangle \in \mathbb{Z} \times \mathbb{Z}^* \exists \langle z, w\rangle \in \mathbb{Z} \times \mathbb{Z}^* (a = \langle \langle x, y\rangle, \langle z, w\rangle\rangle \wedge xw = yz))$$

(Existence follows from using $(\mathbb{Z} \times \mathbb{Z}^*) \times (\mathbb{Z} \times \mathbb{Z}^*)$ as a bounding set.)

Lemma 1.40. *The relation $\asymp$ is an equivalence relation on $\mathbb{Z} \times \mathbb{Z}^*$.*

Proof. Showing $\asymp \subseteq (\mathbb{Z} \times \mathbb{Z}^*) \times (\mathbb{Z} \times \mathbb{Z}^*)$ is straightforward. We start with reflexivity on $\mathbb{Z} \times \mathbb{Z}^*$. Let arbitrary $x \in \mathbb{Z} \times \mathbb{Z}^*$; there exists $\langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = \langle a, b\rangle$. Since $\mathbb{Z}^* \subseteq \mathbb{Z}$, $a, b \in \mathbb{Z}$, so $ab = ba$. It then follows by definition that $\langle x, x\rangle = \langle \langle a, b\rangle, \langle a, b\rangle\rangle \in \asymp$.

Now for symmetry. Let a, b be arbitrary such that $\langle a, b\rangle \in \asymp$. By definition there then exists $\langle x, y\rangle, \langle z, w\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $a = \langle x, y\rangle$, $b = \langle z, w\rangle$, and $xw = yz$. By commutativity we have $zy = wx$, meaning $\langle b, a\rangle = \langle \langle z, w\rangle, \langle x, y\rangle\rangle \in \asymp$.

Finally transitivity. Let a, b, c be arbitrary such that $\langle a, b\rangle, \langle b, c\rangle \in \asymp$. By definition there exists $\langle x, y\rangle, \langle z, w\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ where $a = \langle x, y\rangle$, $b = \langle z, w\rangle$, and $xw = yz$. There also exists $\langle x', y'\rangle, \langle z', w'\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ where $b = \langle x', y'\rangle$, $c = \langle z', w'\rangle$, and $x'w' = y'z'$. It follows that $\langle z, w\rangle = \langle x', y'\rangle$ and $zw' = wz'$. We then have $(xw)w' = (yz)w'$ and $y(zw') = y(wz')$. By associativity we then have $(xw)w' = (yz)w' = y(zw') = y(wz')$. By commutativity and associativity it follows that $(xw')w = (yz')w$. Since $w \in \mathbb{Z}^*$, $w \neq 0$ and by cancellation we have $xw' = yz'$. By definition $\langle a, c\rangle = \langle \langle x, y\rangle, \langle z', w'\rangle\rangle \in \asymp$. □

Definition 1.41. *The set of **rational numbers**, denoted $\mathbb{Q}$, is the quotient set $(\mathbb{Z} \times \mathbb{Z}^*)/\asymp$. That is,*

$$\forall x(x \in \mathbb{Q} \leftrightarrow \exists \langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^* (x = [\langle a, b\rangle]))$$

where

$$\forall \langle a, b\rangle \forall x(x \in [\langle a, b\rangle] \leftrightarrow \langle a, b\rangle \asymp x)$$

Existence of each equivalence class can be shown using $\mathrm{ran}(\asymp)$ as a bounding set. The same can be done for $\mathbb{Q}$ using $\mathscr{P}(\mathrm{ran}(\asymp))$.

Definition 1.42. *The elements $\bar{0}, \bar{1} \in \mathbb{Q}$ are defined by $\bar{0} = [\langle 0, 1\rangle]$ and $\bar{1} = [\langle 1, 1\rangle]$.*

1.3.2 Operations

Definition 1.43. *The binary operation $+$ on $\mathbb{Q}$ is defined as $+ : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$ where*

$$\forall x(x \in + \leftrightarrow \exists \langle a,b\rangle \in \mathbb{Z} \times \mathbb{Z}^* \exists \langle c,d\rangle \in \mathbb{Z} \times \mathbb{Z}^* (x = \langle\langle [\langle a,b\rangle], [\langle c,d\rangle]\rangle, [\langle ad+bc, bd\rangle]\rangle))$$

(Existence follows from using $(\mathbb{Q} \times \mathbb{Q}) \times \mathbb{Q}$ as a bounding set.)

Proof. (Domain/Range bounds, Well-definedness, Functionality) Staring with the domain bound, $\text{dom}(+) = \mathbb{Q} \times \mathbb{Q}$. First let arbitrary $x \in \text{dom}(+)$. By definition there exists y such that $\langle x, y\rangle \in +$. By definition there then exists $\langle a,b\rangle, \langle c,d\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = \langle [\langle a,b\rangle], [\langle c,d\rangle]\rangle \in \mathbb{Q} \times \mathbb{Q}$. Now let arbitrary $\langle x, y\rangle \in \mathbb{Q} \times \mathbb{Q}$. By definition there then exists $\langle a,b\rangle, \langle c,d\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a,b\rangle]$ and $y = [\langle c,d\rangle]$. We then have $\langle\langle [\langle a,b\rangle], [\langle c,d\rangle]\rangle, [\langle ad+bc, bd\rangle]\rangle \in +$. So $\langle x, y\rangle = \langle [\langle a,b\rangle], [\langle c,d\rangle]\rangle \in \text{dom}(+)$.

Next the range bound $\text{ran}(+) \subseteq \mathbb{Q}$. Let arbitrary $y \in \text{ran}(+)$. Then there exists x such that $\langle x, y\rangle \in +$. There then exists $\langle a,b\rangle, \langle c,d\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $y = [\langle ad+bc, bd\rangle]$. We know that $b, d \in \mathbb{Z}^*$. By theorem 1.24 since $b \neq 0$ and $d \neq 0$ we must have $bd \neq 0$. So $bd \in \mathbb{Z} - \{0\} = \mathbb{Z}^*$, and $\langle ad+bc, bd\rangle \in \mathbb{Z} \times \mathbb{Z}^*$. So $y = [\langle ad+bc, bd\rangle] \in \mathbb{Q}$.

Next well-definedness. That is,

$$\begin{aligned}\forall \langle a,b\rangle \in \mathbb{Z} \times \mathbb{Z}^* \forall \langle c,d\rangle \in \mathbb{Z} \times \mathbb{Z}^* \forall \langle a',b'\rangle \in \mathbb{Z} \times \mathbb{Z}^* \forall \langle c',d'\rangle \in \mathbb{Z} \times \mathbb{Z}^* (& \\ ([\langle a,b\rangle] = [\langle a',b'\rangle] \wedge [\langle c,d\rangle] = [\langle c',d'\rangle]) & \\ \to [\langle ad+bc, bd\rangle] = [\langle a'd'+b'c', b'd'\rangle]) &\end{aligned}$$

Let arbitrary $\langle a,b\rangle, \langle c,d\rangle, \langle a',b'\rangle, \langle c',d'\rangle \in \mathbb{Z} \times \mathbb{Z}^*$. Suppose $[\langle a,b\rangle] = [\langle a',b'\rangle]$ and $[\langle c,d\rangle] = [\langle c',d'\rangle]$. We then have $\langle a,b\rangle \asymp \langle a',b'\rangle$ and $\langle c,d\rangle \asymp \langle c',d'\rangle$; so $ab' = ba'$ and $cd' = dc'$. We know $bd, b'd' \in \mathbb{Z}^*$. We then have

$$\begin{aligned}(ad+bc)(b'd') &= (ad)(b'd') + (bc)(b'd') \\ &= (ab')(dd') + (cd')(bb') \\ &= (ba')(dd') + (dc')(bb') \\ &= (bd)(a'd') + (bd)(b'c') \\ &= (bd)(a'd'+b'c')\end{aligned}$$

Which implies $\langle ad+bc, bd\rangle \asymp \langle a'd'+b'c', b'd'\rangle$ and $[\langle ad+bc, bd\rangle] = [\langle a'd'+b'c', b'd'\rangle]$.
(next page)

Finally functionality. Let arbitrary $x \in \mathrm{dom}(+)$. There then exists y such that $\langle x, y\rangle \in +$. By definition there then exists $\langle a, b\rangle, \langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = \langle [\langle a, b\rangle], [\langle c, d\rangle]\rangle$ and $y = [\langle ad + bc, bd\rangle]$. Now let y' be arbitrary such that $\langle x, y'\rangle \in +$. Then there exists $\langle a', b'\rangle, \langle c', d'\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = \langle [\langle a', b'\rangle], [\langle c', d'\rangle]\rangle$ and $y' = [\langle a'd' + b'c', b'd'\rangle]$. Since this implies $[\langle a, b\rangle] = [\langle a', b'\rangle]$ and $[\langle c, d\rangle] = [\langle c', d'\rangle]$, by well definedness we have $y = [\langle ad + bc, bd\rangle] = [\langle a'd' + b'c', b'd'\rangle] = y'$. □

Definition 1.44. *The binary operation $\cdot$ on $\mathbb{Q}$ is defined as $\cdot : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$ where*

$$\forall x(x \in \cdot \leftrightarrow \exists \langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^* \exists \langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^* (x = \langle \langle [\langle a, b\rangle], [\langle c, d\rangle]\rangle, [\langle ac, bd\rangle]\rangle))$$

(Existence follows from using $(\mathbb{Q} \times \mathbb{Q}) \times \mathbb{Q}$ as a bounding set.)

Proof. (Domain/Range bounds, Well-definedness, Functionality) First the domain bound $\mathrm{dom}(\cdot) = \mathbb{Q} \times \mathbb{Q}$. First let arbitrary $x \in \mathrm{dom}(\cdot)$. Then there exists y such that $\langle x, y\rangle \in \cdot$, and $\langle a, b\rangle, \langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $\langle x, y\rangle = \langle [\langle a, b\rangle], [\langle c, d\rangle]\rangle \in \mathbb{Q} \times \mathbb{Q}$. Next let arbitrary $\langle x, y\rangle \in \mathbb{Q} \times \mathbb{Q}$. There then exists $\langle a, b\rangle, \langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b\rangle]$ and $y = [\langle c, d\rangle]$. By definition $\langle \langle [\langle a, b\rangle], [\langle c, d\rangle]\rangle, [\langle ac, bd\rangle]\rangle \in \cdot$; so $\langle x, y\rangle = \langle [\langle a, b\rangle], [\langle c, d\rangle]\rangle \in \mathrm{dom}(\cdot)$.

Next the range bound $\mathrm{ran}(\cdot) \subseteq \mathbb{Q}$. Let arbitrary $y \in \mathrm{ran}(\cdot)$. There then exists x such that $\langle x, y\rangle \in \cdot$. By definition there then exists $\langle a, b\rangle, \langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $y = [\langle ac, bd\rangle]$. Since $b, d \in \mathbb{Z}^*$, we have $b \neq 0$ and $d \neq 0$. By theorem 1.24 we then have $bd \neq 0$. So $\langle ac, bd\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ and therefore $y = [\langle ac, bd\rangle] \in \mathbb{Q}$.

Now for well-definedness. That is

$$\forall \langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^* \forall \langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^* \forall \langle a', b'\rangle \in \mathbb{Z} \times \mathbb{Z}^* \forall \langle c', d'\rangle \in \mathbb{Z} \times \mathbb{Z}^* ($$
$$([\langle a, b\rangle] = [\langle a', b'\rangle] \wedge [\langle c, d\rangle] = [\langle c', d'\rangle]) \to [\langle ac, bd\rangle] = [\langle a'c', b'd'\rangle])$$

Let arbitrary $\langle a, b\rangle, \langle c, d\rangle, \langle a', b'\rangle, \langle c', d'\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $[\langle a, b\rangle] = [\langle a', b'\rangle]$ and $[\langle c, d\rangle] = [\langle c', d'\rangle]$. We then have $ab' = ba'$ and $cd' = dc'$. We know $bd, b'd' \in \mathbb{Z}^*$ and

$$(ac)(b'd') = (ab')(cd') = (ba')(dc') = (bd)(a'c')$$

so $\langle ac, bd\rangle \asymp \langle a'c', b'd'\rangle$ and $[\langle ac, bd\rangle] = [\langle a'c', b'd'\rangle]$.

Finally functionality. Let arbitrary $x \in \mathrm{dom}(\cdot)$. There then exists y such that $\langle x, y\rangle \in \cdot$. There exists $\langle a, b\rangle, \langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $\langle x, y\rangle = \langle \langle [\langle a, b\rangle], [\langle c, d\rangle]\rangle, [\langle ac, bd\rangle]\rangle$. Let y' be arbitrary such that $\langle x, y'\rangle \in \cdot$; there exists $\langle a', b'\rangle, \langle c', d'\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $\langle x, y'\rangle = \langle \langle [\langle a', b'\rangle], [\langle c', d'\rangle]\rangle, [\langle a'c', b'd'\rangle]\rangle$. It follows that $\langle [\langle a, b\rangle], [\langle c, d\rangle]\rangle = \langle [\langle a', b'\rangle], [\langle c', d'\rangle]\rangle$, so by well-definedness we have

$$y = [\langle ac, bd\rangle] = [\langle a'c', b'd'\rangle] = y'$$ □

(next page)

Definition 1.45. *The unary operation $-$ on $\mathbb{Q}$ is defined as $- : \mathbb{Q} \to \mathbb{Q}$ where*

$$\forall x(x \in - \leftrightarrow \exists \langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^*(x = \langle [\langle a, b\rangle], [\langle -a, b\rangle]\rangle))$$

(Existence follows from using $\mathbb{Q} \times \mathbb{Q}$ as a bounding set.)

Proof. (Domain/Range bounds, Well-definedness, Functionality) First the domain bound $\mathrm{dom}(-) = \mathbb{Q}$. First let arbitrary $x \in \mathrm{dom}(-)$. Then there exists y such that $\langle x, y\rangle \in -$. By definition then there exists $\langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b\rangle] \in \mathbb{Q}$. Next let arbitrary $x \in \mathbb{Q}$. Then there exists $\langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b\rangle]$. By definition $\langle [\langle a, b\rangle], [\langle -a, b\rangle]\rangle \in -$ so $x = [\langle a, b\rangle] \in \mathrm{dom}(-)$.

Next the range bound $\mathrm{ran}(-) \subseteq \mathbb{Q}$. Letting arbitrary $y \in \mathrm{ran}(-)$, it follows that there exists x such that $\langle x, y\rangle \in -$ and therefore $\langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $y = [\langle -a, b\rangle]$. Since $\langle -a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^*$, $y \in \mathbb{Q}$.

Now for well-definedness, that is

$$\forall \langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^* \forall \langle a', b'\rangle \in \mathbb{Z} \times \mathbb{Z}^*([\langle a, b\rangle] = [\langle a', b'\rangle] \to [\langle -a, b\rangle] = [\langle -a', b'\rangle])$$

Let arbitrary $\langle a, b\rangle, \langle a', b'\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $[\langle a, b\rangle] = [\langle a', b'\rangle]$. Then by definition $ab' = ba'$ and

$$(-a)b' = -(ab') = -(ba') = b(-a')$$

so $\langle -a, b\rangle \asymp \langle -a', b'\rangle$ and $[\langle -a, b\rangle] = [\langle -a', b'\rangle]$.

Finally functionality. Let arbitrary $x \in \mathrm{dom}(-)$. Then there exists y such that $\langle x, y\rangle \in -$ and $\langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $\langle x, y\rangle = \langle [\langle a, b\rangle], [\langle -a, b\rangle]\rangle$. Let y' be arbitrary such that $\langle x, y'\rangle \in -$. Similarly there then exists $\langle a', b'\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $\langle x, y'\rangle = \langle [\langle a', b'\rangle], [\langle -a', b'\rangle]\rangle$. It follows that $[\langle a, b\rangle] = [\langle a', b'\rangle]$ and by well-definedness $y = [\langle -a, b\rangle] = [\langle -a', b'\rangle] = y'$. □

Definition 1.46. *The unary operation $^{-1}$ on $\mathbb{Q}^*$ is defined as $^{-1} : \mathbb{Q}^* \to \mathbb{Q}^*$ where*

$$\forall x(x \in^{-1} \leftrightarrow \exists \langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^*(x = \langle [\langle a, b\rangle], [\langle b, a\rangle]\rangle \wedge [\langle a, b\rangle] \neq \bar{0}))$$

(Existence follows from using $\mathbb{Q}^* \times \mathbb{Q}^*$ as a bounding set.)

Proof. (Domain/Range bounds, Well-definedness, Functionality) First the domain bound $\mathrm{dom}(^{-1}) = \mathbb{Q}^*$. Let arbitrary $x \in \mathrm{dom}(^{-1})$. There then exists y such that $\langle x, y\rangle \in^{-1}$. By definition there exists $\langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b\rangle] \in \mathbb{Q}$ and $[\langle a, b\rangle] \neq \bar{0}$. It immediately follows that $x \in \mathbb{Q} - \{\bar{0}\} = \mathbb{Q}^*$. Next let arbitrary $x \in \mathbb{Q}^*$. By definition we then have $x \neq \bar{0}$. By definition there exists $\langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b\rangle]$. So $\langle [\langle a, b\rangle], [\langle b, a\rangle]\rangle \in^{-1}$ and $x \in \mathrm{dom}(^{-1})$.

(next page)

We show

$$\forall\langle a,b\rangle \in \mathbb{Z}\times\mathbb{Z}^*([\langle a,b\rangle] \neq \bar{0} \leftrightarrow a \neq 0)$$

Let arbitrary $\langle a,b\rangle \in \mathbb{Z}\times\mathbb{Z}^*$ such that $a = 0$. It follows that $a\cdot 1 = 0 = b\cdot 0$. So $[\langle a,b\rangle] = [\langle 0,1\rangle] = \bar{0}$. Next suppose $[\langle a,b\rangle] = \bar{0}$. It then follows that $a\cdot 1 = b\cdot 0 = 0$ and $a = 0$.

Next the range bound $\mathrm{ran}(^{-1}) \subseteq \mathbb{Q}^*$. Let arbitrary $y \in \mathrm{ran}(^{-1})$. Then there exists x such that $\langle x,y\rangle \in^{-1}$ and $\langle a,b\rangle \in \mathbb{Z}\times\mathbb{Z}^*$ such that $y = [\langle b,a\rangle]$ and $[\langle a,b\rangle] \neq \bar{0}$. It follows then that $a \neq 0$ and $\langle b,a\rangle \in \mathbb{Z}\times\mathbb{Z}^*$. So $y = [\langle b,a\rangle] \in \mathbb{Q}$. Since $b \neq 0$ it also follows that $[\langle b,a\rangle] \neq \bar{0}$ and $y \in \mathbb{Q} - \{\bar{0}\} = \mathbb{Q}^*$.

Now for well-definedness, that is

$$\forall\langle a,b\rangle \in \mathbb{Z}\times\mathbb{Z}^*\forall\langle a',b'\rangle \in \mathbb{Z}\times\mathbb{Z}^*([\langle a,b\rangle] = [\langle a',b'\rangle] \rightarrow [\langle b,a\rangle] = [\langle b',a'\rangle])$$

Let arbitrary $\langle a,b\rangle, \langle a',b'\rangle \in \mathbb{Z}\times\mathbb{Z}^*$ and suppose that $[\langle a,b\rangle] = [\langle a',b'\rangle]$. It follows that $ab' = ba'$, which, by commutativity on integer multiplication, implies $b'a = a'b$; so $\langle b',a'\rangle \asymp \langle b,a\rangle$ and $[\langle b,a\rangle] = [\langle b',a'\rangle]$.

Finally functionality. Let arbitrary $x \in \mathrm{dom}(^{-1})$. There then exists y such that $\langle x,y\rangle \in^{-1}$, and $\langle a,b\rangle \in \mathbb{Z}\times\mathbb{Z}^*$ such that $\langle x,y\rangle = \langle[\langle a,b\rangle],[\langle b,a\rangle]\rangle$. Let y' be arbitrary such that $\langle x,y'\rangle \in^{-1}$. Then there exists $\langle a',b'\rangle \in \mathbb{Z}\times\mathbb{Z}^*$ such that $\langle x,y'\rangle = \langle[\langle a',b'\rangle],[\langle b',a'\rangle]\rangle$. So $[\langle a,b\rangle] = [\langle a',b'\rangle]$. It follows from well-definedness that $y = [\langle b,a\rangle] = [\langle b',a'\rangle] = y'$. □

Definition 1.47. *The binary relation $<$ on $\mathbb{Q}$ is defined such that*

$$\begin{aligned}\forall x(x \in< \leftrightarrow \exists\langle a,b\rangle \in \mathbb{Z}\times\mathbb{Z}^*\exists\langle c,d\rangle \in \mathbb{Z}\times\mathbb{Z}^*(x = \langle[\langle a,b\rangle],[\langle c,d\rangle]\rangle\\ \wedge((ad < bc \wedge ((d > 0 \wedge b > 0) \vee (d < 0 \wedge b < 0)))\\ \vee (bc < ad \wedge ((d > 0 \wedge b < 0) \vee (d < 0 \wedge b > 0))))))\end{aligned}$$

(Existence follows from using $\mathbb{Q}\times\mathbb{Q}$ as a bounding set.)

Lemma 1.48.

$$\begin{aligned}\forall\langle a,b\rangle \in \mathbb{Z}\times\mathbb{Z}^*\forall\langle c,d\rangle \in \mathbb{Z}\times\mathbb{Z}^*\forall\langle a',b'\rangle \in \mathbb{Z}\times\mathbb{Z}^*\forall\langle c',d'\rangle \in \mathbb{Z}\times\mathbb{Z}^*(\\ ([\langle a,b\rangle] = [\langle a',b'\rangle] \wedge [\langle c,d\rangle] = [\langle c',d'\rangle]) \rightarrow (\\ ((ad < bc \wedge ((d > 0 \wedge b > 0) \vee (d < 0 \wedge b < 0)))\\ \vee (bc < ad \wedge ((d > 0 \wedge b < 0) \vee (d < 0 \wedge b > 0)))) \leftrightarrow\\ ((a'd' < b'c' \wedge ((d' > 0 \wedge b' > 0) \vee (d' < 0 \wedge b' < 0)))\\ \vee (b'c' < a'd' \wedge ((d' > 0 \wedge b' < 0) \vee (d' < 0 \wedge b' > 0))))))\end{aligned}$$

Proof. Let arbitrary $\langle a,b\rangle, \langle c,d\rangle, \langle a',b'\rangle, \langle c',d'\rangle \in \mathbb{Z}\times\mathbb{Z}^*$. Suppose $[\langle a,b\rangle] = [\langle a',b'\rangle]$ and $[\langle c,d\rangle] = [\langle c',d'\rangle]$. It follows that $ab' = ba'$ and $cd' = dc'$; so

$$\begin{aligned}(ad)(b'd') = (ba')(dd') = (bd)(a'd')\\ (bc)(b'd') = (dc')(bb') = (bd)(b'c')\end{aligned}$$

(next page)

($\rightarrow$) First suppose

$$(ad < bc \wedge ((d > 0 \wedge b > 0) \vee (d < 0 \wedge b < 0))) \vee \\ (bc < ad \wedge ((d > 0 \wedge b < 0) \vee (d < 0 \wedge b > 0)))$$

Where we are trying to show

$$(a'd' < b'c' \wedge ((d' > 0 \wedge b' > 0) \vee (d' < 0 \wedge b' < 0))) \vee \\ (b'c' < a'd' \wedge ((d' > 0 \wedge b' < 0) \vee (d' < 0 \wedge b' > 0)))$$

We know $b, d, b', d' \in \mathbb{Z}^*$. It then follows easily by contradiction that $bd \neq 0$ and $b'd' \neq 0$. By trichotomy we have four cases:

Case 1: $bd > 0 \wedge b'd' > 0$. It follows from theorem 1.32 that $b < 0 \wedge d < 0$ or $b > 0 \wedge d > 0$. See that either possibility would contradict $(d > 0 \wedge b < 0) \vee (d < 0 \wedge b > 0)$. So by our assumption we must have $ad < bc$. Since $b'd' > 0$, we have

$$(ad)(b'd') < (bc)(b'd')$$

and by our earlier assumption that $[\langle a, b\rangle] = [\langle a', b'\rangle]$ and $[\langle c, d\rangle] = [\langle c', d'\rangle]$, we have

$$(bd)(a'd') < (bd)(b'c')$$

Since $bd > 0$, it follows by theorem 1.35 that $a'd' < b'c'$. Since $b'd' > 0$, it also follows from theorem 1.32 that $(b' < 0 \wedge d' < 0)$ or $(b' > 0 \wedge d' > 0)$.

Case 2: $bd > 0 \wedge b'd' < 0$. Similarly we have $ad < bc$. Since $b'd' < 0$, it follows from theorem 1.34 that

$$(ad)(b'd') > (bc)(b'd')$$

and by our first assumption, that

$$(bd)(a'd') > (bd)(b'c')$$

Since $bd > 0$, theorem 1.35 gives us $a'd' > b'c'$. Since $b'd' < 0$, we also have, by theorem 1.33, $(b' < 0 \wedge d' > 0)$ or $(b' > 0 \wedge d' < 0)$.

Case 3: $bd < 0 \wedge b'd' > 0$. By theorem 1.33, we have $(b < 0 \wedge d > 0)$ or $(b > 0 \wedge d < 0)$. See that either possibility would contradict $(d > 0 \wedge b > 0) \vee (d < 0 \wedge b < 0)$, so we must then have $bc < ad$. Since $b'd' > 0$, we have

$$(bc)(b'd') < (ad)(b'd')$$

and by our first assumption,

$$(bd)(b'c') < (bd)(a'd')$$

Since $bd < 0$ theorem 1.36 allows us to conclude that $a'd' < b'c'$. Since $b'd' > 0$ we also have, by theorem 1.32, $(b' > 0 \wedge d' > 0)$ or $(b' < 0 \wedge d' < 0)$.
(next page)

Case 4: $bd < 0 \wedge b'd' < 0$. Similarly it follows that $bc < ad$. Since $b'd' < 0$, by theorem 1.34,

$$(bc)(b'd') > (ad)(b'd')$$

and by our first assumption,

$$(bd)(b'c') > (bd)(a'd')$$

Since $bd < 0$ theorem 1.36 allows us to conclude that $b'c' < a'd'$. Since $b'd' < 0$, by theorem 1.33 we have $(b' < 0 \wedge d' > 0)$ or $(b' > 0 \wedge d' < 0)$. This proves one direction.

($\leftarrow$) Suppose instead that

$$\begin{aligned}(a'd' < b'c' \wedge ((d' > 0 \wedge b' > 0) \vee (d' < 0 \wedge b' < 0))) \vee \\ (b'c' < a'd' \wedge ((d' > 0 \wedge b' < 0) \vee (d' < 0 \wedge b' > 0)))\end{aligned}$$

We now want to show

$$\begin{aligned}(ad < bc \wedge ((d > 0 \wedge b > 0) \vee (d < 0 \wedge b < 0))) \vee \\ (bc < ad \wedge ((d > 0 \wedge b < 0) \vee (d < 0 \wedge b > 0)))\end{aligned}$$

Our argument for the forward direction proves

$$\begin{aligned}&\forall \langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^* \forall \langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^* \forall \langle a', b'\rangle \in \mathbb{Z} \times \mathbb{Z}^* \forall \langle c', d'\rangle \in \mathbb{Z} \times \mathbb{Z}^* (\\ &\quad ([\langle a, b\rangle] = [\langle a', b'\rangle] \wedge [\langle c, d\rangle] = [\langle c', d'\rangle]) \rightarrow (\\ &\quad\quad ((ad < bc \wedge ((d > 0 \wedge b > 0) \vee (d < 0 \wedge b < 0))) \\ &\quad\quad\quad \vee (bc < ad \wedge ((d > 0 \wedge b < 0) \vee (d < 0 \wedge b > 0)))) \rightarrow \\ &\quad\quad\quad ((a'd' < b'c' \wedge ((d' > 0 \wedge b' > 0) \vee (d' < 0 \wedge b' < 0))) \\ &\quad\quad\quad \vee (b'c' < a'd' \wedge ((d' > 0 \wedge b' < 0) \vee (d' < 0 \wedge b' > 0))))))\end{aligned}$$

The reverse direction immediately follows. □

Lemma 1.49.

$$\begin{aligned}\forall \langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^* \forall \langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^* ([\langle a, b\rangle] < [\langle c, d\rangle] \leftrightarrow \\ ((ad < bc \wedge ((d > 0 \wedge b > 0) \vee (d < 0 \wedge b < 0))) \vee \\ (bc < ad \wedge ((d > 0 \wedge b < 0) \vee (d < 0 \wedge b > 0)))))\end{aligned}$$

Proof. Let arbitrary $\langle a, b\rangle, \langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^*$.

($\rightarrow$) Suppose that $[\langle a, b\rangle] < [\langle c, d\rangle]$. By definition there then exists $\langle a', b'\rangle$, $\langle c', d'\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $[\langle a, b\rangle] = [\langle a', b'\rangle]$ and $[\langle c, d\rangle] = [\langle c', d'\rangle]$, where

$$\begin{aligned}(a'd' < b'c' \wedge ((d' > 0 \wedge b' > 0) \vee (d' < 0 \wedge b' < 0))) \vee \\ (b'c' < a'd' \wedge ((d' > 0 \wedge b' < 0) \vee (d' < 0 \wedge b' > 0)))\end{aligned}$$

The desired conclusion immediately follows from lemma 1.48.
(next page)

($\leftarrow$) Next suppose

$$(ad < bc \wedge ((d > 0 \wedge b > 0) \vee (d < 0 \wedge b < 0))) \vee \\ (bc < ad \wedge ((d > 0 \wedge b < 0) \vee (d < 0 \wedge b > 0)))$$

It immediately follows by definition that $\langle [\langle a, b\rangle], [\langle c, d\rangle]\rangle \in <$. $\square$

Proof. (Range/Domain bound and well-definedness) Showing $<\subseteq \mathbb{Q} \times \mathbb{Q}$ is straighforward. Let arbitrary $x \in <$. Then by definition exists $\langle a, b\rangle, \langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = \langle [\langle a, b\rangle], [\langle c, d\rangle]\rangle$. So $x \in \mathbb{Q} \times \mathbb{Q}$.

Next well-definedness. That is

$$\forall \langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^* \forall \langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^* \forall \langle a', b'\rangle \in \mathbb{Z} \times \mathbb{Z}^* \forall \langle c', d'\rangle \in \mathbb{Z} \times \mathbb{Z}^* (\\ ([\langle a, b\rangle] = [\langle a', b'\rangle] \wedge [\langle c, d\rangle] = [\langle c', d'\rangle] \\ \rightarrow ([\langle a, b\rangle] < [\langle c, d\rangle] \leftrightarrow [\langle a', b'\rangle] < [\langle c', d'\rangle])))$$

Let arbitrary $\langle a, b\rangle, \langle c, d\rangle, \langle a', b'\rangle, \langle c', d'\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ and suppose that $[\langle a, b\rangle] = [\langle a', b'\rangle]$ and $[\langle c, d\rangle] = [\langle c', d'\rangle]$.

($\rightarrow$) Suppose $[\langle a, b\rangle] < [\langle c, d\rangle]$; by lemma 1.49 we have

$$(ad < bc \wedge ((d > 0 \wedge b > 0) \vee (d < 0 \wedge b < 0))) \vee \\ (bc < ad \wedge ((d > 0 \wedge b < 0) \vee (d < 0 \wedge b > 0)))$$

By lemma 1.48 we then have

$$(a'd' < b'c' \wedge ((d' > 0 \wedge b' > 0) \vee (d' < 0 \wedge b' < 0))) \vee \\ (b'c' < a'd' \wedge ((d' > 0 \wedge b' < 0) \vee (d' < 0 \wedge b' > 0)))$$

applying lemma 1.49 again we have $[\langle a', b'\rangle] < [\langle c', d'\rangle]$.

($\leftarrow$) Suppose $[\langle a', b'\rangle] < [\langle c', d'\rangle]$. A similar argument allows us to conclude $[\langle a, b\rangle] < [\langle c, d\rangle]$. $\square$

Definition 1.50. *The binary relation $\leq$ on $\mathbb{Q}$ is defined such that*

$$\forall x (x \in \leq \leftrightarrow \exists \langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^* \exists \langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^* (x = \langle [\langle a, b\rangle], [\langle c, d\rangle]\rangle \wedge \\ ([\langle a, b\rangle] < [\langle c, d\rangle] \vee [\langle a, b\rangle] = [\langle c, d\rangle])))$$

(Existence follows from using $\mathbb{Q} \times \mathbb{Q}$ as a bounding set.)
(next page)

Lemma 1.51.

$$\forall\langle a,b\rangle \in \mathbb{Z}\times\mathbb{Z}^*\forall\langle c,d\rangle \in \mathbb{Z}\times\mathbb{Z}^*([\langle a,b\rangle] \leq [\langle c,d\rangle] \\ \leftrightarrow ([\langle a,b\rangle] < [\langle c,d\rangle] \vee [\langle a,b\rangle] = [\langle c,d\rangle]))$$

Proof. Let arbitrary $\langle a,b\rangle, \langle c,d\rangle \in \mathbb{Z}\times\mathbb{Z}^*$. First suppose $[\langle a,b\rangle] \leq [\langle c,d\rangle]$. By definition there then exists $\langle a',b'\rangle, \langle c',d'\rangle \in \mathbb{Z}\times\mathbb{Z}^*$ such that $[\langle a,b\rangle] = [\langle a',b'\rangle]$, $[\langle c,d\rangle] = [\langle c',d'\rangle]$, and $[\langle a',b'\rangle] < [\langle c',d'\rangle]$ or $[\langle a',b'\rangle] = [\langle c',d'\rangle]$. In the first case where $[\langle a',b'\rangle] < [\langle c',d'\rangle]$ it follows from well-definedness of $<$ that $[\langle a,b\rangle] < [\langle c,d\rangle]$. In the other case it immediately follows that $[\langle a,b\rangle] = [\langle a',b'\rangle] = [\langle c',d'\rangle] = [\langle c,d\rangle]$.

Next suppose $[\langle a,b\rangle] < [\langle c,d\rangle] \vee [\langle a,b\rangle] = [\langle c,d\rangle]$. It immediately follows by definition that $\langle [\langle a,b\rangle], [\langle c,d\rangle]\rangle \in \leq$. □

Proof. (domain/range bound and well-definedness) First we show $\leq \subseteq \mathbb{Q}\times\mathbb{Q}$. Let arbitrary $x \in \leq$. Then by definition there exists $\langle a,b\rangle, \langle c,d\rangle \in \mathbb{Z}\times\mathbb{Z}^*$ such that $x = \langle [\langle a,b\rangle], [\langle c,d\rangle]\rangle$. It immediately follows that $x \in \mathbb{Q}\times\mathbb{Q}$.

Next well definedness, that is

$$\forall\langle a,b\rangle \in \mathbb{Z}\times\mathbb{Z}^*\forall\langle c,d\rangle \in \mathbb{Z}\times\mathbb{Z}^*\forall\langle a',b'\rangle \in \mathbb{Z}\times\mathbb{Z}^*\forall\langle c',d'\rangle \in \mathbb{Z}\times\mathbb{Z}^*(\\ ([\langle a,b\rangle] = [\langle a',b'\rangle] \wedge [\langle c,d\rangle] = [\langle c',d'\rangle] \\ \rightarrow ([\langle a,b\rangle] \leq [\langle c,d\rangle] \leftrightarrow [\langle a',b'\rangle] \leq [\langle c',d'\rangle])))$$

Let arbitrary $\langle a,b\rangle, \langle c,d\rangle, \langle a',b'\rangle, \langle c',d'\rangle \in \mathbb{Z}\times\mathbb{Z}^*$ and suppose that $[\langle a,b\rangle] = [\langle a',b'\rangle]$ and $[\langle c,d\rangle] = [\langle c',d'\rangle]$.

($\rightarrow$) Suppose $[\langle a,b\rangle] \leq [\langle c,d\rangle]$. By lemma 1.51 we then have $[\langle a,b\rangle] < [\langle c,d\rangle]$ or $[\langle a,b\rangle] = [\langle c,d\rangle]$. In the case where $[\langle a,b\rangle] < [\langle c,d\rangle]$ it follows from the well-definedness of $<$ that $[\langle a',b'\rangle] < [\langle c',d'\rangle]$. In the other case we have $[\langle a,b\rangle] = [\langle a',b'\rangle] = [\langle c',d'\rangle] = [\langle c,d\rangle]$. So $[\langle a',b'\rangle] < [\langle c',d'\rangle] \vee [\langle a',b'\rangle] = [\langle c',d'\rangle]$. By lemma 1.51 we then have $[\langle a',b'\rangle] \leq [\langle c',d'\rangle]$.

The reverse implication follows from the same argument. □

1.3.3 Properties

Theorem 1.52. *Let $x, y, z \in \mathbb{Q}$.*

1. $(x + y) + z = x + (y + z)$ *(Associative Law for Addition)*
2. $x + y = y + x$ *(Commutative Law for Addition)*
3. $x + \bar{0} = x$ *(Identity Law for Addition)*
4. $x + (-x) = \bar{0}$ *(Inverses Law for Addition)*
5. $(xy)z = x(yz)$ *(Associative Law for Multiplication)*
6. $xy = yx$ *(Commutative Law for Multiplication)*
7. $x \cdot \bar{1} = x$ *(Identity Law for Multiplication)*
8. *If* $r \neq \bar{0}$, *then* $r \cdot r^{-1} = \bar{1}$ *(Inverses Law for Multiplication)*
9. $x(y + z) = xy + xz$ *(Distributive Law)*
10. *Precisely one of* $x < y$ *or* $x = y$ *or* $x > y$ *holds (Trichotomy Law)*
11. *If* $x < y$ *and* $y < z$, *then* $x < z$ *(Transitive Law)*
12. *If* $x < y$ *then* $x + z < y + z$ *(Addition Law for Order)*
13. *If* $x < y$ *and* $z > \bar{0}$, *then* $xz < yz$ *(Multiplication Law for Order)*
14. $\bar{0} \neq \bar{1}$ *(Non-Triviality)*

Proof. (1) We want to show

$$\forall x \in \mathbb{Q} \forall y \in \mathbb{Q} \forall z \in \mathbb{Q}((x + y) + z = x + (y + z))$$

Let arbitrary $x, y, z \in \mathbb{Q}$. Then there exists $\langle a, b\rangle, \langle c, d\rangle, \langle e, f\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b\rangle]$, $y = [\langle c, d\rangle]$, and $z = [\langle e, f\rangle]$. We then have

$$\begin{aligned}
(x + y) + z &= ([\langle a, b\rangle] + [\langle c, d\rangle]) + [\langle e, f\rangle] \\
&= [\langle ad + bc, bd\rangle] + [\langle e, f\rangle] \\
&= [\langle (ad + bc)f + (bd)e, (bd)f\rangle] \\
&= [\langle ((ad)f + (bc)f) + (bd)e, (bd)f\rangle] \\
&= [\langle a(df) + (b(cf) + b(de)), b(df)\rangle] \\
&= [\langle a(df) + b(cf + de), b(df)\rangle] \\
&= [\langle a, b\rangle] + [\langle cf + de, df\rangle] \\
&= [\langle a, b\rangle] + ([\langle c, d\rangle] + [\langle e, f\rangle]) \\
&= x + (y + z)
\end{aligned}$$

□

(next page)

Lemma 1.53.

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}^* (x = 0 \leftrightarrow [\langle x, y \rangle] = \bar{0})$$

Proof. Let arbitrary $x \in \mathbb{Z}$ and $y \in \mathbb{Z}^*$. First suppose that $x = 0$. It follows that $x \cdot 1 = 0 \cdot 1 = 0 = y \cdot 0$. So $\langle x, y \rangle \asymp \langle 0, 1 \rangle$ and $[\langle x, y \rangle] = [\langle 0, 1 \rangle] = \bar{0}$. Next suppose $[\langle x, y \rangle] = \bar{0}$. Then since $\bar{0} = [\langle 0, 1 \rangle]$, we have $\langle x, y \rangle \asymp \langle 0, 1 \rangle$ so $x = x \cdot 1 = y \cdot 0 = 0$. □

Lemma 1.54.

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}^* (x = y \leftrightarrow [\langle x, y \rangle] = \bar{1})$$

Proof. Let arbitrary $x \in \mathbb{Z}$ and $y \in \mathbb{Z}^*$. First suppose that $x = y$. Then it follows that $x \cdot 1 = y \cdot 1$, meaning $\langle x, y \rangle \asymp \langle 1, 1 \rangle$ and $[\langle x, y \rangle] = [\langle 1, 1 \rangle] = \bar{1}$. Next suppose $[\langle x, y \rangle] = \bar{1}$. It follows that $\langle x, y \rangle \asymp \langle 1, 1 \rangle$, so $x = x \cdot 1 = y \cdot 1 = y$. □

Proof. (2) We want to show

$$\forall x \in \mathbb{Q} \forall y \in \mathbb{Q} (x + y = y + x)$$

Let arbitrary $x, y \in \mathbb{Q}$. Then there exists $\langle a, b \rangle, \langle c, d \rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b \rangle]$ and $y = [\langle c, d \rangle]$. We have

$$\begin{aligned} x + y &= [\langle a, b \rangle] + [\langle c, d \rangle] \\ &= [\langle ad + bc, bd \rangle] \\ &= [\langle cb + da, db \rangle] \\ &= [\langle c, d \rangle] + [\langle a, b \rangle] = y + x \end{aligned}$$

□

Proof. (3) We want to show

$$\forall x \in \mathbb{Q} (x + \bar{0} = x)$$

Let arbitrary $x \in \mathbb{Q}$. Then there exists $\langle a, b \rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b \rangle]$. We then have

$$\begin{aligned} x + \bar{0} &= [\langle a, b \rangle] + [\langle 0, 1 \rangle] \\ &= [\langle a \cdot 1 + b \cdot 0, b \cdot 1 \rangle] \\ &= [\langle a, b \rangle] = x \end{aligned}$$

□

Proof. (4) We want to show

$$\forall x \in \mathbb{Q} (x + (-x) = \bar{0})$$

Let arbitrary $x \in \mathbb{Q}$. Then there exists $\langle a, b \rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b \rangle]$. By definition $-x = [\langle -a, b \rangle]$. So

$$\begin{aligned} x + (-x) &= [\langle a, b \rangle] + [\langle -a, b \rangle] \\ &= [\langle ab + b(-a), bb \rangle] = [\langle 0, bb \rangle] \end{aligned}$$

Since $b \neq 0$, $bb \in \mathbb{Z}^*$. So by lemma 1.53, $x + (-x) = [\langle 0, bb \rangle] = \bar{0}$. □

(next page)

Proof. (5) We want to show

$$\forall x \in \mathbb{Q} \forall y \in \mathbb{Q} \forall z \in \mathbb{Q}((xy)z = x(yz))$$

Let arbitrary $x, y, z \in \mathbb{Q}$. Then there exists $\langle a, b\rangle, \langle c, d\rangle, \langle e, f\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b\rangle]$, $y = [\langle c, d\rangle]$, and $z = [\langle e, f\rangle]$. We then have

$$\begin{aligned}(xy)z &= ([\langle a, b\rangle] \cdot [\langle c, d\rangle]) \cdot [\langle e, f\rangle] \\ &= [\langle ac, bd\rangle] \cdot [\langle e, f\rangle] \\ &= [\langle (ac)e, (bd)f\rangle] \\ &= [\langle a(ce), b(df)\rangle] \\ &= [\langle a, b\rangle] \cdot [\langle ce, df\rangle] \\ &= [\langle a, b\rangle] \cdot ([\langle c, d\rangle] \cdot [\langle e, f\rangle]) = x(yz)\end{aligned}$$

□

Proof. (6) We want to show

$$\forall x \in \mathbb{Q} \forall y \in \mathbb{Q}(xy = yx)$$

Let arbitrary $x, y \in \mathbb{Q}$. Then there exists $\langle a, b\rangle, \langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b\rangle]$ and $y = [\langle c, d\rangle]$. We then have

$$xy = [\langle a, b\rangle] \cdot [\langle c, d\rangle] = [\langle ac, bd\rangle] = [\langle ca, db\rangle] = [\langle c, d\rangle] \cdot [\langle a, b\rangle] = yx$$

□

Proof. (7) We want to show

$$\forall x \in \mathbb{Q}(x \cdot \bar{1} = x)$$

Let arbitrary $x \in \mathbb{Q}$. Then there exists $\langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b\rangle]$. We then have

$$x \cdot \bar{1} = [\langle a, b\rangle] \cdot [\langle 1, 1\rangle] = [\langle a \cdot 1, b \cdot 1\rangle] = [\langle a, b\rangle] = x$$

□

Proof. (8) We want to show

$$\forall x \in \mathbb{Q}(x \neq \bar{0} \to x \cdot x^{-1} = \bar{1})$$

Let arbitrary $x \in \mathbb{Q}$. Then there exists $\langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b\rangle]$. Suppose $x \neq \bar{0}$, meaning $x \in \mathbb{Q}^*$; by definition we then have $x^{-1} = [\langle a, b\rangle]^{-1} = [\langle b, a\rangle]$. So

$$x \cdot x^{-1} = [\langle a, b\rangle] \cdot [\langle b, a\rangle] = [\langle ab, ba\rangle] = [\langle ab, ab\rangle]$$

We know $[\langle a, b\rangle] \neq \bar{0}$, so by lemma 1.53 $a \neq 0$. Since $b \neq 0$ we then know $ab \neq 0$ and $ab \in \mathbb{Z}^*$. So by lemma 1.54, $x \cdot x^{-1} = [\langle ab, ab\rangle] = \bar{1}$. □

(next page)

Proof. (9) We want to show

$$\forall x \in \mathbb{Q} \forall y \in \mathbb{Q} \forall z \in \mathbb{Q}(x(y+z) = xy + xz)$$

Let arbitrary $x, y, z \in \mathbb{Q}$. Then there exists $\langle a, b\rangle, \langle c, d\rangle, \langle e, f\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b\rangle]$, $y = [\langle c, d\rangle]$, and $z = [\langle e, f\rangle]$. We then have

$$\begin{aligned} x(y+z) &= [\langle a, b\rangle] \cdot ([\langle c, d\rangle] + [\langle e, f\rangle]) \\ &= [\langle a, b\rangle] \cdot [\langle cf + de, df\rangle] \\ &= [\langle a(cf + de), b(df)\rangle] \\ &= [\langle a(cf) + a(de), b(df)\rangle] \end{aligned}$$

We know $b \in \mathbb{Z}^*$, so by lemma 1.54, $[\langle b, b\rangle] = \bar{1}$. By (7) we have $[\langle a(cf) + a(de), b(df)\rangle] = [\langle a(cf) + a(de), b(df)\rangle] \cdot \bar{1}$. So

$$\begin{aligned} [\langle a(cf) + a(de), b(df)\rangle] &= [\langle a(cf) + a(de), b(df)\rangle] \cdot \bar{1} \\ &= [\langle a(cf) + a(de), b(df)\rangle] \cdot [\langle b, b\rangle] \\ &= [\langle (a(cf) + a(de))b, (b(df))b\rangle] \\ &= [\langle (ac)(bf) + (bd)(ae), (bd)(bf)\rangle] \\ &= [\langle ac, bd\rangle] + [\langle ae, bf\rangle] \\ &= [\langle a, b\rangle] \cdot [\langle c, d\rangle] + [\langle a, b\rangle] \cdot [\langle e, f\rangle] \\ &= xy + xz \end{aligned}$$

So $x(y+z) = [\langle a(cf) + a(de), b(df)\rangle] = xy + xz$. □

Proof. (10) We want to show trichotomy:

$$\begin{aligned} \forall x \in \mathbb{Q} \forall y \in \mathbb{Q}((&x < y \vee x = y \vee y < x) \\ &\wedge \neg(x < y \wedge x = y) \wedge \neg(x < y \wedge y < x) \wedge \neg(x = y \wedge y < x)) \end{aligned}$$

Let arbitrary $x, y \in \mathbb{Q}$. Then there exists $\langle a, b\rangle, \langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b\rangle]$ and $y = [\langle c, d\rangle]$. We start with the latter three statements. First suppose for contradiction that $x < y \wedge x = y$. The former implies $ad < bc$ or $bc < ad$, while the latter $ad = bc$; either case contradicts trichotomy on $\mathbb{Z}$.

Next suppose that $x < y$ and $y < x$. We know $bd \neq 0$ so we have 2 cases. First suppose $bd > 0$. Then it follows that $(b < 0 \wedge d < 0) \vee (b > 0 \wedge d > 0)$. $x < y$ would then imply $ad < bc$ while $y < x$ would imply $cb < da$ and $bc < ad$, contradicting trichotomy on $\mathbb{Z}$. Next suppose $bd < 0$. We then have $(b > 0 \wedge d < 0) \vee (b < 0 \wedge d > 0)$, where $x < y$ would then imply $bc < ad$ while $y < x$ would imply $da < cb$ and $ad < bc$, contradicting trichotomy.

Next suppose $x = y \wedge y < x$. The former implies $ad = bc$ while the latter $bc < ad$ or $ad < bc$. Either case contradicts trichotomy.
(next page)

Finally we prove the first statement. Trichotomy on $\mathbb{Z}$ gives us $ad < bc \vee ad = bc \vee bc < ad$. We know $b \neq 0$ and $d \neq 0$. We have the following cases:

Case 1a: $ad < bc$, $b < 0$, $d < 0$. Then $[\langle a, b\rangle] < [\langle c, d\rangle]$ and $x < y$.
Case 1b: $ad < bc$, $b < 0$, $d > 0$. Then $[\langle c, d\rangle] < [\langle a, b\rangle]$ and $y < x$.
Case 1c: $ad < bc$, $b > 0$, $d < 0$. Then $[\langle c, d\rangle] < [\langle a, b\rangle]$ and $y < x$.
Case 1d: $ad < bc$, $b > 0$, $d > 0$. Then $[\langle a, b\rangle] < [\langle c, d\rangle]$ and $x < y$.
Case 2: $ad = bc$. Then $[\langle a, b\rangle] = [\langle c, d\rangle]$ and $x = y$.
Case 3a: $bc < ad$, $b < 0$, $d < 0$. Then $[\langle c, d\rangle] < [\langle a, b\rangle]$ and $y < x$.
Case 3b: $bc < ad$, $b < 0$, $d > 0$. Then $[\langle a, b\rangle] < [\langle c, d\rangle]$ and $x < y$.
Case 3c: $bc < ad$, $b > 0$, $d < 0$. Then $[\langle a, b\rangle] < [\langle c, d\rangle]$ and $x < y$.
Case 3d: $bc < ad$, $b > 0$, $d > 0$. Then $[\langle c, d\rangle] < [\langle a, b\rangle]$ and $y < x$. □

Proof. (11) We want to show

$$\forall x \in \mathbb{Q} \forall y \in \mathbb{Q} \forall z \in \mathbb{Q}((x < y \wedge y < z) \rightarrow x < z)$$

Let arbitrary $x, y, z \in \mathbb{Q}$. Then there exists $\langle a, b\rangle, \langle c, d\rangle, \langle e, f\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b\rangle]$, $y = [\langle c, d\rangle]$, and $z = [\langle e, f\rangle]$. Suppose $x < y$ and $y < z$. $x < y$ implies

$$(ad < bc \wedge ((d > 0 \wedge b > 0) \vee (d < 0 \wedge b < 0))) \\ \vee (bc < ad \wedge ((d > 0 \wedge b < 0) \vee (d < 0 \wedge b > 0)))$$

and $y < z$ implies

$$(cf < de \wedge ((f > 0 \wedge d > 0) \vee (f < 0 \wedge d < 0))) \\ \vee (de < cf \wedge ((f > 0 \wedge d < 0) \vee (f < 0 \wedge d > 0)))$$

We have 8 cases.

Case 1a: $(ad < bc \wedge ((d > 0 \wedge b > 0) \vee (d < 0 \wedge b < 0)))$, $(cf < de \wedge ((f > 0 \wedge d > 0) \vee (f < 0 \wedge d < 0)))$, $b > 0$. We must have $d > 0$ and then $f > 0$. So $(ad)f < (bc)f$ and $b(cf) < b(de)$, which, by integer transitivity, implies $d(af) < d(be)$ and therefore $af < be$. This, together with $f > 0$ and $b > 0$, implies $[\langle a, b\rangle] < [\langle e, f\rangle]$ and $x < z$.

Case 1b: $(ad < bc \wedge ((d > 0 \wedge b > 0) \vee (d < 0 \wedge b < 0)))$, $(cf < de \wedge ((f > 0 \wedge d > 0) \vee (f < 0 \wedge d < 0)))$, $b < 0$. We must have $d < 0$ and then $f < 0$. So $(ad)f > (bc)f$ and $b(cf) > b(de)$, which, by integer transitivity, implies $d(af) > d(be)$ and therefore $af < be$. This, together with $f < 0$ and $b < 0$, implies $[\langle a, b\rangle] < [\langle e, f\rangle]$ and $x < z$.

Case 2a: $(ad < bc \wedge ((d > 0 \wedge b > 0) \vee (d < 0 \wedge b < 0)))$, $(de < cf \wedge ((f > 0 \wedge d < 0) \vee (f < 0 \wedge d > 0)))$, $b > 0$. We must have $d > 0$ and then $f < 0$. So $(ad)f > (bc)f$ and $b(de) < b(cf)$, which, by integer transitivity, implies $d(be) < d(af)$ and therefore $be < af$. This, together with $f < 0$ and $b > 0$, implies $[\langle a, b\rangle] < [\langle e, f\rangle]$ and $x < z$.

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Case 2b: $(ad < bc \wedge ((d > 0 \wedge b > 0) \vee (d < 0 \wedge b < 0)))$, $(de < cf \wedge ((f > 0 \wedge d < 0) \vee (f < 0 \wedge d > 0)))$, $b < 0$. We must have $d < 0$ and then $f > 0$. So $(ad)f < (bc)f$ and $b(de) > b(cf)$, which, by integer transitivity, implies $d(af) < d(be)$ and therefore $be < af$. This, together with $f > 0$ and $b < 0$, implies $[\langle a, b\rangle] < [\langle e, f\rangle]$ and $x < z$.

Case 3a: $(bc < ad \wedge ((d > 0 \wedge b < 0) \vee (d < 0 \wedge b > 0)))$, $(cf < de \wedge ((f > 0 \wedge d > 0) \vee (f < 0 \wedge d < 0)))$, $b > 0$. We must have $d < 0$ and then $f < 0$. So $(bc)f > (ad)f$ and $b(cf) < b(de)$, which, by integer transitivity, implies $d(af) < d(be)$ and therefore $be < af$. This, together with $f < 0$ and $b > 0$, implies $[\langle a, b\rangle] < [\langle e, f\rangle]$ and $x < z$.

Case 3b: $(bc < ad \wedge ((d > 0 \wedge b < 0) \vee (d < 0 \wedge b > 0)))$, $(cf < de \wedge ((f > 0 \wedge d > 0) \vee (f < 0 \wedge d < 0)))$, $b < 0$. We must have $d > 0$ and then $f > 0$. So $(bc)f < (ad)f$ and $b(cf) > b(de)$, which, by integer transitivity, implies $d(af) > d(be)$ and therefore $be < af$. This, together with $f > 0$ and $b < 0$, implies $[\langle a, b\rangle] < [\langle e, f\rangle]$ and $x < z$.

Case 4a: $(bc < ad \wedge ((d > 0 \wedge b < 0) \vee (d < 0 \wedge b > 0)))$, $(de < cf \wedge ((f > 0 \wedge d < 0) \vee (f < 0 \wedge d > 0)))$, $b > 0$. We must then have $d < 0$ and then $f > 0$. So $(bc)f < (ad)f$ and $b(de) < b(cf)$, which, by integer transitivity, implies $d(af) > d(be)$ and therefore $af < be$. This, together with $f > 0$ and $b > 0$, implies $[\langle a, b\rangle] < [\langle e, f\rangle]$ and $x < z$.

Case 4b: $(bc < ad \wedge ((d > 0 \wedge b < 0) \vee (d < 0 \wedge b > 0)))$, $(de < cf \wedge ((f > 0 \wedge d < 0) \vee (f < 0 \wedge d > 0)))$, $b < 0$. We must then have $d > 0$ and then $f < 0$. So $(bc)f > (ad)f$ and $b(de) > b(cf)$, which, by integer transitivity, implies $d(af) < d(be)$ and therefore $af < be$. This, together with $f < 0$ and $b < 0$, implies $[\langle a, b\rangle] < [\langle e, f\rangle]$ and $x < z$. □

Proof. (12) We want to show

$$\forall x \in \mathbb{Q} \forall y \in \mathbb{Q} \forall z \in \mathbb{Q} (x < y \rightarrow x + z < y + z)$$

Let arbitrary $x, y, z \in \mathbb{Q}$. Then there exists $\langle a, b\rangle, \langle c, d\rangle, \langle e, f\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b\rangle]$, $y = [\langle c, d\rangle]$, and $z = [\langle e, f\rangle]$. Suppose $x < y$, which implies

$$(ad < bc \wedge ((d > 0 \wedge b > 0) \vee (d < 0 \wedge b < 0))) \\ \vee (bc < ad \wedge ((d > 0 \wedge b < 0) \vee (d < 0 \wedge b > 0)))$$

First suppose $bd > 0$. It follows that $(b > 0 \wedge d > 0) \vee (b < 0 \wedge d < 0)$ and then that $ad < bc$. We know $f \neq 0$ so $f > 0$ or $f < 0$, where in either case $ff > 0$. So

$$(af)(df) = (ad)(ff) < (bc)(ff) = (bf)(cf)$$

and

$$(af + be)(df) = (af)(df) + (be)(df) < (bf)(cf) + (be)(df) = (bf)(cf + de)$$

(next page)

We know $(af + be)(df) < (bf)(cf + de)$. We have four cases:

Case 1: $(b > 0 \wedge d > 0)$, $f > 0$. Then $bf > 0$ and $df > 0$. This, combined with the above, implies $[\langle af + be, bf\rangle] < [\langle cf + de, df\rangle]$. We have

$$x+z = [\langle a, b\rangle]+[\langle e, f\rangle] = [\langle af+be, bf\rangle] < [\langle cf+de, df\rangle] = [\langle c, d\rangle]+[\langle e, f\rangle] = y+z$$

Case 2: $(b > 0 \wedge d > 0)$, $f < 0$. Then $bf < 0$ and $df < 0$, implying $[\langle af + be, bf\rangle] < [\langle cf + de, df\rangle]$ and, similarly, $x + z < y + z$.

Case 3: $(b < 0 \wedge d < 0)$, $f > 0$. Then $bf < 0$ and $df < 0$, where an identical argument to case 2 implies $x + z < y + z$.

Case 4: $(b < 0 \wedge d < 0)$, $f < 0$. Then $bf > 0$ and $df > 0$, where an identical argument to case 1 implies $x + z < y + z$.

Next suppose $bd < 0$. It follows that $(d > 0 \wedge b < 0) \vee (d < 0 \wedge b > 0)$, and then $bc < ad$. We then have

$$(bf)(cf) = (bc)(ff) < (ad)(ff) = (af)(df)$$

and

$$(bf)(cf + de) = (bf)(cf) + (bf)(de) < (af)(df) + (bf)(de) = (af + be)(df)$$

Again we have four cases.

Case 1: $(b < 0 \wedge d > 0)$, $f > 0$. Then $bf < 0$ and $df > 0$. This, combined with the fact that $(bf)(cf+de) < (af+be)(df)$, implies $[\langle af+be, bf\rangle] < [\langle cf+de, df\rangle]$, which means $x + z < y + z$.

Case 2: $(b < 0 \wedge d > 0)$, $f < 0$. Then $bf > 0$ and $df < 0$. This also implies $[\langle af + be, bf\rangle] < [\langle cf + de, df\rangle]$, which means $x + z < y + z$.

Case 3: $(b > 0 \wedge d < 0)$, $f > 0$. Then $bf > 0$ and $df < 0$, where an identical argument to case 2 implies $x + z < y + z$.

Case 4: $(b > 0 \wedge d < 0)$, $f < 0$. Then $bf < 0$ and $df > 0$, where an identical argument to case 1 implies $x + z < y + z$. □

(next page)

Proof. (13) We want to show

$$\forall x \in \mathbb{Q} \forall y \in \mathbb{Q} \forall z \in \mathbb{Q}((x < y \wedge z > \bar{0}) \rightarrow xz < yz)$$

Let arbitrary $x, y, z \in \mathbb{Q}$. Then there exists $\langle a, b\rangle, \langle c, d\rangle, \langle e, f\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b\rangle]$, $y = [\langle c, d\rangle]$, and $z = [\langle e, f\rangle]$. Suppose $x < y$, which implies

$$(ad < bc \wedge ((d > 0 \wedge b > 0) \vee (d < 0 \wedge b < 0))) \\ \vee (bc < ad \wedge ((d > 0 \wedge b < 0) \vee (d < 0 \wedge b > 0)))$$

Suppose $z > \bar{0}$. It follows by definition that

$$(0 < e \wedge ((f > 0 \wedge 1 > 0) \vee (f < 0 \wedge 1 < 0))) \vee \\ (e < 0 \wedge ((f > 0 \wedge 1 < 0) \vee (f < 0 \wedge 1 > 0)))$$

Which is equivalent to

$$(0 < e \wedge f > 0) \vee (e < 0 \wedge f < 0)$$

First suppose $bd > 0$. It follows that $(b > 0 \wedge d > 0) \vee (b < 0 \wedge d < 0)$ and then that $ad < bc$. We have 2 cases. First suppose $f > 0$. This immediately implies that $e > 0$, so $ef > 0$ and

$$(ae)(df) = (ad)(ef) < (bc)(ef) = (bf)(ce)$$

We also have $(bf > 0 \wedge df > 0) \vee (bf < 0 \wedge df < 0)$. By definition this implies $[\langle ae, bf\rangle] < [\langle ce, df\rangle]$, and

$$xz = [\langle a, b\rangle] \cdot [\langle e, f\rangle] = [\langle ae, bf\rangle] < [\langle ce, df\rangle] = [\langle c, d\rangle] \cdot [\langle e, f\rangle] = yz$$

Next suppose $f < 0$. This implies $e < 0$, so $ef > 0$. A similar argument as in the first case allows us to conclude $(ae)(df) < (bf)(ce)$. Again we also have $(bf > 0 \wedge df > 0) \vee (bf < 0 \wedge df < 0)$, so $[\langle ae, bf\rangle] < [\langle ce, df\rangle]$ and $xz < yz$.

Next suppose $bd < 0$. It follows that $(d > 0 \wedge b < 0) \vee (d < 0 \wedge b > 0)$, and then that $bc < ad$. Again we have 2 cases. First suppose $f > 0$. Then $e > 0$ and $ef > 0$. So

$$(bf)(ce) = (bc)(ef) < (ad)(ef) = (ae)(df)$$

We also have $(df > 0 \wedge bf < 0) \vee (df < 0 \wedge bf > 0)$. By definition this means $[\langle ae, bf\rangle] < [\langle ce, df\rangle]$, and, similarly to the previous cases, $xz < yz$. Next suppose $f < 0$. Then $e < 0$ and $ef > 0$. A similar argument as in the first case allows us to conclude $(bf)(ce) < (ae)(df)$. Again we also have $(df > 0 \wedge bf < 0) \vee (df < 0 \wedge bf > 0)$, so $[\langle ae, bf\rangle] < [\langle ce, df\rangle]$ and $xz < yz$. □

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Proof. (14) We want to show $\bar{0} \neq \bar{1}$. Suppose for contradiction that they are equal. Then $[\langle 0, 1\rangle] = [\langle 1, 1\rangle]$, implying $0 = 0 \cdot 1 = 1 \cdot 1 = 1$, which contradicts integer nontriviality. □

Terminology
The properties of the rational numbers listed here are satisfied by other number systems as well (such as the reals), and are sufficiently important to merit special terminology. Any set with two binary operations that satisfies parts (1)-(9) is called a "field", and any field that also has a relation that satisfies parts (10)-(14) is called an "ordered field".

1.3.4 Joining the integers and the rationals

Theorem 1.55. *Let $i : \mathbb{Z} \to \mathbb{Q}$ be defined by $i(x) = [\langle x, 1\rangle]$ for all $x \in \mathbb{Z}$. That is,*

$$\forall x(x \in i \leftrightarrow \exists a \in \mathbb{Z}(x = \langle a, [\langle a, 1\rangle]\rangle))$$

1. *The function $i : \mathbb{Z} \to \mathbb{Q}$ is injective.*
2. *$i(0) = \bar{0}$ and $i(1) = \bar{1}$.*
3. *Let $x, y \in \mathbb{Z}$. Then*
   (a) *$i(x + y) = i(x) + i(y)$*
   (b) *$i(-x) = -i(x)$*
   (c) *$i(xy) = i(x)i(y)$*
   (d) *$x < y$ if and only if $i(x) < i(y)$*
4. *For each $x \in \mathbb{Q}$ there are $a, b \in \mathbb{Z}$ such that $b \neq 0$ and $x = i(a)(i(b))^{-1}$.*

Proof. (1) We start with the domain bound $\mathrm{dom}(i) = \mathbb{Z}$. Let arbitrary $x \in \mathrm{dom}(i)$. Then there exists y such that $\langle x, y\rangle \in i$. By definition there then exists $a \in \mathbb{Z}$ such that $x = a$. Next let $x \in \mathbb{Z}$. Then by definition $\langle x, [\langle x, 1\rangle]\rangle \in i$. So $x \in \mathrm{dom}(i)$.

Next the range bound $\mathrm{ran}(i) \subseteq \mathbb{Q}$. Let arbitary $y \in \mathrm{ran}(i)$; then there exists x such that $\langle x, y\rangle \in i$. By definition there then exists $a \in \mathbb{Z}$ such that $y = [\langle a, 1\rangle]$. Since $\langle a, 1\rangle \in \mathbb{Z} \times \mathbb{Z}^*$, $y \in \mathbb{Q}$.

Next, functionality. Let $x \in \mathrm{dom}(i)$. Then there exists y such that $\langle x, y\rangle \in i$, and then $a \in \mathbb{Z}$ such that $\langle x, y\rangle = \langle a, [\langle a, 1\rangle]\rangle$. Let y' be arbitrary such that $\langle x, y'\rangle \in i$. Then there exists $a' \in \mathbb{Z}$ such that $\langle x, y'\rangle = \langle a', [\langle a', 1\rangle]\rangle$. It follows then that $a = x = a'$, and therefore $a \cdot 1 = 1 \cdot a'$. This means $\langle a, 1\rangle \asymp \langle a', 1\rangle$ and $y = [\langle a, 1\rangle] = [\langle a', 1\rangle] = y'$.

Finally injectivity. Let arbitrary $y \in \mathrm{ran}(i)$. There then exists x such that $\langle x, y\rangle \in i$, and therefore $a \in \mathbb{Z}$ such that $\langle x, y\rangle = \langle a, [\langle a, 1\rangle]\rangle$. Let x' be arbitrary such that $\langle x', y\rangle \in i$. Then there exists $a' \in \mathbb{Z}$ such that $\langle x', y\rangle = \langle a', [\langle a', 1\rangle]\rangle$. We know $[\langle a, 1\rangle] = y = [\langle a', 1\rangle]$, so $x = a = a \cdot 1 = 1 \cdot a' = a' = x'$. □

Proof. (2) By definition $\langle 0, [\langle 0, 1\rangle]\rangle \in i$. So $i(0) = [\langle 0, 1\rangle] = \bar{0}$. We also have by definition that $\langle 1, [\langle 1, 1\rangle]\rangle \in i$. So $i(1) = [\langle 1, 1\rangle] = \bar{1}$. □

Proof. (3a) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}(i(x + y) = i(x) + i(y))$$

Let arbitrary $x, y \in \mathbb{Z}$. By definition $\langle x + y, [\langle x + y, 1\rangle]\rangle \in i$. It then follows by definition of $+$ that

$$i(x + y) = [\langle x + y, 1\rangle] = [\langle x, 1\rangle] + [\langle y, 1\rangle] = i(x) + i(y)$$ □

(next page)

Proof. (3b) We want to show

$$\forall x \in \mathbb{Z}(i(-x) = -i(x))$$

Let arbitrary $x \in \mathbb{Z}$. By definition $\langle -x, [\langle -x, 1\rangle]\rangle \in i$. So

$$i(-x) = [\langle -x, 1\rangle] = -[\langle x, 1\rangle] = -i(x)$$ □

Proof. (3c) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}(i(xy) = i(x)i(y))$$

Let arbitrary $x, y \in \mathbb{Z}$. By definition $\langle xy, [\langle xy, 1\rangle]\rangle \in i$. So

$$i(xy) = [\langle xy, 1\rangle] = [\langle x, 1\rangle] \cdot [\langle y, 1\rangle] = i(x)i(y)$$ □

Proof. (3d) We want to show

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}(x < y \leftrightarrow i(x) < i(y))$$

Let arbitrary $x, y \in \mathbb{Z}$. By definition $i(x) < i(y)$ is equivalent to $[\langle x, 1\rangle] < [\langle y, 1\rangle]$, which is equivalent to

$$\begin{aligned}(x < y \wedge ((1 > 0 \wedge 1 > 0) \vee (1 < 0 \wedge 1 < 0)) \\ \vee (y < x \wedge ((1 > 0 \wedge 1 < 0) \vee (1 < 0 \wedge 1 > 0))\end{aligned}$$

Since $1 < 0$ is a contradiction, this statement is equivalent to $x < y$. □

Proof. (4) We want to show

$$\forall x \in \mathbb{Q} \exists a \in \mathbb{Z} \exists b \in \mathbb{Z}(b \neq 0 \wedge x = i(a)(i(b))^{-1})$$

Let arbitrary $x \in \mathbb{Q}$. Then there exists $\langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b\rangle]$. Since $b \in \mathbb{Z}$, $b \neq 0$. By lemma 1.53 we have $[\langle b, 1\rangle] \neq \bar{0}$. By definition, $\langle [\langle b, 1\rangle], [\langle 1, b\rangle]\rangle \in^{-1}$, so

$$x = [\langle a, b\rangle] = [\langle a, 1\rangle] \cdot [\langle 1, b\rangle] = [\langle a, 1\rangle] \cdot [\langle b, 1\rangle]^{-1} = i(a)(i(b))^{-1}$$ □

1.3.5 More operations

Definition 1.56. *The binary operation* $- : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$ *is defined as*

$$\forall x(x \in - \leftrightarrow \exists a \in \mathbb{Q} \exists b \in \mathbb{Q}(x = \langle \langle a, b \rangle, a + (-b) \rangle))$$

Proof. (domain/range/functionality) First we show $\mathrm{dom}(-) = \mathbb{Q} \times \mathbb{Q}$. Let arbitrary $x \in -$. Then there exists y such that $\langle x, y \rangle \in -$. There then exists $a, b \in \mathbb{Q}$ such that $x = \langle a, b \rangle$. So $x \in \mathbb{Q} \times \mathbb{Q}$. Next let arbitrary $x \in \mathbb{Q} \times \mathbb{Q}$. Then there exists $a, b \in \mathbb{Q}$ such that $x = \langle a, b \rangle$. By definition $\langle \langle a, b \rangle, a + (-b) \rangle \in -$. So $x = \langle a, b \rangle \in \mathrm{dom}(-)$.

Next we show $\mathrm{ran}(-) \subseteq \mathbb{Q}$. Let arbitrary $y \in \mathrm{ran}(-)$. Then there exists x such that $\langle x, y \rangle \in -$, and $a, b \in \mathbb{Q}$ such that $y = a + (-b)$. So $y \in \mathbb{Q}$.

Finally functionality. Let arbitrary $x \in \mathrm{dom}(-)$. There exists y, and there exists $a, b \in \mathbb{Q}$ such that $\langle x, y \rangle = \langle \langle a, b \rangle, a + (-b) \rangle \in -$. Let y' be arbitrary such that $\langle x, y' \rangle \in -$. Then there exists $a', b' \in \mathbb{Q}$ such that $\langle x, y' \rangle = \langle \langle a', b' \rangle, a' + (-b') \rangle \in -$. We have $\langle a, b \rangle = x = \langle a', b' \rangle$. So $y = a + (-b) = a' + (-b') = y'$. □

Definition 1.57. *The binary operation* $\div : \mathbb{Q} \times \mathbb{Q}^* \to \mathbb{Q}$ *is defined as*

$$\forall x(x \in \div \leftrightarrow \exists a \in \mathbb{Q} \exists b \in \mathbb{Q}^*(x = \langle \langle a, b \rangle, ab^{-1} \rangle))$$

The number $a \div b$ *is also denoted* $\frac{a}{b}$.

Proof. (domain/range/functionality) First we show $\mathrm{dom}(\div) = \mathbb{Q} \times \mathbb{Q}^*$. Let arbitrary $x \in \mathrm{dom}(\div)$. Then there exists y such that $\langle x, y \rangle \in \div$. Then there also exists $a \in \mathbb{Q}$ and $b \in \mathbb{Q}^*$ such that $x = \langle a, b \rangle$. So $x \in \mathbb{Q} \times \mathbb{Q}^*$. Next let arbitrary $x \in \mathbb{Q} \times \mathbb{Q}^*$. Then there exists $a \in \mathbb{Q}$ and $b \in \mathbb{Q}^*$ such that $x = \langle a, b \rangle$. By definition $\langle \langle a, b \rangle, ab^{-1} \rangle \in \div$. So $x = \langle a, b \rangle \in \mathrm{dom}(\div)$.

Next we show $\mathrm{ran}(\div) \subseteq \mathbb{Q}$. Let arbitrary $y \in \mathrm{ran}(\div)$. By definition there exists x such that $\langle x, y \rangle \in \div$, and therefore $a \in \mathbb{Q}$ and $b \in \mathbb{Q}^*$ such that $y = ab^{-1}$. Since $\cdot : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$, $y = ab^{-1} \in \mathbb{Q}$.

Finally functionality. Letting arbitary $x \in \mathrm{dom}(\div)$, there exists y such that $\langle x, y \rangle \in \div$, as well as $a \in \mathbb{Q}$ and $b \in \mathbb{Q}^*$ such that $\langle x, y \rangle = \langle \langle a, b \rangle, ab^{-1} \rangle$. Let y' be arbitary such that $\langle x, y' \rangle \in \div$. Then there exists $a' \in \mathbb{Q}$ and $b' \in \mathbb{Q}^*$ such that $\langle x, y' \rangle = \langle \langle a', b' \rangle, a'(b')^{-1} \rangle$. It follows that $\langle a, b \rangle = x = \langle a', b' \rangle$ and then $y = ab^{-1} = a'(b')^{-1} = y'$. □

(next page)

Theorem 1.58. *Let $a, c \in \mathbb{Z}$ and $b, d \in \mathbb{Z}^*$. We had defined $i : \mathbb{Z} \to \mathbb{Q}$ as*

$$\forall x(x \in i \leftrightarrow \exists a \in \mathbb{Z}(x = \langle a, [\langle a, 1\rangle]\rangle))$$

We then have

1. $\frac{i(a)}{i(b)} = \frac{i(c)}{i(d)}$ *if and only if* $ad = bc$
2. $\frac{i(a)}{i(b)} + \frac{i(c)}{i(d)} = \frac{i(a)i(d)+i(b)i(c)}{i(b)i(d)}$
3. $-\frac{i(a)}{i(b)} = \frac{-i(a)}{i(b)}$
4. $\frac{i(a)}{i(b)} \cdot \frac{i(c)}{i(d)} = \frac{i(a)i(c)}{i(b)i(d)}$
5. *If* $a \neq 0$, *then* $\left(\frac{i(a)}{i(b)}\right)^{-1} = \frac{i(b)}{i(a)}$
6. *If* $b > 0$ *and* $d > 0$ *or if* $b < 0$ *and* $d < 0$, *then* $\frac{i(a)}{i(b)} < \frac{i(c)}{i(d)}$ *if and only if* $ad < bc$;
   If $b > 0$ *and* $d < 0$ *or if* $b < 0$ *and* $d > 0$, *then* $\frac{i(a)}{i(b)} < \frac{i(c)}{i(d)}$ *if and only if* $ad > bc$.

Proof. (1) Let arbitrary $a, c \in \mathbb{Z}$ and $b, d \in \mathbb{Z}^*$. By definition

$$\begin{aligned}\frac{i(a)}{i(b)} = \frac{i(c)}{i(d)} &\leftrightarrow [\langle a, 1\rangle] \cdot [\langle b, 1\rangle]^{-1} = [\langle c, 1\rangle] \cdot [\langle d, 1\rangle]^{-1} \\ &\leftrightarrow [\langle a, b\rangle] = [\langle c, d\rangle] \\ &\leftrightarrow ad = bc\end{aligned}$$

□

Proof. (2) Let arbitrary $a, c \in \mathbb{Z}$ and $b, d \in \mathbb{Z}^*$. By definition

$$\begin{aligned}\frac{i(a)}{i(b)} + \frac{i(c)}{i(d)} &= [\langle a, 1\rangle] \cdot [\langle b, 1\rangle]^{-1} + [\langle c, 1\rangle] \cdot [\langle d, 1\rangle]^{-1} \\ &= [\langle a, b\rangle] + [\langle c, d\rangle] \\ &= [\langle ad + bc, bd\rangle] \\ &= [\langle ad + bc, 1\rangle] \cdot [\langle 1, bd\rangle] \\ &= i(ad + bc)(i(bd))^{-1} \\ &= \frac{i(ad) + i(bc)}{i(bd)} \\ &= \frac{i(a)i(d) + i(b)i(c)}{i(b)i(d)}\end{aligned}$$

□

(next page)

Proof. (3) Let arbitrary $a \in \mathbb{Z}$ and $b \in \mathbb{Z}^*$. By definition

$$\begin{aligned} -\frac{i(a)}{i(b)} &= -([\langle a, 1\rangle] \cdot [\langle b, 1\rangle]^{-1}) \\ &= -[\langle a, b\rangle] \\ &= [\langle -a, b\rangle] \\ &= [\langle -a, 1\rangle] \cdot [\langle 1, b\rangle] \\ &= i(-a)(i(b)^{-1}) \\ &= \frac{i(-a)}{i(b)} = \frac{-i(a)}{i(b)} \end{aligned}$$

□

Proof. (4) Let arbitrary $a, c \in \mathbb{Z}$ and $b, d \in \mathbb{Z}^*$. By definition

$$\begin{aligned} \frac{i(a)}{i(b)} \cdot \frac{i(c)}{i(d)} &= [\langle a, b\rangle] \cdot [\langle c, d\rangle] \\ &= [\langle ac, bd\rangle] \\ &= [\langle ac, 1\rangle] \cdot [\langle 1, bd\rangle] \\ &= [\langle ac, 1\rangle] \cdot [\langle bd, 1\rangle]^{-1} \\ &= i(ac)(i(bd))^{-1} \\ &= \frac{i(ac)}{i(bd)} = \frac{i(a)i(c)}{i(b)i(d)} \end{aligned}$$

□

Proof. (5) Let arbitrary $a \in \mathbb{Z}$ and $b \in \mathbb{Z}^*$. Suppose $a \neq 0$. By definition we have

$$\begin{aligned} \left(\frac{i(a)}{i(b)}\right)^{-1} &= [\langle a, b\rangle]^{-1} \\ &= [\langle b, a\rangle] \\ &= [\langle b, 1\rangle] \cdot [\langle a, 1\rangle]^{-1} \\ &= i(b)(i(a))^{-1} = \frac{i(b)}{i(a)} \end{aligned}$$

□

Proof. (6) Let arbitrary $a, c \in \mathbb{Z}$ and $b, d \in \mathbb{Z}^*$. First suppose $(b > 0 \wedge d > 0) \vee (b < 0 \wedge d < 0)$. Suppose $\frac{i(a)}{i(b)} < \frac{i(c)}{i(d)}$. We then have

$$[\langle a, b\rangle] = \frac{i(a)}{i(b)} < \frac{i(c)}{i(d)} = [\langle c, d\rangle]$$

Which, along with our first assumption, immediately implies $ad < bc$. Next suppose $ad < bc$. This, and our first assumption, similarly implies $[\langle a, b\rangle] < [\langle c, d\rangle]$, so

$$\frac{i(a)}{i(b)} = [\langle a, b\rangle] < [\langle c, d\rangle] = \frac{i(c)}{i(d)}$$

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Next suppose $(b > 0 \wedge d < 0) \vee (b < 0 \wedge d > 0)$. Suppose $\frac{i(a)}{i(b)} < \frac{i(c)}{i(d)}$. We then have

$$[\langle a, b\rangle] = \frac{i(a)}{i(b)} < \frac{i(c)}{i(d)} = [\langle c, d\rangle]$$

Which, along with our assumption, immediately implies $bc < ad$. Next suppose $bc < ad$. Similarly this implies $[\langle a, b\rangle] < [\langle c, d\rangle]$, so

$$\frac{i(a)}{i(b)} = [\langle a, b\rangle] < [\langle c, d\rangle] = \frac{i(c)}{i(d)} \qquad \square$$

1.3.6 More properties

Lemma 1.59.

$$\forall x \in \mathbb{Z} \forall y \in \mathbb{Z}^* (\bar{0} < [\langle x, y \rangle] \leftrightarrow 0 < xy)$$

Proof. Let arbitrary $x \in \mathbb{Z}$ and $y \in \mathbb{Z}^*$. We have

$$[\langle 0, 1 \rangle] = \bar{0} < [\langle x, y \rangle] \leftrightarrow ((0 < x \wedge 0 < y) \vee (0 > x \wedge 0 > y)) \leftrightarrow 0 < xy \qquad \square$$

Theorem 1.60. *Let $x, y, z, w \in \mathbb{Q}$.*

1. $-1 < 0 < 1$
2. *If $x < y$ then $-y < -x$*
3. $x \cdot 0 = 0$
4. *If $x > 0$ and $y > 0$. Then $x + y > 0$ and $xy > 0$*
5. *If $x > 0$, then $\frac{1}{x} > 0$*
6. *If $0 < x < y$, then $\frac{1}{y} < \frac{1}{x}$*
7. *If $0 < x < y$ and $0 < z < w$, then $xz < yw$*

Proof. (1) We want to show

$$-\bar{1} < \bar{0} < \bar{1}$$

We know $1 \cdot 1 = 1 > 0$. So by lemma 1.59 we have $\bar{0} < [\langle 1, 1 \rangle] = \bar{1}$. Since $\bar{0} < \bar{1}$, we have

$$-\bar{1} = \bar{0} + (-\bar{1}) < \bar{1} + (-\bar{1}) = \bar{0} \qquad \square$$

Proof. (2) We want to show

$$\forall x \in \mathbb{Q} \forall y \in \mathbb{Q} (x < y \to -y < -x)$$

Let arbitrary $x, y \in \mathbb{Q}$ and suppose $x < y$. We have

$$x - y < y - y = 0$$

and

$$-y = ((-x) + x) - y = (-x) + (x - y) < (-x) + 0 = -x \qquad \square$$

Proof. (3) We want to show

$$\forall x \in \mathbb{Q} (x \cdot \bar{0} = \bar{0})$$

Let arbitrary $x \in \mathbb{Q}$. Then there exists $\langle a, b \rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b \rangle]$. We then have

$$x \cdot \bar{0} = [\langle a, b \rangle] \cdot [\langle 0, 1 \rangle] = [\langle 0, b \rangle] = \bar{0} \qquad \square$$

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Proof. (4) We want to show

$$\forall x \in \mathbb{Q} \forall y \in \mathbb{Q}((x > 0 \wedge y > 0) \to (x + y > 0 \wedge xy > 0))$$

Let arbitrary $x, y \in \mathbb{Q}$. Suppose $x > 0$ and $y > 0$. It follows that $x+y > 0+y = y$, where by transitivity $x + y > 0$. It also follows that $xy > 0 \cdot y = 0$. □

Proof. (5) We want to show

$$\forall x \in \mathbb{Q} \left(x > \bar{0} \to \frac{1}{x} > \bar{0} \right)$$

Let arbitrary $x \in \mathbb{Q}$. Then there exists $\langle a, b \rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b \rangle]$. Suppose that $x > \bar{0}$. By trichotomy we then have $x \neq \bar{0}$ and therefore $a \neq 0$. So $a \in \mathbb{Z}^*$ and $x \in \mathbb{Q}^*$. We know $[\langle a, b \rangle] > \bar{0}$, so by lemma 1.59 $ba = ab > 0$, and again by the same lemma $x^{-1} = [\langle b, a \rangle] > \bar{0}$. So

$$\frac{1}{x} = 1 \cdot x^{-1} = x^{-1} = [\langle b, a \rangle] > \bar{0}$$ □

Proof. (6) We want to show

$$\forall x \in \mathbb{Q} \forall y \in \mathbb{Q} \left(0 < x < y \to \frac{1}{y} < \frac{1}{x} \right)$$

Let arbitrary $x, y \in \mathbb{Q}$ and suppose that $0 < x < y$. By (5) we have $\frac{1}{x} = x^{-1} > 0$ and $\frac{1}{y} = y^{-1} > 0$. By trichotomy we also know $x, y \neq 0$. So

$$1 = xx^{-1} < yx^{-1}$$

and

$$\frac{1}{y} = 1 \cdot y^{-1} = y^{-1} \cdot 1 < y^{-1}(yx^{-1}) = (y^{-1}y)x^{-1} = 1 \cdot x^{-1} = \frac{1}{x}$$ □

Proof. (7) We want to show

$$\forall x \in \mathbb{Q} \forall y \in \mathbb{Q} \forall z \in \mathbb{Q} \forall w \in \mathbb{Q}((0 < x < y \wedge 0 < z < w) \to xz < yw)$$

Let arbitrary $x, y, z, w \in \mathbb{Q}$. Suppose $0 < x < y$ and $0 < z < w$. We then have $xz < yz$ and $yz < yw$, where by transitivity it follows that $xz < yw$. □

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Theorem 1.61. *Defining 2 as* $1+1$*,*

1. $1 < 2$
2. *For* $x, y \in \mathbb{Q}$, $\frac{x+y}{2} \in \mathbb{Q}$*, and if* $x < y$ *then* $x < \frac{x+y}{2} < y$*.*

Proof. (1) We know $0 < 1$. We then have

$$1 = 0 + 1 < 1 + 1 = 2$$ □

Proof. (2) We want to show

$$\forall x \in \mathbb{Q} \forall y \in \mathbb{Q} \left(\frac{x+y}{2} \in \mathbb{Q} \wedge \left(x < y \to x < \frac{x+y}{2} < y \right) \right)$$

Let arbitrary $x, y \in \mathbb{Q}$. We know $2 > 1 > 0$ so by trichotomy $2 \neq 0$. Since $+ : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$ and $\div : \mathbb{Q} \times \mathbb{Q}^* \to \mathbb{Q}$ it follows that $\frac{x+y}{2} \in \mathbb{Q}$.

Suppose $x < y$. We have

$$x + y < y + y = y \cdot 1 + y \cdot 1 = y(1+1) = y \cdot 2$$

and since $2 > 0$ we know $2^{-1} = \frac{1}{2} > 0$. So

$$\frac{x+y}{2} = (x+y)2^{-1} < (y \cdot 2)2^{-1} = y(2 \cdot 2^{-1}) = y \cdot 1 = y$$

Similarly we have

$$x \cdot 2 = x + x < x + y$$

and then

$$x = x(2 \cdot 2^{-1}) = (x \cdot 2)2^{-1} < (x+y)2^{-1} = \frac{x+y}{2}$$ □

Lemma 1.62.

$$\forall x \in \mathbb{Q} \forall a \in \mathbb{Z} \forall b \in \mathbb{Z} ((x > 0 \wedge x = \frac{i(a)}{i(b)} \wedge b \neq 0) \to ((a > 0 \wedge b > 0) \vee (a < 0 \wedge b < 0)))$$

Proof. Let arbitrary $x \in \mathbb{Q}$ and $a, b \in \mathbb{Z}$. Suppose $x > 0$, $x = \frac{i(a)}{i(b)}$ and $b \neq 0$. By trichotomy on the integers we have two cases.

Case 1: $b > 0$. Then since $\frac{i(a)}{i(b)} = x > 0 = [\langle 0, 1 \rangle] = \frac{i(0)}{i(1)}$ we have $0 = 0 \cdot b < 1 \cdot a = a$. So $a > 0 \wedge b > 0$.

Case 2: $b < 0$. $\frac{i(a)}{i(b)} > \frac{i(0)}{i(1)}$ then implies $0 = 0 \cdot b > 1 \cdot a = a$. So $a < 0$ and $b < 0$. □

(next page)

Lemma 1.63.

$$\forall x \in \mathbb{Q}(x > 0 \to \exists a \in \mathbb{Z} \exists b \in \mathbb{Z}(x = \frac{i(a)}{i(b)} \wedge a > 0 \wedge b > 0))$$

Proof. Let arbitrary $x \in \mathbb{Q}$. Suppose $x > 0$. By definition there exists $\langle a, b \rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b \rangle] = \frac{i(a)}{i(b)}$. By the previous lemma it follows that $(a > 0 \wedge b > 0) \vee (a < 0 \wedge b < 0)$. In the former case there is nothing more to prove, so suppose $a < 0$ and $b < 0$. We then know $-a > 0$ and $-b > 0$. Since $(-a)b = (-b)a$ we then have

$$\frac{i(-a)}{i(-b)} = \frac{i(a)}{i(b)} = x > 0$$ □

Lemma 1.64.

$$\forall x \in \mathbb{Z}((x > 0 \to x \geq 1) \wedge (x < 0 \to x \leq -1))$$

Proof. Let arbitrary $x \in \mathbb{Z}$. Suppose $x > 0$. For contradiction suppose $x < 1$. It follows then that $0 < x < 1$, which contradicts integer discreteness, so by trichotomy $x \geq 1$. Suppose $x < 0$. For contradiction suppose $x > -1$. Similarly we then have $-1 < x < 0 = (-1) + 1$, the contradiction. So by trichotomy $x \leq -1$. □

Lemma 1.65.

$$\forall x \in \mathbb{Q} \forall y \in \mathbb{Q}((x > 0 \wedge y > 0) \to \exists n \in \mathbb{N}(y < i(j(n))x))$$

Where i refers to the integer-rational embedding in theorem 1.55 and j the natural-integer embedding in theorem 1.27.

Proof. Let arbitrary $x, y \in \mathbb{Q}$. Suppose $x > 0$ and $y > 0$. By lemma 1.63 there exists $a, b, c, d \in \mathbb{Z}$ where $a, b, c, d > 0$, $x = \frac{i(a)}{i(b)}$, and $y = \frac{i(c)}{i(d)}$. We know $cb > 0$, so $cb + 1 > 0$. We also know $cb + 1 \in \mathbb{Z}$, so by theorem 1.27 there then exists $n \in \mathbb{N}$ such that $j(n) = cb + 1$. We then know $i(j(n)) = i(cb + 1)$, and

$$i(j(n))x = i(cb + 1) \cdot \frac{i(a)}{i(b)} = \frac{i(cb + 1)i(a)}{i(b)} = \frac{i((cb + 1)a)}{i(b)}$$

To show $\frac{i((cb+1)a)}{i(b)} = i(j(n))x > y = \frac{i(c)}{i(d)}$. We know $b > 0$ and $d > 0$, so it is then sufficient to show $cb < d((cb + 1)a)$. We know $ad > 0$ so $ad \geq 1$ and

$$d((cb + 1)a) = (ad)(cb + 1) \geq cb + 1 > cb$$ □

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Lemma 1.66.

$$\forall x \in \mathbb{Q} \left(x > 0 \to \exists n \in \mathbb{N} \left(\frac{1}{i(j(n))} < x \right) \right)$$

Proof. Let arbitrary $x \in \mathbb{Q}$. Suppose $x > 0$. We know $1 > 0$ so by the previous lemma there exists $n \in \mathbb{N}$ such that $i(j(n))x > 1$. By theorem 1.27 we know that $j(n) > 0$, so $i(j(n)) \neq 0$ and $i(j(n)) \in \mathbb{Q}^*$, so $(i(j(n)))^{-1}$ is defined. Since $j(n) > 0$ we also have $0 = i(0) < i(j(n))$, so $(i(j(n)))^{-1} = \frac{1}{i(j(n))} > 0$. We then have

$$\frac{1}{i(j(n))} = 1 \cdot (i(j(n)))^{-1} < (i(j(n)) \cdot x) \cdot (i(j(n)))^{-1} = x \cdot (i(j(n)) \cdot (i(j(n)))^{-1}) = x$$

and we are done. □

Lemma 1.67. *For each $x \in \mathbb{Q}$, let x^2 denote $x \cdot x$.*

$$\forall x \in \mathbb{Q} \forall y \in \mathbb{Q} \left((x > 0 \wedge y > 0 \wedge x^2 < y) \to \exists n \in \mathbb{N} \left(\left(x + \frac{1}{i(j(n))} \right)^2 < y \right) \right)$$

Proof. Let arbitrary $x, y \in \mathbb{Q}$. Suppose $x, y > 0$ and $x^2 < y$. By the latter assumption we have $y - x^2 > 0$. We also have $2x + 1 > 0$, so $(2x+1)^{-1} = \frac{1}{2x+1} > 0$ is defined. We then have $(y - x^2)(2x+1)^{-1} > 0$, so by the previous lemma there exists $n \in \mathbb{N}$ such that

$$\frac{1}{i(j(n))} < (y - x^2)(2x+1)^{-1}$$

Since integer $j(n) > 0$, we know $j(n) \geq 1$ and therefore $i(j(n)) \geq 1$. Since $i(j(n)) > 0$ we also have $1 \geq \frac{1}{i(j(n))}$ and $\frac{1}{i(j(n))} > 0$. So

$$\frac{1}{i(j(n))} \geq \left(\frac{1}{i(j(n))} \right)^2$$

We then have

$$\begin{aligned} \left(x + \frac{1}{i(j(n))} \right)^2 &= x^2 + 2x\frac{1}{i(j(n))} + \left(\frac{1}{i(j(n))} \right)^2 \leq x^2 + 2x\frac{1}{i(j(n))} + \frac{1}{i(j(n))} \\ &= x^2 + (2x+1)\frac{1}{i(j(n))} \end{aligned}$$

Where since we know $(2x+1)\frac{1}{i(j(n))} < y - x^2$,

$$\left(x + \frac{1}{i(j(n))} \right)^2 \leq x^2 + (2x+1)\frac{1}{i(j(n))} < x^2 + (y - x^2) = y$$ □

(next page)

Theorem 1.68.

$$\forall x \in \mathbb{Q}(-(-x) = x)$$

Proof. Let arbitrary $x \in \mathbb{Q}$. There exists $\langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b\rangle]$. We have $-x = [\langle -a, b\rangle]$ and

$$-(-x) = [\langle -(-a), b\rangle] = [\langle a, b\rangle] = x \qquad \square$$

Theorem 1.69.

$$\forall x \in \mathbb{Q}^* \forall y \in \mathbb{Q}^* \left(\left(\frac{x}{y} \right)^{-1} = \frac{y}{x} \right)$$

Proof. Let arbitrary $x, y \in \mathbb{Q}^*$. There exists $\langle a, b\rangle, \langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b\rangle]$ and $y = [\langle c, d\rangle]$. y^{-1} is defined as $[\langle d, c\rangle]$. We have

$$\frac{x}{y} = xy^{-1} = [\langle a, b\rangle] \cdot [\langle d, c\rangle] = [\langle ad, bc\rangle]$$

Since $x \neq 0$ we know $a \neq 0$, so $ad \neq 0$, $\frac{x}{y} \neq 0$, and $\left(\frac{x}{y}\right)^{-1}$ is defined. So

$$\begin{aligned} \left(\frac{x}{y} \right)^{-1} &= [\langle ad, bc\rangle]^{-1} = [\langle bc, ad\rangle] = [\langle cb, da\rangle] = [\langle c, d\rangle] \cdot [\langle b, a\rangle] \\ &= [\langle c, d\rangle] \cdot [\langle a, b\rangle]^{-1} = yx^{-1} = \frac{y}{x} \end{aligned} \qquad \square$$

Theorem 1.70.

$$\forall x \in \mathbb{Q}^*((x^{-1})^{-1} = x)$$

Proof. Let arbitrary $x \in \mathbb{Q}^*$. There exists $\langle a, b\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b\rangle]$. We have $x^{-1} = [\langle b, a\rangle]$ and

$$(x^{-1})^{-1} = [\langle b, a\rangle]^{-1} = [\langle a, b\rangle] = x \qquad \square$$

Theorem 1.71.

$$\forall a \in \mathbb{Z} \forall b \in \mathbb{Z}^* \left([\langle a, b\rangle] = \frac{i(a)}{i(b)} \right)$$

Proof. Let arbitrary $a \in \mathbb{Z}$ and $b \in \mathbb{Z}^*$. By definition $i(a) = [\langle a, 1\rangle]$ and $i(b) = [\langle b, 1\rangle]$. Since $b \neq 0$, $i(b) \neq 0$ and $i(b) \in \mathbb{Q}^*$. So

$$\frac{i(a)}{i(b)} = [\langle a, 1\rangle] \cdot [\langle b, 1\rangle]^{-1} = [\langle a, 1\rangle] \cdot [\langle 1, b\rangle] = [\langle a \cdot 1, 1 \cdot b\rangle] = [\langle a, b\rangle] \qquad \square$$

(next page)

Theorem 1.72. *For $x, y, z \in \mathbb{Q}$,*

1. $x \leq y \to x + z \leq y + z$
2. $x \leq y \wedge z > 0 \to xz \leq yz$
3. $x \leq y \to -y \leq -x$
4. $0 < x \leq y \to \frac{1}{y} \leq \frac{1}{x}$

Proof. (1) Suppose $x \leq y$. In the case where $x = y$ we have $x+z = y+z$. In the case where $x < y$, we have $x + z < y + z$. Either case implies $x + z \leq y + z$. □

Proof. (2) Suppose $x \leq y$ and $z > 0$. In the case where $x = y$ we have $xz = yz$. In the case where $x < y$, since $z > 0$ we have $xz < yz$. Either case implies $xz \leq yz$. □

Proof. (3) Suppose $x \leq y$. In the case where $x = y$ we have $-x = -y$. In the case where $x < y$, we have $-y < -x$. Either case implies $-y \leq -x$. □

Proof. (4) Suppose $0 < x \leq y$. In the case where $x = y$ we have $\frac{1}{x} = \frac{1}{y}$. In the case where $x < y$, since $0 < x < y$, we have $\frac{1}{y} < \frac{1}{x}$. Either case implies $\frac{1}{y} \leq \frac{1}{x}$. □

Theorem 1.73.

$$\forall x \in \mathbb{Q} \forall y \in \mathbb{Q}(-(xy) = (-x)y)$$

Proof. Let arbitrary $x, y \in \mathbb{Q}$. There exists $\langle a, b\rangle, \langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b\rangle]$ and $y = [\langle c, d\rangle]$. We have $xy = [\langle ac, bd\rangle]$ and therefore

$$-(xy) = [\langle -(ac), bd\rangle] = [\langle (-a)c, bd\rangle] = [\langle -a, b\rangle] \cdot [\langle c, d\rangle] = (-x)y \qquad \square$$

Theorem 1.74.

$$\forall x \in \mathbb{Q} \forall y \in \mathbb{Q}(-(x + y) = (-x) + (-y))$$

Proof. Let arbitrary $x, y \in \mathbb{Q}$. There exists $\langle a, b\rangle, \langle c, d\rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $x = [\langle a, b\rangle]$ and $y = [\langle c, d\rangle]$. We have $x + y = [\langle ad + bc, bd\rangle]$ and therefore

$$\begin{aligned} -(x + y) &= [\langle -(ad + bc), bd\rangle] = [\langle -(ad) + (-(bc)), bd\rangle] = [\langle (-a)d + b(-c), bd\rangle] \\ &= [\langle -a, b\rangle] + [\langle -c, d\rangle] = (-x) + (-y) \end{aligned}$$ □

(next page)

Lemma 1.75.

$$\forall a \in \mathbb{Q}^* \forall b \in \mathbb{Q}^* (ab \in \mathbb{Q}^* \wedge a^{-1}b^{-1} = (ab)^{-1})$$

Proof. Let arbitrary $a, b \in \mathbb{Q}^*$. Then there exists $\langle a_1, a_2 \rangle, \langle b_1, b_2 \rangle \in \mathbb{Z} \times \mathbb{Z}^*$ such that $a = [\langle a_1, a_2 \rangle]$ and $b = [\langle b_1, b_2 \rangle]$. Since $a, b \neq 0$, we have $a_1, b_1 \neq 0$ and $a_1 b_1 \neq 0$, meaning $ab = [\langle a_1 b_1, a_2 b_2 \rangle] \neq 0$ and $ab \in \mathbb{Q}^*$. Since $a_2, b_2 \neq 0$ we also have $a_2 b_2 \neq 0$. So

$$a^{-1}b^{-1} = [\langle a_2, a_1 \rangle] \cdot [\langle b_2, b_1 \rangle] = [\langle a_2 b_2, a_1 b_1 \rangle] = [\langle a_1 b_1, a_2 b_2 \rangle]^{-1} = (ab)^{-1} \quad \square$$

Theorem 1.76. *Let $a, c \in \mathbb{Q}$ and $b, d \in \mathbb{Q}^*$.*

1. $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$
2. $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$

Proof. (1) We have

$$\begin{aligned} \frac{a}{b} + \frac{c}{d} &= ab^{-1} + cd^{-1} = (ab^{-1} + cd^{-1})((bd)(bd)^{-1}) \\ &= ((ab^{-1} + cd^{-1})(bd))(bd)^{-1} = ((ab^{-1})(bd) + (cd^{-1})(bd))(bd)^{-1} \\ &= (ad + bc)(bd)^{-1} = \frac{ad + bc}{bd} \end{aligned}$$

$\square$

Proof. (2) We have

$$\frac{a}{b} \cdot \frac{c}{d} = (ab^{-1})(cd^{-1}) = (ac)(b^{-1}d^{-1}) = (ac)(bd)^{-1} = \frac{ac}{bd} \quad \square$$

Theorem 1.77. $0_{\mathbb{Q}} = -0_{\mathbb{Q}}$

Proof. We have

$$0 = 0 \cdot (-1) = (-0) \cdot 1 = -0 \quad \square$$

1.4 Dedekind Cuts

1.4.1 Definitions

Definition 1.78. *Let $A \subseteq \mathbb{Q}$ be a set. The set A is a **Dedekind cut** if the following three properties hold.*

1. *$A \neq \emptyset$ and $A \neq \mathbb{Q}$.*
2. *Let $x \in A$. If $y \in \mathbb{Q}$ and $y \geq x$, then $y \in A$.*
3. *Let $x \in A$. Then there is some $y \in A$ such that $y < x$.*

Lemma 1.79. *Let $r \in \mathbb{Q}$. The set*

$$D = \{x \in \mathbb{Q} \mid x > r\}$$

That is, the set D where

$$\forall x(x \in D \leftrightarrow (x \in \mathbb{Q} \wedge x > r))$$

is a Dedekind cut.

Proof. We need to show that D satisfies the three properties defining a dedekind cut. We know $r + 1 \in \mathbb{Q}$. Since $r + 1 > r$ by definition $r + 1 \in D$, so $D \neq \emptyset$. We also know $r \not< r$, so $r \notin D$, meaning $D \neq \mathbb{Q}$.

Next let arbitrary $x \in D$. By definition $x \in \mathbb{Q}$ and $x > r$. Let $y \in \mathbb{Q}$ be arbitrary such that $y \geq x$. We have $x < y$ or $x = y$. In the former case $y > r$ follows by transitivity. Since $y > r$ in both cases, by definition $y \in D$.

Finally let arbitrary $x \in D$. By definition $x \in \mathbb{Q}$ and $x > r$. We know that $\frac{x+r}{2} \in \mathbb{Q}$ and $r < \frac{x+r}{2} < x$. So $\frac{x+r}{2} \in D$ and $\frac{x+r}{2} < x$. □

Definition 1.80. *Let $r \in \mathbb{Q}$. The **rational cut** at r, denoted D_r, is the Dedekind cut*

$$D_r = \{x \in \mathbb{Q} \mid x > r\}$$

*An **irrational cut** is a Dedekind cut that is not a rational cut at any rational number.*

Lemma 1.81. *Let $A \subseteq \mathbb{Q}$ be a Dedekind cut.*

1. *$\mathbb{Q} - A = \{x \in \mathbb{Q} \mid x < a$ for all $a \in A\}$*
2. *Let $x \in \mathbb{Q} - A$. If $y \in \mathbb{Q}$ and $y \leq x$, then $y \in \mathbb{Q} - A$*

Proof. (1) Let arbitrary $x \in \mathbb{Q} - A$. Let arbitrary $a \in A$. By trichotomy $x > a \vee x = a \vee x < a$. In the case where $x = a$ we have $x \in A$, which is a contradiction. In the case where $x > a$, since A is a dedekind cut this would imply $x \in A$, again a contradiction. We must then have $x < a$.

Next let arbitrary $x \in \{x \in \mathbb{Q} \mid x < a$ for all $a \in A\}$, meaning $\forall a \in A(x < a)$. Suppose for contradiction that $x \in A$. Since A is a dedekind cut there then exists $y \in A$ such that $y < x$, contradicting $x < y$. □

(next page)

Proof. (2) We have arbitrary $x \in \mathbb{Q} - A$, and arbitrary $y \in \mathbb{Q}$ such that $y \leq x$. Suppose for contradiction that $y \in A$. Since A is a dedekind cut, the fact that $y \leq x$ would imply that $x \in A$, the contradiction. □

Lemma 1.82.

$$\forall A \forall B(A \neq B \to (A - B \neq \emptyset \vee B - A \neq \emptyset))$$

Proof. Let A, B be arbitrary and suppose that $A \neq B$. This means $\neg\forall x(x \in A \leftrightarrow x \in B)$, so there then exists x such that $x \in A \wedge x \notin B$ or $x \in B \wedge x \notin A$, meaning $x \in A - B$ or $x \in B - A$. □

Lemma 1.83. *Let $A, B \subseteq \mathbb{Q}$ be Dedekind cuts. Then precisely one of $A \subsetneqq B$ or $A = B$ or $B \subsetneqq A$ holds. (where $A \subsetneqq B \to (A \subseteq B \wedge A \neq B)$)*

Proof. If $A = B$ there is nothing to prove, so suppose $A \neq B$. We then have two cases. First suppose there exists $x \in A - B$. It follows that $x \in \mathbb{Q} - B$ and by lemma 1.81 that $\forall b \in B(x < b)$. Letting arbitrary $b \in B$, we will then have $x < b$. Since A is a dedekind cut, it then follows that $b \in A$ and therefore $B \subseteq A$. Since $A \neq B$ we then have $B \subsetneqq A$.

Next suppose there exists $x \in B - A$. Similarly it follows that $x \in \mathbb{Q} - A$ and $\forall a \in A(x < a)$. Letting arbitrary $a \in A$, we have $x < a$ and by definition $a \in B$. So $A \subsetneqq B$.

If $A = B$ by definition neither $A \subsetneqq B$ nor $B \subsetneqq A$ can be true. Suppose $A \subsetneqq B$ and $B \subsetneqq A$. It follows that $A \neq B$. However $A \subseteq B$ and $B \subseteq A$ together imply $A = B$, a contradiction. □

Lemma 1.84. *Let A be a non-empty family of subsets of $\mathbb{Q}$ and suppose that for all $X \in A$ X is a dedekind cut. If $\bigcup A \neq \mathbb{Q}$, then $\bigcup A$ is a dedekind cut.*

Proof. Suppose $\bigcup A \neq \mathbb{Q}$. It is straightforward to show that $\bigcup A \subseteq \mathbb{Q}$. Next see that since A is nonempty then there exists $X \in A$ that is, by assumption, a dedekind cut. It follows then that $X \neq \emptyset$ and that $\bigcup A$, where

$$\forall x(x \in \bigcup A \leftrightarrow \exists X \in A(x \in X))$$

is nonempty. By assumption we know $\bigcup A \neq \mathbb{Q}$.

Next let arbitrary $x \in \bigcup A$. Let arbitrary $y \in \mathbb{Q}$ and suppose that $y \geq x$. By definition there exists $X \in A$ such that $x \in X$. Since by assumption X is a dedekind cut, it then follows that $y \in X$, and then $y \in \bigcup A$.

Finally let arbitrary $x \in \bigcup A$. Again by definition there exists $X \in A$ such that $x \in X$, where X is a dedekind cut. So there exists $y \in X$ such that $y < x$, and by definition $y \in \bigcup A$. □

1.4.2 More properties

Lemma 1.85. *Let $A, B \subseteq \mathbb{Q}$ be Dedekind cuts.*

1. *The set*
$$\{r \in \mathbb{Q} \mid r = a + b \text{ for some } a \in A \text{ and } b \in B\}$$
is a Dedekind cut.

2. *The set*
$$\{r \in \mathbb{Q} \mid -r < c \text{ for some } c \in \mathbb{Q} - A\}$$
is a Dedekind cut.

3. *Suppose $0 \in \mathbb{Q} - A$ and $0 \in \mathbb{Q} - B$. The set*
$$\{r \in \mathbb{Q} \mid r = ab \text{ for some } a \in A \text{ and } b \in B\}$$
is a Dedekind cut.

4. *Suppose that there is some $q \in \mathbb{Q} - A$ such that $q > 0$. The set*
$$\{r \in \mathbb{Q} \mid r > 0 \text{ and } \frac{1}{r} < c \text{ for some } c \in \mathbb{Q} - A\}$$
is a Dedekind cut.

Proof. (1) Let $D = \{r \in \mathbb{Q} \mid r = a + b \text{ for some } a \in A \text{ and } b \in B\}$, that is, the set such that

$$\forall x(x \in D \leftrightarrow x \in \mathbb{Q} \wedge \exists a \in A \exists b \in B(x = a + b))$$

It is straightforward to show that $D \subseteq \mathbb{Q}$. Since A and B are dedekind cuts there exists $x \in A$ and $y \in B$; so by definition $x + y \in D$ and D is nonempty. We also know $A, B \neq \mathbb{Q}$. Since $A, B \subseteq \mathbb{Q}$, we must then have $A - \mathbb{Q}, B - \mathbb{Q} = \emptyset$, and therefore $\mathbb{Q} - A, \mathbb{Q} - B \neq \emptyset$. There then exists $x \in \mathbb{Q} - A = \{x \in \mathbb{Q} \mid \forall a \in A(x < a)\}$ and $y \in \mathbb{Q} - B = \{x \in \mathbb{Q} \mid \forall b \in B(x < b)\}$. We show $\neg \exists a \in A \exists b \in B(x = a + b)$. Let arbitrary $a \in A$ and $b \in B$. By definition we have $x < a$ and $y < b$, so $x + y < a + b$, $x + y \neq a + b$, and $x + y \notin D$. Since $x + y \in \mathbb{Q}$ it follows that $D \neq \mathbb{Q}$.

Next let arbitrary $x \in D$ and $y \in \mathbb{Q}$. Suppose $y \geq x$. By definition there exists $a \in A$ and $b \in B$ such that $x = a + b$. We have $(a + b) - x = 0$ and $y = ((a + b) - x) + y = (a + (y - x)) + b$. Since $y \geq x = a + b$ we then have $a + (y - x) \geq a$, so since A is a dedekind cut $a + (y - x) \in A$. So by definition $y = (a + (y - x)) + b \in D$.

Finally let arbitrary $x \in D$. There exists $a \in A$ and $b \in B$ such that $x = a + b$. Since $a \in A$ there exists $a' \in A$ such that $a' < a$. It follows that $a' + b < a + b = x$, where by definition $a' + b \in D$. □

(next page)

Proof. (2) Let $D = \{r \in \mathbb{Q} \mid -r < c \text{ for some } c \in \mathbb{Q} - A\}$. That is, the set such that

$$\forall x(x \in D \leftrightarrow x \in \mathbb{Q} \wedge \exists c \in \mathbb{Q} - A(-x < c))$$

It is straightforward to show that $D \subseteq \mathbb{Q}$. Since A is a dedekind cut, $A \subseteq \mathbb{Q}$ and $A \neq \mathbb{Q}$. So $\mathbb{Q} - A \neq \emptyset$ and there exists $x \in \mathbb{Q} - A$. Since $x - 1 < x$, by definition $-(x-1) \in D$ and $D \neq \emptyset$.

Next, since A is nonempty, there exists $x \in A$, so $-(-x) \in A$ and so $-(-x) \notin \mathbb{Q} - A$. Let arbitrary $c \in \mathbb{Q} - A$. We then have $\neg(-(-x) \leq c)$, meaning $\neg(-(-x) < c)$. So $-x \notin D$ and $-x \in \mathbb{Q}$, meaning $D \neq \mathbb{Q}$.

Next let arbitrary $x \in D$ and $y \in \mathbb{Q}$. By definition there exists $c \in \mathbb{Q} - A$ such that $-x < c$. Suppose $x \leq y$. It follows that $-y \leq -x$ and $-y < c$. So $y \in D$.

Finally, let arbitrary $x \in D$, where by definition there exists $c \in \mathbb{Q} - A$ such that $-x < c$. We then have $-c < x$ and $-c < \frac{-c+x}{2} < x$. So $-\left(\frac{-c+x}{2}\right) < c$, meaning $\frac{-c+x}{2} \in D$, while $\frac{-c+x}{2} < x$. □

Proof. (3) Suppose $0 \in \mathbb{Q} - A$ and $0 \in \mathbb{Q} - B$. Let $D = \{r \in \mathbb{Q} \mid r = ab \text{ for some } a \in A \text{ and } b \in B\}$, that is, the set such that

$$\forall x(x \in D \leftrightarrow x \in \mathbb{Q} \wedge \exists a \in A \exists b \in B(x = ab))$$

It is straightforward to show $D \subseteq \mathbb{Q}$. Since A and B are dedekind cuts there exists $a \in A$ and $b \in B$, where by definition $ab \in D$. So $D \neq \emptyset$. Let arbitrary $a \in A$ and $b \in B$. Since $0 \in \mathbb{Q} - A$ and $0 \in \mathbb{Q} - B$ we know $a > 0$ and $b > 0$. So $ab > 0$, where by trichotomy $ab \neq 0$ and $0 \notin D$, so $D \neq \mathbb{Q}$.

Next let arbitrary $x \in D$, so there exists $a \in A$ and $b \in B$ such that $x = ab$. Let arbitrary $y \in \mathbb{Q}$ and suppose that $y \geq x$. In the case where $y = x$ it immediately follows that $y \in D$. In the case where $y > x$ we have $y > ab$. Since $0 \in \mathbb{Q} - A$ and $a \in A$, we have $0 < a$. It also follows that $a^{-1} > 0$ is defined. Then $\frac{y}{a} > b$ and by definition $\frac{y}{a} \in B$. Since $y = a \cdot \frac{y}{a}$, $y \in D$.

Finally let arbitrary $x \in D$, where there exists $a \in A$ and $b \in B$ such that $x = ab$. Since A is a dedekind cut there exist $a' \in A$ such that $a' < a$. Since $0 \in \mathbb{Q} - B$ and $b \in B$, we have $0 < b$. So $a'b < ab = x$, where $a'b \in D$. □

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Proof. (4) Suppose there exists $q \in \mathbb{Q} - A$ such that $q > 0$. Let $D = \{r \in \mathbb{Q} \mid r > 0 \text{ and } \frac{1}{r} < c \text{ for some } c \in \mathbb{Q} - A\}$. That is,

$$\forall x \left(x \in D \leftrightarrow x \in \mathbb{Q} \wedge x > 0 \wedge \exists c \in \mathbb{Q} - A \left(\frac{1}{x} < c \right) \right)$$

It is straightforward to show that $D \subseteq \mathbb{Q}$. We know $q > 0$, so $q > \frac{q}{2} > 0$. We know $q^{-1} > 0$ so $\frac{2}{q} = 2q^{-1} > 0$. We also have $q > \frac{q}{2} = \left(\frac{2}{q}\right)^{-1} = 1/(\frac{2}{q})$. So $\frac{2}{q} \in D$ and $D \neq \emptyset$. We know $0 \in \mathbb{Q}$ and $0 \not< 0$, so $0 \notin D$ and $D \neq \mathbb{Q}$.

Next let arbitrary $x \in D$. It follows that $x > 0$ and that there exists $c \in \mathbb{Q} - A$ such that $\frac{1}{x} < c$. Let arbitrary $y \in \mathbb{Q}$ and suppose that $y \geq x$. In the case where $y = x$ it immediately follows that $y \in D$. In the case where $y > x$ we have $\frac{1}{y} < \frac{1}{x} < c$ and $y > 0$, so $y \in D$.

Finally let arbitrary $x \in D$, where then $x > 0$ and that there exists $c \in \mathbb{Q} - A$ such that $\frac{1}{x} < c$. We know $\frac{1}{x} > 0$, so $c > 0$ and $0 < \frac{1}{c} < x$. We then know

$$0 < \frac{1}{c} < \frac{x + \frac{1}{c}}{2} < x$$

where

$$1/\left(\frac{x + \frac{1}{c}}{2}\right) < c$$

So $\frac{x+\frac{1}{c}}{2} \in D$. □

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Theorem 1.86. *Let $A \subseteq \mathbb{Q}$ be a Dedekind cut. Let $y \in \mathbb{Q}$.*

1. *Suppose that $y > 0$. Then there are $u \in A$ and $v \in \mathbb{Q} - A$ such that $y = u - v$, and $v < e$ for some $e \in \mathbb{Q} - A$.*

2. *Suppose that $y > 1$, and that there is some $q \in \mathbb{Q} - A$ such that $q > 0$. Then there are $r \in A$ and $s \in \mathbb{Q} - A$ such that $s > 0$, and $y > \frac{r}{s}$, and $s < g$ for some $g \in \mathbb{Q} - A$.*

Proof. (1) We want to show

$$\exists u \in A \exists v \in \mathbb{Q} - A(y = u - v \wedge \exists e \in \mathbb{Q} - A(v < e))$$

Since A is a dedekind cut, there exists $w \in A$ and $z \in \mathbb{Q} - A$. By lemma 1.81 we have $z < w$, so $w - z > 0$. We know $y > 0$, so by lemma 1.65 there exists $n \in \mathbb{N}$ such that $w - z < i(j(n))y$. We then have $w + i(j(n))(-y) = w - (i(j(n))y) < z$. By lemma 1.81 it follows that $w + i(j(n))(-y) \in \mathbb{Q} - A$. We now define G as

$$G = \{p \in \mathbb{N} \mid w + i(j(p))(-y) \in \mathbb{Q} - A\}$$

By definition $n \in G$, so $G \neq \emptyset$ and by well-ordering there exists $m \in G$ such that for $\forall g \in G(m \leq g)$. By definition $w + i(j(m))(-y) \in \mathbb{Q} - A$.

We first show $w + (i(j(m)) - 1)(-y) \notin \mathbb{Q} - A$. Suppose for contradiction that this were true. We know $i(j(m)) - 1 = i(j(m)) + i(-1) = i(j(m) - 1)$. We have 3 cases. First suppose $j(m) - 1 < 0$. In this case it follows that $0 < j(m) < 1$, contradicting integer discreteness. Next suppose $j(m) - 1 = 0$. Then $i(j(m) - 1) = 0$, and $w = w + i(j(m) - 1)(-y) = w + (i(j(m)) - 1)(-y) \in \mathbb{Q} - A$. This contradicts $w \in A$. Finally suppose $j(m) - 1 > 0$. Then there exists $m' \in \mathbb{N}$ such that $j(m') = j(m) - 1$. Since $j(m) - 1 < j(m)$, $j(m') < j(m)$ and $m' < m$. We have $w + i(j(m'))(-y) = w + i(j(m) - 1)(-y) = w + (i(j(m)) - 1)(-y) \in \mathbb{Q} - A$ and therefore $m' \in G$. Since $m' < m$ this contradicts m being the smallest element in G.

So $w + (i(j(m)) - 1)(-y) \notin \mathbb{Q} - A$. We must then have $w + (i(j(m)) - 1)(-y) \in A$. We know $w + i(j(m))(-y) \in \mathbb{Q} - A$. Finally we have $(w + (i(j(m)) - 1)(-y)) - (w + i(j(m))(-y)) = y$. We have 2 cases now.

First suppose $\exists e \in \mathbb{Q} - A(w + i(j(m))(-y) < e)$. Taking $w + (i(j(m)) - 1)(-y)$ and $w + i(j(m))(-y)$ as witnesses, our desired conclusion follows.

Next suppose this were not the case. Then $\forall e \in \mathbb{Q} - A(w + i(j(m))(-y) \geq e)$. We know $\frac{y}{2} > 0$. So $(w + i(j(m))(-y)) + \frac{y}{2} > w + i(j(m))(-y)$. Then by our case assumption we must have $(w + i(j(m))(-y)) + \frac{y}{2} \notin \mathbb{Q} - A$ and $(w + i(j(m))(-y)) + \frac{y}{2} \in A$. We know $-\frac{y}{2} < 0$, so $(w + i(j(m))(-y)) - \frac{y}{2} < w + i(j(m))(-y) \in \mathbb{Q} - A$ and by lemma 1.81 we have $(w + i(j(m))(-y)) - \frac{y}{2} \in \mathbb{Q} - A$. Taking $e = w + i(j(m))(-y)$, $u = (w + i(j(m))(-y)) + \frac{y}{2}$, and $v = (w + i(j(m))(-y)) - \frac{y}{2}$, we have $u - v = y$ and $v < e$. □

(next page)

Proof. (2) We want to show

$$\exists r \in A \exists s \in \mathbb{Q} - A(s > 0 \wedge y > \frac{r}{s} \wedge \exists g \in \mathbb{Q} - A(s < g))$$

We know $y > 1$ and $q > 0$. So $y - 1 > 0$ and $(y-1)\frac{q}{2} > 0$. By part 1 there then exists $u \in A$ and $v, e \in \mathbb{Q} - A$ such that $(y-1)\frac{q}{2} = u - v$ and $v < e$. We have 2 cases.

First suppose $v > \frac{q}{2}$. It follows that $(y-1)v > (y-1)\frac{q}{2} = u - v$. So $yv > u$ and, since $v > \frac{q}{2} > 0$, $y > \frac{u}{v}$. We know $v < e$. So using u, v, e as witnesses we have our desired result.

Next suppose $v \leq \frac{q}{2}$. We know $2 < 3 < 4$, so $\frac{1}{2} < \frac{3}{4} < \frac{4}{4} = 1$. Denoting $r = u + (\frac{3q}{4} - v)$ and $s = \frac{3q}{4}$, we know $v \leq \frac{q}{2} < \frac{3q}{4}$, so $\frac{3q}{4} - v > 0$ and $u + (\frac{3q}{4} - v) > u \in A$. Since A is a dedekind cut $r = u + (\frac{3q}{4} - v) \in A$. We also have $\frac{3q}{4} < q \in \mathbb{Q} - A$, so by lemma 1.81, $s = \frac{3q}{4} \in \mathbb{Q} - A$. This gives us

$$r - s = (u + (\frac{3q}{4} - v)) - \frac{3q}{4} = u - v = (y-1)\frac{q}{2} < (y-1)\frac{3q}{4} = (y-1)s$$

so $ys > r$. Since $s = \frac{3q}{4} > 0$, $y > \frac{r}{s}$; where $s < q \in \mathbb{Q} - A$. □

1.5 Construction of the Reals

1.5.1 Definitions and Operations

Definition 1.87. *The set of* ***real numbers****, denoted* $\mathbb{R}$*, is defined by*

$$\mathbb{R} = \{A \subseteq \mathbb{Q} \mid A \text{ is a Dedekind cut}\}$$

Definition 1.88. *The binary operation* $+$ *on* $\mathbb{R}$ *is defined for* $A, B \in \mathbb{R}$ *as*

$$A + B = \{r \in \mathbb{Q} \mid r = a + b \text{ for some } a \in A \text{ and } b \in B\}$$

That is, the function $+ : (\mathbb{R} \times \mathbb{R}) \to \mathbb{R}$*,*

$$\begin{aligned}&\forall x(x \in + \leftrightarrow \exists A \in \mathbb{R} \exists B \in \mathbb{R}(\\ &\quad x = \langle\langle A, B\rangle, \{r \in \mathbb{Q} \mid r = a + b \text{ for some } a \in A \text{ and } b \in B\}\rangle))\end{aligned}$$

Where $\{r \in \mathbb{Q} \mid r = a + b \text{ for some } a \in A \text{ and } b \in B\}$ *is the set* D *such that*

$$\forall x(x \in D \leftrightarrow x \in \mathbb{Q} \wedge \exists a \in A \exists b \in B(x = a + b))$$

(Existence follows from using $(\mathbb{R} \times \mathbb{R}) \times \mathbb{R}$ as a bounding set.)

Proof. (Range/Domain bounds, Functionality) First the domain bound $\text{dom}(+) = \mathbb{R} \times \mathbb{R}$. Let arbitrary $x \in \text{dom}(+)$. Then there exists y such that $\langle x, y\rangle \in +$. So by definition there exists $A, B \in \mathbb{R}$ such that $x = \langle A, B\rangle$. So $x \in \mathbb{R} \times \mathbb{R}$. Next let arbitrary $x \in \mathbb{R} \times \mathbb{R}$. Then there exists $A, B \in \mathbb{R}$ such that $x = \langle A, B\rangle$. The set $D = \{r \in \mathbb{Q} \mid r = a + b \text{ for some } a \in A \text{ and } b \in B\}$ exists by separation, and by definition $\langle\langle A, B\rangle, D\rangle \in +$. So $x = \langle A, B\rangle \in \text{dom}(+)$.

Next the range bound $\text{ran}(+) \subseteq \mathbb{R}$. Let arbitrary $y \in \text{ran}(+)$. Then there exists x such that $\langle x, y\rangle \in +$. By definition there exists $A, B \in \mathbb{R}$ such that $y = \{r \in \mathbb{Q} \mid r = a + b \text{ for some } a \in A \text{ and } b \in B\}$, which, since A, B are dedekind cuts, is a dedekind cut by lemma 1.85. Therefore $y \in \mathbb{R}$.

Finally functionality. Let $x \in \text{dom}(+)$. There then exists y where $\langle x, y\rangle \in +$, and therefore $A, B \in \mathbb{R}$ such that $\langle x, y\rangle = \langle\langle A, B\rangle, D\rangle$, where $D = \{r \in \mathbb{Q} \mid r = a + b \text{ for some } a \in A \text{ and } b \in B\}$. Let y' be arbitrary such that $\langle x, y'\rangle \in +$. Then there exists A', B' such that $\langle x, y'\rangle = \langle\langle A', B'\rangle, D'\rangle$, where $D' = \{r \in \mathbb{Q} \mid r = a + b \text{ for some } a \in A' \text{ and } b \in B'\}$. We know $\langle A', B'\rangle = x = \langle A, B\rangle$, so since D is unique by extensionality and D' is defined by the same condition, $y = D = D' = y'$. □

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Definition 1.89. *The unary operation $-$ on $\mathbb{R}$ is defined for $A \in \mathbb{R}$ as*

$$-A = \{r \in \mathbb{Q} \mid -r < c \text{ for some } c \in \mathbb{Q} - A\}$$

That is, the function $- : \mathbb{R} \to \mathbb{R}$,

$$\forall x(x \in - \leftrightarrow \exists A \in \mathbb{R} \exists D(\\ x = \langle A, D\rangle \wedge \forall r(r \in D \leftrightarrow r \in \mathbb{Q} \wedge \exists c \in \mathbb{Q} - A(-r < c))))$$

Where existence follows from using $\mathbb{R} \times \mathbb{R}$ as a bounding set.

Proof. (Range/Domain bounds, Functionality) First the domain bound $\text{dom}(-) = \mathbb{R}$. Let arbitrary $x \in \text{dom}(-)$. There exists y such that $\langle x, y\rangle \in -$. By definition then there exists $A \in \mathbb{R}$ such that $x = A$. So $x \in \mathbb{R}$. Next let arbitrary $x \in \mathbb{R}$. It follows that x is a dedekind cut. The set D, where

$$\forall r(r \in D \leftrightarrow r \in \mathbb{Q} \wedge \exists c \in \mathbb{Q} - x(-r < c))$$

exists by separation. By definition $\langle x, D\rangle \in -$. So $x \in \text{dom}(-)$.

Next the range bound $\text{ran}(-) \subseteq \mathbb{R}$. Let arbitrary $y \in \text{ran}(-)$. There exists x such that $\langle x, y\rangle \in -$. By definition there then exists $A \in \mathbb{R}$ and D such that $D = \{r \in \mathbb{Q} \mid -r < c \text{ for some } c \in \mathbb{Q} - A\}$, where $\langle x, y\rangle = \langle A, D\rangle$. By theorem 1.85, D is a dedekind cut. So $y = D \in \mathbb{R}$.

Finally functionality. Let arbitrary $x \in \text{dom}(-)$. There exists y such that $\langle x, y\rangle \in -$. So again by definition there exists $A \in \mathbb{R}$ and D such that $D = \{r \in \mathbb{Q} \mid -r < c \text{ for some } c \in \mathbb{Q} - A\}$, where $\langle x, y\rangle = \langle A, D\rangle$. Now let y' be arbitrary such that $\langle x, y'\rangle \in -$. Then there exists $A' \in \mathbb{R}$ and D' such that $D' = \{r \in \mathbb{Q} \mid -r < c \text{ for some } c \in \mathbb{Q} - A'\}$, where $\langle x, y'\rangle = \langle A', D'\rangle$. We know $A = x = A'$. So since D is unique by extensionality and D' is defined by the same condition, $y = D = D' = y'$. □

Definition 1.90. *For $A, B \in \mathbb{R}$, the relation $<$ on $\mathbb{R}$ is defined by $A < B$ iff $B \subsetneq A$. That is,*

$$\forall x(x \in < \leftrightarrow \exists A \in \mathbb{R} \exists B \in \mathbb{R}(x = \langle A, B\rangle \wedge B \subsetneq A))$$

The relation $\leq$ on $\mathbb{R}$ is defined by $A \leq B$ iff $B \subseteq A$. That is,

$$\forall x(x \in \leq \leftrightarrow \exists A \in \mathbb{R} \exists B \in \mathbb{R}(x = \langle A, B\rangle \wedge B \subseteq A))$$

It is straightforward to show $< \subseteq \mathbb{R} \times \mathbb{R}$ and $\leq \subseteq \mathbb{R} \times \mathbb{R}$. This justifies existence by separation and the domain/range bound.
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Lemma 1.91. *Let $A \subseteq \mathbb{Q}$ be a dedekind cut. Let $r \in \mathbb{Q}$.*

1. *$A \subsetneq D_r$ if and only if there is some $q \in \mathbb{Q} - A$ such that $r < q$.*
2. *$A \subseteq D_r$ if and only if $r \in \mathbb{Q} - A$ if and only if $r < a$ for all $a \in A$.*

Proof. (1) We want to show

$$A \subsetneq D_r \leftrightarrow \exists q \in \mathbb{Q} - A(r < q)$$

Where D_r is the rational cut at r:

$$D_r = \{x \in \mathbb{Q} \mid x > r\}$$

First suppose $A \subsetneq D_r$. This means $A \subseteq D_r$ and $A \neq D_r$. It follows that $D_r \nsubseteq A$ and that there exists $x \in D_r$ such that $x \notin A$. So we have $x \in \mathbb{Q} - A$, where by definition $r < x$.

Next suppose there exists $q \in \mathbb{Q} - A$ such that $r < q$. It follows by definition that $q \in D_r$. Since $q \notin A$, we must then have $D_r \neq A$. Let arbitrary $x \in A$. Since $q \in \mathbb{Q} - A$, by lemma 1.81 we have $q < x$ and therefore $r < x$. By definition $x \in D_r$ and $A \subset D_r$. So $A \subsetneq D_r$. □

Proof. (2) We want to show

$$A \subseteq D_r \leftrightarrow r \in \mathbb{Q} - A \leftrightarrow \forall a \in A(r < a)$$

The second equivalence follows from lemma 1.81. We just need to show the first. Suppose $A \subseteq D_r$. We know $r \in \mathbb{Q}$. Suppose for contradiction that $r \in A$. It then follows that $r \in D_r$ and that $r < r$, the contradiction. So $r \in \mathbb{Q} - A$.

Suppose $r \in \mathbb{Q} - A$. Let arbitrary $x \in A$. By lemma 1.81 we have $r < x$. So by definition $x \in D_r$. □

Lemma 1.92. *Let $A \in \mathbb{R}$ and let $r \in \mathbb{Q}$.*

1. *$A > D_r$ if and only if there is some $q \in \mathbb{Q} - A$ such that $q > r$.*
2. *$A \geq D_r$ if and only if $r \in \mathbb{Q} - A$ if and only if $a > r$ for all $a \in A$.*
3. *If $A < D_0$ then $-A \geq D_0$.*

Proof. (1) We want to show

$$A > D_r \leftrightarrow \exists q \in \mathbb{Q} - A(q > r)$$

Suppose $A > D_r$. Then by definition $A \subsetneq D_r$. Since A is real, by lemma 1.91 there exists $q \in \mathbb{Q} - A$ such that $q > r$.

Next suppose there exists $q \in \mathbb{Q} - A$ such that $q > r$. Then since A is real, by lemma 1.91 we have $A \subsetneq D_r$. Then by definition since $A, D_r \in \mathbb{R}$, $D_r < A$. □

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Proof. (2) Since A is real, by lemma 1.91,

$$A \subseteq D_r \leftrightarrow r \in \mathbb{Q} - A \leftrightarrow \forall a \in A(r < a)$$

Where by definition since $A, D_r \in \mathbb{R}$,

$$A \subseteq D_r \leftrightarrow D_r \leq A$$ □

Proof. (3) Suppose $A < D_0$. By definition then $D_0 \subsetneq A$, and there exists $q \in A$ such that $q \notin D_0$. Since $q \notin D_0$ we have $q \leq 0$. Since A is a dedekind cut we then have $0 \in A$ and then $0 \notin \mathbb{Q} - A$. Suppose for contradiction that $0 \in -A$. Then by definition there exists $c \in \mathbb{Q} - A$ such that $0 = -0 < c$, which by lemma 1.81 would imply $0 \in \mathbb{Q} - A$, the contradiction. So $0 \in \mathbb{Q} - (-A)$, which by part 2 of this lemma implies $-A \geq D_0$. □

Definition 1.93. *Where $\mathbb{R}_{\geq 0} = \{x \in \mathbb{R} \mid x \geq D_0\}$, we define the relation $\odot$ as the binary operation from $\mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0}$ into $\mathbb{R}$ where*

$$\begin{aligned}\forall x(x \in \odot \leftrightarrow \exists A \in \mathbb{R} \exists B \in \mathbb{R} \exists C \in \mathbb{R}(x = \langle \langle A, B \rangle, C \rangle \\ \wedge \forall y(y \in C \leftrightarrow \exists a \in A \exists b \in B(y = ab))))\end{aligned}$$